

Learned representations and kernel corrections for radiative-transfer emulation

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Abstract

We compare neural networks, exact Matérn kernel regression and their combinations as emulators of a six-dimensional radiative-transfer table on the 285-band EMIT grid, and we study how their forward accuracy carries over to reflectance retrieval. Over ten partitions of 23,313 states, a kernel with one length scale per input reaches 0.095% mean relative radiance error and a network 0.387%; a kernel on the concatenated hidden features of five networks reaches 0.076%. Forward and inverse rankings disagree. A convex stack improves on a wide feature kernel by only 0.0006 radiance percentage points, yet its 95th-percentile reflectance error is larger at every partition, three times larger on average. We prove that on the physical domain the constrained retrieval error is at most the retrieval-weighted component error divided by the transmission, with constant one, that the rate this implies is sharp, and that errors proportional to the transmission lose no rate. [[TQ-ABSTRACT]]

1 Introduction

Atmospheric correction for imaging spectroscopy relates the radiance measured at the sensor to the reflectance of the surface through radiative-transfer calculations that depend on the state of the atmosphere and on the viewing geometry. Learned emulators replace repeated simulator evaluations with a fitted map from that state to the spectral quantities the retrieval needs. Neural emulators have been built into reflectance retrieval [1, 2], Gaussian-process emulators have been developed for the OCO-2 forward model [3], and physics-guided multi-fidelity networks use a cheaper code as a second source of information [4]. Two questions follow: which representation gives the most accurate forward emulator, and which of its errors matter once the emulator is inverted.

We study exact kernel regression, fully connected networks and two post-hoc combinations of the two. In residual correction the network supplies a mean and a kernel fits its training residual. In feature regression the network supplies a frozen representation and a kernel fits the target on that representation [12]. Both differ from deep kernel learning, where representation and kernel are trained jointly [10], and from regularizing the inner layers of a network with kernel flows [11]. The comparison complements work on kernel methods for operator learning [5], on neural operators such as DeepONet and Fourier neural operators [6, 7], and classical kernel regression [8].

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The data are 23,313 tabulated radiative-transfer evaluations on the 285-band EMIT grid [13]. The state has six coordinates and each output component is well described by a few dozen principal coefficients, so an exact kernel solve on all 18,884 training rows is affordable. On ten common partitions, the kernel with one length scale per input is four times more accurate than the network on radiance, a residual kernel reduces the network’s error by a factor of almost three, and an exact kernel on the network’s hidden features is about as accurate as the input-scaled kernel. A wider network is a worse emulator but a better feature map: from width 512 to width 2,000 the network’s own radiance error doubles while the error of a kernel on its features falls by a fifth. Concatenating the hidden features of five networks and solving one kernel on them improves on the features of a single network at every partition and gives the lowest radiance error of all the ten-partition comparisons.

The central empirical result concerns the inverse. The stack built on the wide network improves mean radiance error over its own feature kernel by 0.0006 percentage points, while its mean within-split 95th-percentile reflectance error rises from 7.57 to 22.71 points, and the tail is worse at all ten partitions. Small forward gains can therefore coexist with large changes in the tail of the retrieval error. The reversal is not produced by the few entries where the inverse is undefined: it persists on the physical domain of the retrieval, with coverage and failure counts reported, [[TQ-INTRO]] and on a fresh partition evaluated once under a hypothesis fixed in advance.

The analysis has three levels. Exact finite-perturbation identities relate radiance and reflectance errors through the inverse denominator. A transfer inequality with constant one bounds a physically constrained retrieval by the component error relative to the transmission. Tail and conditional-error assumptions then give rates, including a sharp worst-case exponent and conditions under which no rate is lost; the central quantity is the profile of the conditional error against the transmission, not the error alone. On test data the transmission-weighted term of the bound ranks nine emulators exactly as their 95th-percentile retrieval errors, which the unweighted component error does not. For residual correction, a Hilbert-space identity gives the exact criterion for improvement, and a principal-component decomposition separates truncation error from the error of the kernel step.

Section 2 reviews related work and Section 3 describes the data and the protocol. Sections 4 and 5 give the models and the main comparison, and Section 6 the physical domain, the training targets and the fresh partition. Section 7 varies training size, output rank, width and tuning. Sections 8 and 9 develop the residual and retrieval analyses, and Sections 10–13 report the associated diagnostics. Proofs, extended tables, a comparison on OCO-2 and ten further emulation problems are in the appendices.

2 Related work

Radiative transfer and its surrogates. Relevant radiative-transfer codes include 6S [14], libRadtran [15, 16] and MODTRAN [17]; lookup-table calculations enter atmospheric correction [2] and optimal estimation [18]. The carbon-dioxide product of OCO-2 [19, 20] rests on the forward model and inverse method of [21, 22]. Surrogates enter because the forward model is the cost. Neural emulation of longwave radiation goes back to [23] and was carried into climate models by [24]; later work measures how accurate such emulators can be made [25, 26] and how learned parameterizations behave when the host model is stepped forward [27, 28]. In imaging spectroscopy and land-surface inference the same idea appears as emulator toolboxes for

radiative-transfer models [29, 30], Gaussian-process emulation for parameter retrieval [31], and the wider use of Gaussian processes in earth observation [32]; [1] places an emulator inside a reflectance retrieval and [3] fits a cross-validated Gaussian process to the OCO-2 forward model. Replacing an expensive simulator by a statistical model is a classical problem [33], as is using a cheap model to inform an expensive one [34, 35, 36], surveyed for propagation and inference in [37]; the network of [4] is a multi-fidelity instance on the corpus rescored in Section 13. The measurement requirements behind the tables used here are those of the EMIT mission [13] and of the surface-imaging missions that follow it [38].

Kernel regression. The estimators are classical: interpolation in a reproducing-kernel Hilbert space [39], smoothing splines and the representer theorem [40, 41, 42], ridge regularization [43] read as spectral filtering of an ill-posed problem [44], and the Matérn family, whose smoothness index fixes the differentiability of the sample paths [45, 8]. Rates under source and capacity conditions are given in [46, 47, 48, 49]; the textbook accounts are [50, 51, 52]. Each component here is a vector of principal coefficients, so the multi-output formulation is the relevant one [53, 54]. Kernels also serve as solvers rather than regressors: Bayesian numerical homogenization [55], operator-adapted wavelets [56] and the Gaussian-process treatment of nonlinear partial differential equations [57] use the same estimator, and cross-validating the kernel itself gives the kernel-flow family [58, 59, 11].

Cost. An exact solve is cubic in the number of training rows, which has produced a long line of approximations: the Nyström method and its analysis [60, 61, 62], random features [63] and their operator-learning counterpart [64], preconditioned solvers at large scale [65, 66], inducing-point and variational Gaussian processes [67, 68, 69], and structured interpolation with hardware-aware linear algebra [70, 71]. At $n = 18,884$ the exact factorization is affordable, and every full-data EMIT fit below is an exact solve.

Choosing and combining kernels. A kernel can be built rather than picked: spectral mixtures [72], additive structure [73], compositional search over a grammar [74], and per-coordinate relevance fitted by marginal likelihood [8] or by cross-validation [58]. Several kernels can be combined instead of one being chosen, by semidefinite programming [75], by the algorithms surveyed in [76], or by alignment with the target Gram matrix [77, 78]. Combining predictors rather than kernels is older still [79, 80, 81]. Selection on held-out rows is classical [82, 83, 84]; Section 9.2 records the finite-class guarantee it provides when the selection sample is independent of every earlier fitting decision.

Networks, and the kernels they induce. Feedforward networks are universal approximators [85]; those trained here use the standard adaptive optimizer [86] with decoupled weight decay [87] and early stopping [88]; dropout [89] is another standard regularizer. In the infinite-width limit a network is a Gaussian process [90, 91, 92] whose kernel is the arc-cosine kernel of [93], and gradient descent on a wide network follows the neural tangent kernel [94, 95]; both enter the kernel comparison of Section 11 as heads. The limit is not the whole story, since a finite network learns its representation [97], and the comparison between kernel machines and trained networks is itself unsettled [96]; a learned representation under a kernel is what deep kernel learning [10], the Gaussian process with a neural-operator mean [9] and the construction of [12] exploit. Kolmogorov–Arnold networks [98, 99] are the architecture behind the multi-fidelity model of [4].

Operator learning supplies the other comparison: DeepONet [6], Fourier neural operators [7] and the general formulation [100], with fair-comparison protocols [101], physics-informed training [102], the cost-accuracy trade-off [103], kernel competitors [5], and the benchmark collections [104, 105, 106] from which Appendix D draws.

3 Data and protocol

The data are $n = 23,313$ tabulated radiative-transfer evaluations provided by the Jet Propulsion Laboratory as a lookup table for EMIT atmospheric correction. Each input $x \in \mathbb{R}^6$ collects the atmospheric and viewing state: an aerosol optical depth in $[0.0500, 1.2000]$, a surface elevation in $[-0.4997, 5.9998]$ km, a water-vapour column in $[0.0502, 6.9999]$ cm, a relative azimuth in $[0.0029, 179.9947]^\circ$, a solar zenith angle in $[0.0012, 64.9981]^\circ$, and a view zenith angle in $[150.0003, 179.9991]^\circ$ from the downward vertical, that is, within 30° of nadir; these are the sampled extrema, over the ranges on which such tables are built [2, 17]. Each output consists of four spectral components on the 285-point EMIT grid spanning 381–2493 nm: path radiance Y_1 , direct flux Y_2 , diffuse flux Y_3 and spherical albedo Y_4 (Figure 1). The quantities used downstream are the radiance at a surface reflectance ρ ,

$$L(\rho) = Y_1 + \rho \frac{Y_2 + Y_3}{1 - \rho Y_4}, \quad (1)$$

and the reflectance recovered by inverting (1) with predicted components at the true radiance,

$$\hat{\rho} = \frac{L - \hat{Y}_1}{\hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4(L - \hat{Y}_1)}, \quad (2)$$

both evaluated at $\rho = 0.7$ in the main comparison. The table carries the states, the four components and the wavelength grid; the simulator version and its configuration are not part of it, and nothing below depends on them.

Protocol. The table is split into a test block of 2,331 rows and a remaining block of 20,982, of which 2,098 rows are held out for validation and 18,884 are used for fitting; the split is repeated with seeds 101–110. Standardization and principal components are fitted on the training rows of each split, hyperparameters, checkpoints, coordinatewise selections and combination weights are chosen on its validation rows, and every family is evaluated on its test rows. Unless stated otherwise, tables report the mean and sample standard deviation over the ten splits.

The model families and ablations were developed on this table with its test errors in view. The ten splits therefore measure how the comparison varies with the partition; their test blocks overlap, and their standard deviations are descriptive rather than confidence intervals. Section 5.1 reports paired differences on common splits. Two evaluations in Section 6 were specified before they were run: an experiment on the training targets, whose arms, seeds and scoring were recorded in a protocol before its first fit, and a fresh partition with a single hypothesis fixed in advance and evaluated once.

Structure of the outputs. Three properties of the table shape the comparison. First, the outputs are low rank. On standardized training outputs, 99.99% of the variance of every component lies in the first 11–25 principal components and 99.9999% in at most 44 (Figure 2,

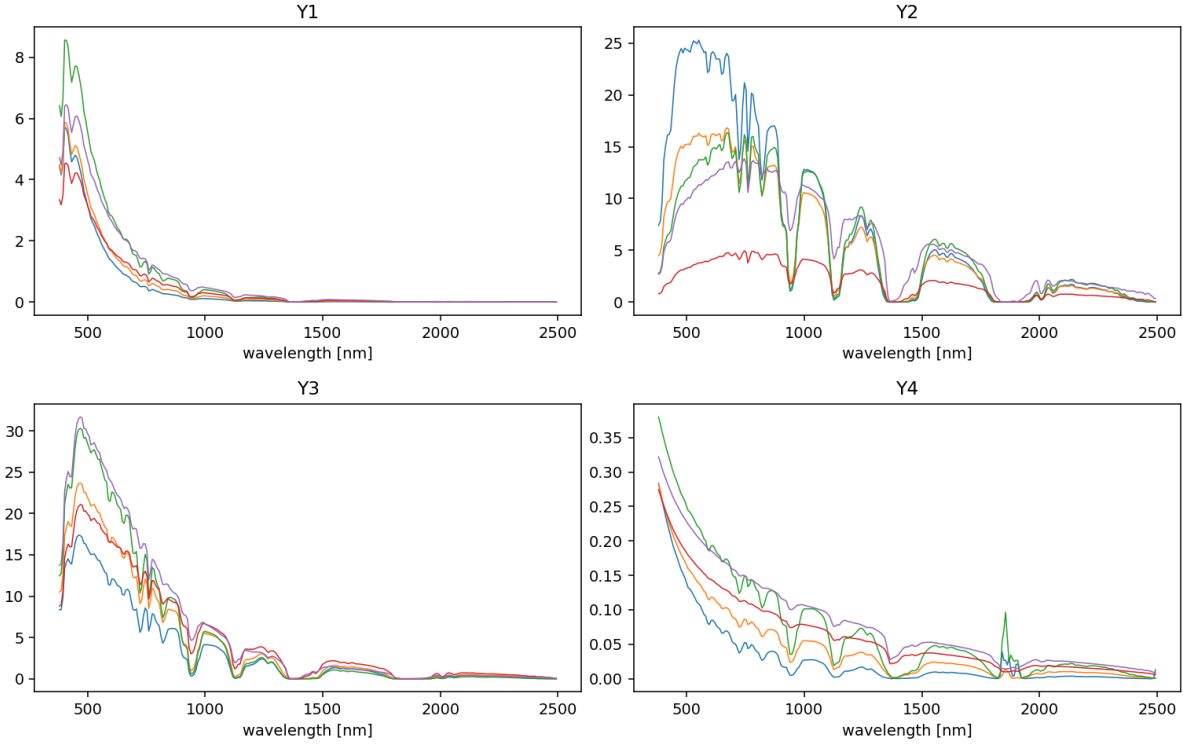


Figure 1: Example output curves across the EMIT wavelength grid, five samples per panel. The curves are smooth, share the deep water-vapor absorption features near 1380 and 1900 nm, and differ mainly by smooth rescalings along the state axes.

left), and the smallest correlation between adjacent bands is 0.994952. All regressors below act on the first 64 principal coefficients, whose unexplained training variance is below 10^{-6} , except the networks, which are trained on the 285 standardized bands. A small variance tail is an aggregate statement; Section 8 states the projection error explicitly.

Second, the inputs matter unequally. Shuffling the aerosol coordinate reduces the validation R^2 of a linear model by 0.89, the solar zenith angle by 0.54, and elevation and water vapor by about 0.1 each; the two remaining angles have smaller effects (Figure 2, right). This motivates a separate length scale for each input.

Third, low-degree surrogates are already accurate. Linear regression on the standardized state reaches $R^2 = 0.83$, against 0.74 for distance-weighted five-nearest-neighbors, and cubic ridge regression has a mean relative component error of about two percent (Table 1).

Conditioning of the inverse. In deep absorption bands the transmitted flux $Y_2 + Y_3$ can be as small as 10^{-21} , and at a fully opaque band the radiance does not depend on the surface reflectance at all, so the inverse is not identifiable there. Even the exact tabulated components do not invert exactly in floating point: over the whole table the round trip has reflectance RMSE 7.4×10^{-3} , all of it from the entries with the smallest transmissions (Section 6). We therefore report both the center and the tail of the reflectance error. In the all-band statistics a non-finite inverse is assigned zero and counted; none occurs in the main comparison.

Differentiating (2) separates the components. Write $D = \hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4(L - \hat{Y}_1)$ for its

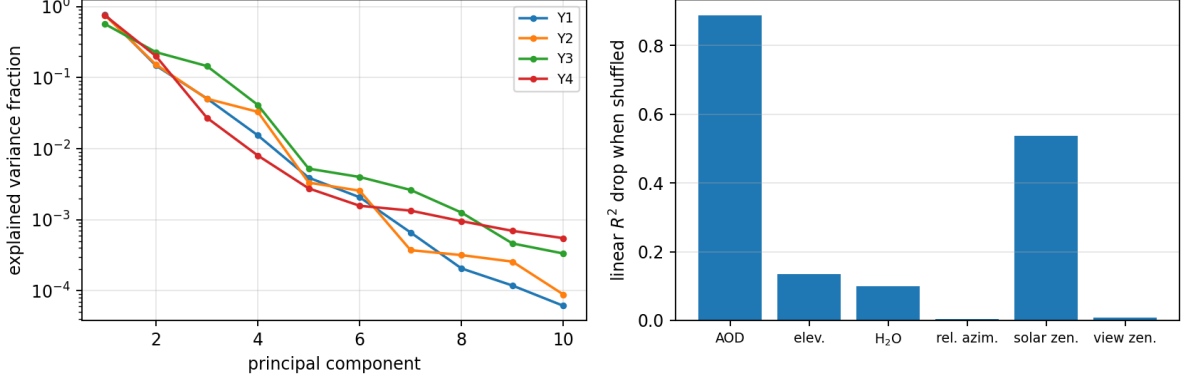


Figure 2: Left: explained-variance spectra of the standardized outputs. Every component concentrates 99.99% of its variance in 11–25 principal components out of 285 bands. Right: drop in linear validation R^2 when a single standardized input coordinate is shuffled. Aerosol optical depth and solar zenith angle dominate; the azimuth and view-zenith coordinates are almost inert at first order.

denominator. Then

$$\frac{\partial \hat{\rho}}{\partial \hat{Y}_4} = -\hat{\rho}^2, \quad \frac{\partial \hat{\rho}}{\partial \hat{Y}_2} = \frac{\partial \hat{\rho}}{\partial \hat{Y}_3} = -\frac{\hat{\rho}}{D}, \quad \frac{\partial \hat{\rho}}{\partial \hat{Y}_1} = -\frac{\hat{Y}_2 + \hat{Y}_3}{D^2}. \quad (3)$$

At exact identifiable components $\hat{\rho} = \rho = 0.7$, so the albedo derivative is -0.49 whatever the transmitted flux, while the path-radiance and flux derivatives grow without bound as D decreases. Errors in path radiance and flux are therefore the ones that matter near opaque bands.

3.1 Finite-perturbation control of the inversion

The derivatives in (3) describe local conditioning. The following elementary identities give a finite-error test and distinguish forward accuracy from inversion stability. For one sample and band, write $a = Y_1$, $t = Y_2 + Y_3$, $s = Y_4$ and $L = a + \rho t / (1 - \rho s)$. Set $e_a = \hat{a} - a$, $e_t = \hat{t} - t$, $e_s = \hat{s} - s$.

Proposition 1 (Conditional radiance and reflectance stability). *Suppose $0 \leq \rho \leq \rho_{\max}$, $t > 0$ and $q = 1 - \rho s > 0$. If also $\hat{q} = 1 - \rho \hat{s} > 0$, the forward error satisfies the exact identity*

$$\hat{L} - L = e_a + \frac{\rho e_t}{\hat{q}} + \frac{\rho^2 t e_s}{q \hat{q}}. \quad (4)$$

In particular, if $q, \hat{q} \geq \kappa > 0$,

$$|\hat{L} - L| \leq |e_a| + \rho_{\max} |e_t| / \kappa + \rho_{\max}^2 |t e_s| / \kappa^2.$$

For inversion at the true radiance, put $u = L - a = \rho t / q$, $D = t / q$, and $\hat{D} = \hat{t} + \hat{s}(L - \hat{a})$. Whenever $\hat{D} \neq 0$,

$$\hat{\rho} - \rho = -\frac{(1 - \rho \hat{s})e_a + \rho e_t + \rho u e_s}{\hat{D}}, \quad \hat{D} - D = e_t + u e_s - \hat{s} e_a. \quad (5)$$

If $B = |e_t| + |ue_s| + |\widehat{se}_a| \leq \eta D$ for some $0 \leq \eta < 1$, then $\widehat{D} \geq (1 - \eta)D > 0$ and

$$|\widehat{\rho} - \rho| \leq \frac{|1 - \rho\widehat{s}||e_a| + \rho|e_t| + \rho|ue_s|}{(1 - \eta)D}. \quad (6)$$

Proof in Appendix A. When $\widehat{q} \neq 0$, the same calculation gives the exact forward-inverse link

$$\widehat{\rho} - \rho = -\frac{\widehat{q}}{\widehat{D}}(\widehat{L} - L). \quad (7)$$

Consequently, on a common set where $|\widehat{q}| \leq Q$ and $|\widehat{D}| \geq \delta > 0$, the corresponding unweighted L^2 errors obey $\|\widehat{\rho} - \rho\|_2 \leq (Q/\delta)\|\widehat{L} - L\|_2$. Without the denominator condition there is no uniform implication from small spectral relative radiance error to small reflectance error. For example, take two bands with $a = s = 0$, $t = (1, \varepsilon)$ and $0 < \rho < 1$. A predictor with $\widehat{t} = t$, $\widehat{s} = 0$ and $\widehat{a} = (0, c\varepsilon)$, where $0 < c < \rho$, has relative spectral radiance error $c\varepsilon/(\rho\sqrt{1 + \varepsilon^2}) \rightarrow 0$ but retrieves $(\rho, \rho - c)$, so its reflectance RMSE remains $c/\sqrt{2}$. Every denominator is positive for $\varepsilon > 0$ and both retrieved reflectances remain physical. This example isolates the loss of inversion stability as the second-band flux vanishes.

The forward bound need not deteriorate when t tends to zero, whereas the inverse bound requires error control relative to $D = t/q$. Thus a small spectral radiance error does not certify stable reflectance recovery. At $t = 0$, $L = a$ is independent of ρ wherever the forward formula is defined, so reflectance is unidentifiable from this band. The identities are exact-arithmetic statements; floating-point cancellation in $L - a$ can further affect an implemented inverse.

An empirical comparison should therefore fix its passband and conditioning threshold before the models are evaluated. The scorer used in Section 6 takes a positive flux scale from the training rows and a mask computed from the true components alone, common to all families, so that a failing model cannot change its own coverage; it reports the coverage, the non-finite inversions and the denominator failures separately from the error on the successful inversions.

4 Models

All models are trained per component and evaluated in physical units after undoing the standardization. Unless stated otherwise they regress the 64 leading principal coefficients of the standardized component. Every hyperparameter named below is selected on the validation rows of the split at hand. The comparisons of Sections 5–7.1 run on CPUs; most diagnostics of Sections 10–13 train their networks on one T4 GPU.

Reference family. Cubic polynomial ridge regression [43] on all monomials of degree at most three in the six standardized inputs, with the ridge weight selected from $\{10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$.

Exact kernel ridge regression. Three constructions, all Matérn kernels [45] on the standardized inputs, with the length scale expressed as a multiple of the median pairwise distance of the fit set. The nugget γ enters the system as $(K + n\gamma I)\alpha = Z$. The subset baseline is a Matérn-5/2 kernel fitted on a uniformly drawn subset of 4,000 training points, the landmark choice of a Nyström approximation [60, 61], with multiplier in $\{1, 2\}$ and nugget in $\{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. The second is the exact solve on every training row: smoothness $\nu \in \{3/2, 5/2\}$, multiplier in

$\{0.5, 1, 2, 4\}$ and the same nugget grid, tuned on validation from a 6,000-row training subsample, with the winner refit on all 18,884 rows by one Cholesky factorization [107]. The third gives each input its own length scale, an automatic relevance determination [90, 8]: starting from the isotropic winner, a coordinate search over per-input multipliers in $\{1/4, 1/2, 1, 2, 4\}$ runs for two sweeps scored on validation, after which smoothness, scale and nugget are re-tuned at the metric found and the solve is refit on every row.

Fully connected networks. Three ELU hidden layers of 512 units per component, trained on the 285 standardized bands with AdamW [86, 87] at learning rate 10^{-3} , cosine decay, batch size 1024 and weight decay 10^{-6} , for up to 150 epochs with early stopping on validation loss [88] (patience 25). Five networks are trained per component and split from different initializations; the tables report the first and the mean of the five.

Kernel corrections of the neural mean. The construction of [12], with every kernel stage an exact solve on all training rows. (i) *Residual correction*: an exact Matérn regression, on the standardized inputs, of the network’s training residuals in coefficient space, added back to the network’s projection into that space, with smoothness, scale and nugget selected by the validation error of the corrected prediction; it is applied to the single network and to the five-network mean. (ii) *Kernel on features*: the same exact regression with the input metric replaced by the network’s last hidden layer, standardized. This is the post-hoc counterpart of deep kernel learning [10], in which representation and kernel are trained together, and of the kernel-flow regularization of a network’s inner layers [11]; here the representation is fixed first and the kernel is solved exactly on it. (iii) *Per-coordinate selection*: for each of the 64 output coefficients, the validation winner among the exact kernel, the network, the ensemble, the two corrected predictors and the feature kernel. (iv) *Convex stack*: per component, a convex combination of the same six heads with weights fitted on the validation rows in the physical relative metric, in the manner of stacked generalization [80] and stacked regression [81].

Sample size, rank and width. Section 7 repeats the isotropic exact kernel, the network, its residual correction and the feature kernel at nested training sizes, at other output ranks and at other widths; the main table is the full-size, rank-64, width-512 cell of that grid.

5 Results

Table 1 compares the model families at 18,884 training rows, output rank 64 and network width 512.

Polynomial and subset-kernel baselines. Cubic ridge regression gives 1.86% mean component error and 0.760% radiance error. The 4,000-point Matérn fit reduces these to 0.88% and 0.332%. Its albedo error depends strongly on the split: about 1.9% on seven splits and 3.3–4.3% on the other three.

Full-data and input-scaled kernels. The isotropic kernel on all training rows reduces the radiance error to 0.239% and the path-radiance error to 0.08%, against 0.17% for the subset fit. Its albedo error is 2.3–2.9%: steadier than the subset fit, but larger than that fit on its seven best splits. A separate length scale for each input brings the radiance error to 0.095%, the mean

model	component rel. L^2 [%]				radiance	reflectance
	Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $ \hat{\rho} - \rho $ [pp]
Cubic ridge	1.16±0.02	2.18±0.04	1.11±0.03	2.98±0.06	0.760±0.011	0.471±0.010
Matérn KRR, 4000-point fit	0.17±0.01	0.61±0.04	0.29±0.02	2.44±0.87	0.332±0.029	0.178±0.030
Matérn KRR, exact on all rows	0.08±0.01	0.42±0.03	0.20±0.00	2.62±0.18	0.239±0.016	0.141±0.018
Matérn KRR, per-input scales	0.08±0.01	0.16±0.02	0.08±0.01	2.18±0.16	0.095±0.009	0.036±0.004
FC-DNN (3×512)	0.48±0.02	0.97±0.04	0.41±0.02	3.02±0.18	0.387±0.012	0.245±0.008
Five-network mean	0.37±0.01	0.67±0.03	0.28±0.01	2.51±0.09	0.251±0.006	0.163±0.003
DNN + residual KRR	0.10±0.00	0.30±0.02	0.16±0.01	2.43±0.08	0.141±0.006	0.089±0.008
Ensemble + residual KRR	0.08±0.00	0.27±0.03	0.13±0.00	2.42±0.17	0.123±0.007	0.080±0.008
Kernel on network features	0.18±0.01	0.17±0.01	0.10±0.01	1.99±0.08	0.098±0.007	0.054±0.004
Per-coordinate selection	0.19±0.01	0.44±0.10	0.13±0.02	2.25±0.18	0.193±0.022	0.123±0.023
Convex stack	0.07±0.00	0.16±0.01	0.09±0.00	1.94±0.06	0.087±0.006	0.048±0.003

Table 1: Test errors under the protocol of Section 3, mean $\pm$ sample standard deviation over ten splits. Component and radiance columns report mean relative L^2 error; radiance is evaluated at $\rho = 0.7$. The last column is the median absolute reflectance error in percentage points, over all bands, deep absorption windows included. Table 3 gives the corresponding 95th percentiles and RMSE.

component error to 0.63% and the median reflectance error to 0.036 percentage points. The fitted scales reflect the weak azimuth dependence of the flux components: the relative-azimuth length scale is sixteen to thirty-two times the others, with median factors of sixteen for direct flux and albedo and thirty-two for diffuse flux, on every split for the flux components and on nine of ten for albedo. The view-zenith scales are two to sixteen times longer, and the path-radiance metric is nearly isotropic.

Networks and kernel corrections. The network has 1.22% mean component error and 0.387% radiance error; averaging five initializations gives 0.96% and 0.251%. Residual kernel regression reduces the radiance error of the single network to 0.141% and that of the ensemble to 0.123%. The feature kernel has 0.61% mean component error and 0.098% radiance error, level with the input-scaled kernel: it is lower on six of ten splits, though its mean is slightly higher. Its median reflectance error, 0.054 points against 0.036, is larger on every split.

Selection and stacking. Both combiners use the isotropic full-data kernel, the network, its five-member ensemble, the two residual-corrected predictors and the feature kernel; the input-scaled kernel is not in the pool. Coordinatewise selection gives 0.193% radiance error, worse than several of its members. The convex stack gives 0.087% radiance error, 0.57% mean component error and 0.048 points of median reflectance error, about 8% below the input-scaled kernel and 11% below its best member, the feature kernel, on radiance. On mean component error it improves on the input-scaled kernel at nine splits, by 0.06 ± 0.05 points, and on the feature kernel at all ten, by 0.047 ± 0.006 points.

The residual second moments show why a combination can help. On path radiance, the residuals of the isotropic kernel and of the five-network ensemble have correlation 0.27 ± 0.09 , below their error ratio of about 0.34 at every split; for a quadratic loss this is the classical condition under which a convex combination improves on its better member [79, 12]. The stack minimizes a mean relative norm instead, and the reflectance is a nonlinear function of the components, so the condition is suggestive rather than decisive for those metrics. The two

candidate / reference	mean gain $\pm$ SD [pp]	gain range [pp]	wins/10
Input-scaled / isotropic KRR	0.1436 $\pm$ 0.0122	[0.1196, 0.1585]	10
Residual correction / network	0.2464 $\pm$ 0.0090	[0.2365, 0.2615]	10
Narrow stack / input-scaled KRR	0.0083 $\pm$ 0.0066	[-0.0054, 0.0185]	9
Wide feature kernel / input-scaled KRR	0.0165 $\pm$ 0.0066	[0.0054, 0.0295]	10
Wide / narrow feature kernel	0.0191 $\pm$ 0.0079	[0.0096, 0.0352]	10
Wide stack / wide feature kernel	0.0006 $\pm$ 0.0026	[-0.0053, 0.0037]	8

Table 2: Paired radiance contrasts on seeds 101–110. Gain is the reference error minus the candidate error, in percentage points (pp). SD is the sample standard deviation of the ten paired differences; the range and the win count describe the same ten differences.

residual-corrected predictors are far more strongly correlated, about 0.88 on path radiance and 0.99 on albedo.

Spherical albedo. Albedo has the largest relative error for every family: its small values in the short-wave infrared make relative errors sensitive to small absolute ones. The width-512 stack reaches 1.94%. The learning curves of Section 7 show little improvement with sample size on this component; Section 6 tests whether the out-of-domain training targets contribute to that plateau.

5.1 Paired comparisons and the radiance–reflectance trade-off

Paired differences compare two families on the same split. Table 2 reports them for the main and wide pipelines, with positive gains favoring the candidate. The wide feature kernel improves on the input-scaled kernel at all ten splits, by 0.0165 ± 0.0066 radiance percentage points. The wide stack’s improvement over its own feature kernel is 0.0006 ± 0.0026 points, with eight wins and two losses: small against the split-to-split variation of the difference itself.

Radiance and reflectance order the families differently. Table 3 reports the all-band reflectance tails beside the medians; each entry averages a within-split statistic, so the mean of ten 95th percentiles is not a pooled 95th percentile. For the width-512 pipeline the convex stack lowers the radiance error relative to the feature kernel but raises the mean 95th percentile of the reflectance error from 7.75 to 29.44 points, and the tail is worse at all ten splits, by factors between 2.63 and 4.94. The width-2000 pipeline repeats the reversal: 7.57 points for the feature kernel and 22.71 for the stack, worse at all ten splits by factors between 2.36 and 3.79. The near-tie at 0.079% against 0.078% radiance error hides a threefold difference in the inverse.

The input-scaled kernel shows the same split between center and tail: its median error is only 0.0365 points but its mean 95th percentile is 31.16 points, while the feature kernels have slightly larger medians and much smaller tails. The kernel on concatenated features has the smallest tail of all, 5.93 points. The identities of Section 3.1 tie the inverse error to the predicted denominator as well as to the forward error, which is why optimizing a component or radiance error alone need not control the reflectance tail.

model	radiance [%]	median [pp]	95th percentile [pp]	RMSE [pp]
Cubic ridge	0.760±0.011	0.471±0.010	70.68±0.02	4.18e+03
Isotropic KRR	0.239±0.016	0.141±0.018	69.53±0.42	4.18e+03
Input-scaled KRR	0.095±0.009	0.036±0.004	31.16±6.21	1.34e+04
Narrow feature kernel	0.098±0.007	0.054±0.004	7.75±1.13	1.78e+03
Narrow convex stack	0.087±0.006	0.048±0.003	29.44±5.89	1.25e+03
Concatenated features, 5×512	0.076±0.007	0.039±0.004	5.93±0.47	1.06e+04
Wide feature kernel	0.079±0.009	0.045±0.006	7.57±0.95	1.40e+03
Wide convex stack	0.078±0.011	0.043±0.005	22.71±4.42	5.81e+03

Table 3: All-band retrieval metrics over the ten splits of the main, concatenated-feature and wide pipelines. Radiance is the mean relative L^2 error; the reflectance median, 95th percentile and RMSE are in percentage points at $\rho = 0.7$. The first three columns show mean $\pm$ sample standard deviation across splits and RMSE its mean. The inverse is neither clipped nor restricted to the physical domain; every inverse in these runs is finite, and the RMSE is dominated by large finite values near zeros of the denominator. Section 6 gives the same comparison on the physical domain.

6 The physical domain, the training targets and a fresh partition

The all-band scores of Table 3 mix three things: the error of the emulator, entries where the retrieval is not defined, and entries where it is defined but badly conditioned. This section separates them. It describes the physical domain of the retrieval and the entries of the table that fall outside it, scores the main pipeline on that domain with its coverage and failures, asks whether the out-of-domain entries matter when they are used as training targets, and repeats the forward–inverse comparison on a fresh partition fixed in advance.

6.1 The physical domain

Write $t = Y_2 + Y_3$, $s = Y_4$ and $q = 1 - \rho s$, with $\rho = 0.7$. The retrieval analysis of Section 9 holds on

$$\mathcal{A} = \{t > 0, 0 \leq s < 1\},$$

which is a sufficient domain for the physical retrieval rather than the largest set on which the rational formulas are defined: a negative albedo with $t > 0$ gives $q > 0$ and a well-defined inverse, and $q < 0$ violates the physical hypothesis without being a zero of a denominator. At $q = 0$ the forward formula is undefined, and at $t = 0$, wherever it is defined, $L = Y_1$ does not depend on the reflectance, so the band carries no information about it. A small positive transmission makes the inverse ill-conditioned but not undefined.

Table 4 counts the entries of the table outside $\mathcal{A}$. There are 438 nonpositive transmissions, the smallest -4.5×10^{-20} , and 32,130 transmissions below 10^{-12} ; 9,858 negative albedos, the smallest -0.0724 ; and 454 albedos at or above one, 214 of which equal 2 exactly, which can only be a fill value. At 448 entries $q \leq 0$, the smallest -0.4 . In all, 10,356 entries (0.156%) lie outside $\mathcal{A}$. They occupy 227 bands but are concentrated near 1841–1908 and 2485–2493 nm, and 18,859 of the 23,313 states are admissible at every band.

A stricter domain adds a conditioning floor,

$$\mathcal{C} = \mathcal{A} \cap \{t \geq \tau, q \geq \kappa\}, \quad \tau = 10^{-12}, \quad \kappa = 0.3,$$

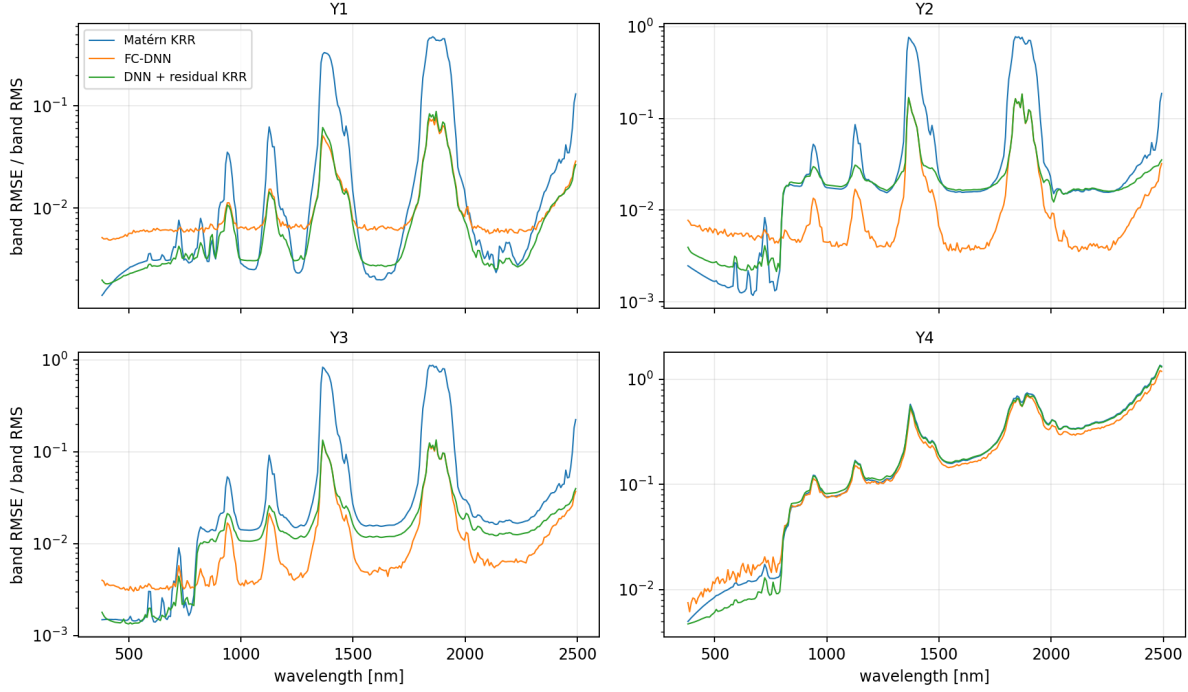


Figure 3: Where along the spectrum each family makes its error: band RMSE divided by the band’s root-mean-square value over the test rows, against wavelength, for the isotropic Matérn kernel, the 3×512 network and the network with its residual kernel. The kernel’s error follows the band structure, rising by two orders of magnitude in the water-vapour bands near 940, 1140, 1380 and 1870 nm; the network’s is nearly flat across the spectrum and higher in the windows; the residual correction keeps the network’s floor in the windows and most of the kernel’s advantage outside them. On the spherical albedo Y_4 the three families agree to within a factor of two at every wavelength, and all of them degrade towards 2500 nm. The bandwise relative RMSE is a different aggregate from the per-sample relative norm of Table 1.

with τ in the flux units of the table; at $\rho = 0.7$ the albedo condition already gives $q > 0.3$. Inverting the exact tabulated components returns the reflectance to within floating-point error, and that error is informative:

domain	round-trip RMSE (reflectance)	coverage
all entries	7.38×10^{-3}	100%
$\mathcal{A}$	4.27×10^{-3}	99.844%
$\mathcal{C}$	1.08×10^{-9}	99.516%

In exact arithmetic the round trip is exact wherever it is defined. In floating point, adding $\rho t/q$ to Y_1 and subtracting Y_1 again loses the added term when it is small relative to the spacing of numbers near Y_1 , which is what happens at the smallest transmissions. Removing 0.33% of the entries takes the round-trip error from 4×10^{-3} to 10^{-9} ; what remains of the all-band reflectance tail on $\mathcal{C}$ belongs to the emulators.

condition on the tabulated truth	entries	share
transmitted flux $t = Y_2 + Y_3 \leq 0$	438	0.0066%
$t < 10^{-12}$	32,130	0.4836%
albedo $Y_4 < 0$	9,858	0.1484%
albedo $Y_4 \geq 1$	454	0.0068%
albedo exactly 2 (reported fill marker)	214	0.0032%
admissible $\mathcal{A} = \{t > 0, 0 \leq Y_4 < 1\}$	6,633,849	99.8441%

Table 4: The table against the physical domain, over all $23,313 \times 285 = 6,644,205$ state–band entries. The categories overlap. Small positive transmission is a conditioning property, not a violation of $t > 0$; an albedo outside $[0, 1)$ violates the physical domain without necessarily leaving the algebraic domain of the inverse.

family	radiance [%]	all bands p_{95} [pp]	$\tau_0 = 10^{-12}$ failed [%] p_{95} [pp]	$\tau_0 = 10^{-3}$ failed [%] p_{95} [pp]	$\tau_0 = 10^{-2}$ failed [%] p_{95} [pp]
cubic ridge (3×512)	0.754	70.67	4.39	45.78	1.46
isotropic kernel (3×512)	0.213	69.06	2.85	17.70	0.42
input-scaled kernel (3×512)	0.088	23.53	1.78	5.19	0.07
network (3×512)	0.376	65.96	2.52	10.16	0.11
network + residual kernel (3×512)	0.139	46.38	2.05	6.68	0.04
kernel on features (3×512)	0.105	8.26	1.41	2.62	0.00
convex stack (3×512)	0.093	22.91	1.69	4.26	0.01

Table 5: The main pipeline on the physical domain: the raw arm, mean $\pm$ sample standard deviation over [[TQ-SEEDS]] splits. Radiance is the mean relative error in percent, and the all-band column is the unrestricted inverse of Table 3. At each floor τ_0 , the failed column is the percentage of covered entries at which the inversion fails, and p_{95} is the 95th percentile of the absolute reflectance error over the successful inversions, in percentage points.

6.2 The main pipeline on the physical domain

The families of the main comparison were refitted on the ten splits at both network widths with their float64 test predictions kept; these are the raw arm of the experiment in Section 6.3. Each family is scored on the test entries that satisfy

$$t \geq \tau_0 T_{\text{train}}, \quad 0 \leq s < 1, \quad q \geq 0.3,$$

where T_{train} is the median positive transmission of the split’s training block (4.67 at split 101) and $\tau_0 \in \{10^{-12}, 10^{-3}, 10^{-2}\}$; the three floors were fixed before the first fit, and at $\rho = 0.7$ the albedo condition already gives $q > 0.3$. An inversion fails when its predicted denominator is not positive or its value is not finite. Failures are counted, and the percentiles are taken over the successful inversions. Coverage depends on the truth alone and is the same for every family.

[[TQ-DOMAIN]]

6.3 Training on physically admissible targets

The training pipeline standardizes, compresses and fits the targets as they come, including the 0.156% of entries outside $\mathcal{A}$ and the fill values among them. Scoring on $\mathcal{A}$ cannot tell whether those entries shaped the fitted representations, the principal components or the albedo plateau

family	arm	Δ radiance [pp]	lower	Δ albedo [%]	Δ all-band p_{95} [pp]	Δ physical p_{95} [pp]	lower
cubic ridge (3×512)	admissible	0.0297	0/1	-0.915	0.01	2.78	0/1
	matched	0.0014	0/1	0.157	0.00	0.13	0/1
isotropic kernel (3×512)	admissible	0.0279	0/1	-0.689	1.21	5.93	0/1
	matched	0.0365	0/1	0.323	0.55	4.77	0/1
input-scaled kernel (3×512)	admissible	-0.0044	1/1	-0.582	17.84	1.40	0/1
	matched	0.0061	0/1	0.158	13.39	0.66	0/1
network (3×512)	admissible	0.0250	0/1	-1.241	2.38	1.38	0/1
	matched	0.0205	0/1	-0.976	1.36	0.30	0/1
network + residual kernel (3×512)	admissible	0.0016	0/1	-0.725	13.45	1.89	0/1
	matched	-0.0019	1/1	-0.592	4.91	0.84	0/1
kernel on features (3×512)	admissible	-0.0213	1/1	-0.489	2.44	0.49	0/1
	matched	-0.0083	1/1	-0.402	1.90	0.49	0/1
convex stack (3×512)	admissible	-0.0171	1/1	-0.469	14.54	0.79	0/1
	matched	-0.0081	1/1	-0.377	5.30	0.29	0/1

Table 6: Training on admissible targets: paired differences from the raw arm, arm minus raw, mean $\pm$ sample standard deviation over [[TQ-SEEDS]] splits. Radiance differences are in percentage points of relative error, albedo differences in percentage points of the relative albedo error, and the reflectance percentiles in percentage points; the physical p_{95} uses the floor $\tau_0 = 10^{-12}$. The columns headed lower count the splits on which the arm is below the raw arm.

width	arm	stack – features radiance [pp]	lower	tail ratio all bands	> 1	tail ratio $\tau_0 = 10^{-12}$	tail ratio $\tau_0 = 10^{-3}$	tail ratio $\tau_0 = 10^{-2}$	> 1
3×512	raw	-0.0113	1/1	2.77	1/1	1.63	1.29	1.15	1/1
	admissible	-0.0071	1/1	3.50	1/1	1.62	1.46	1.22	1/1
	matched	-0.0111	1/1	2.78	1/1	1.46	1.36	1.18	1/1

Table 7: The convex stack against the kernel on features in each arm: the radiance difference, stack minus feature kernel, in percentage points, and the ratio of their 95th-percentile reflectance errors over all bands and on the physical domain at the three floors; mean $\pm$ sample standard deviation over [[TQ-SEEDS]] splits. The columns headed > 1 count the splits on which the stack has the larger tail, over all bands and at $\tau_0 = 10^{-12}$.

of Section 7; only refitting can. We therefore ran the pipeline under three training policies: the original one (*raw*); removal of every training state with an entry outside $\mathcal{A}$ at any band (*admissible*), with no clipping or imputation; and an unfiltered random subset of the raw training rows of the same size as the admissible set (*matched*), which separates the effect of the filter from that of losing rows. Whole states are removed so that every remaining spectrum is complete for the principal-component step; this changes the training distribution, and the matched arm measures how much of any difference is only the loss of 19% of the rows. The validation rows, the validation loss and the test blocks are those of the raw pipeline in all three arms, and every arm is scored on the same test entries with a flux scale taken from the original training block.

The arms, the two network configurations of Table 1 and Table 12, the seeds 101–110, the scoring and the summaries reported below were fixed in a written protocol before the first run, and every run was completed and is reported. Each run records the digests of its data and of its split indices, and saves the float64 predictions of every family. Splits 101–104 ran on one machine and splits 105–110 on another, and the three arms of a split always ran on the same machine, so every paired difference below is taken within one computing environment (Table 19).

[[TQ-POLICY]]

6.4 Where the reflectance error sits

Figure 4 shows where along the spectrum the errors of one split sit. The component and radiance residuals are smooth in wavelength and widen where the transmitted flux is small, so the families differ mainly in the amplitude of a shared shape. The reflectance residual does not inherit that shape. Over all bands its outer envelopes open into vertical walls, bands where the 95th percentile of the absolute error exceeds two points, and the walls have two parts. A common core of 54 bands, at 1126–1134, 1342–1469, 1796–1952 and 2412–2493 nm, is a wall for every family, the most accurate included: there the denominator, not the emulator, sets the width. Outside the core each family adds bands in proportion to its inaccuracy: none for the kernel on features, 3 for the stack, 12 for the input-scaled kernel, 22 for the corrected network, 29 for the network and 79 for the cubic ridge, of a 133-band union. Restricting to the physical domain changes none of these counts, and neither does the stricter floor $\tau_0 = 10^{-3}$, which removes 6.3% of the entries: the walls come from transmissions that are small but far above either floor, where the retrieval divides a small component error by a small denominator. The feature kernel’s advantage in the tail is that it adds no wall of its own.

Theorem 6 locates the walls in advance. Its item (i) holds at every entry, so at each band the 95th percentile of the constrained retrieval error is at most the 95th percentile of $\min\{R, e_R/t\}$, which uses only the component errors and the true transmission. On split 101, with $R = 0.9$ and S the largest training albedo not above one, the constrained retrieval on the physical domain has exactly the walls of Figure 4, family by family, and no entry violates the pointwise inequality. For every family each wall lies in the set of bands where the bound’s 95th percentile exceeds two points; that set is larger by 8 to 21 bands, and on the walls the bound exceeds the realized 95th percentile by a median factor of 1.30 to 1.38, depending on the family, and never by more than 4.5. The bands marked by the bound for all six families number 64 and include the whole common core of 54.

6.5 A fresh partition, fixed in advance

The comparisons above use partitions that were in view while the families were developed. To test the reversal against that selection, a simplified pipeline was frozen together with one hypothesis and evaluated once on a new partition. Its six families are the cubic ridge, a 3×512 network (GELU activations, Adam) trained on the 285 standardized bands, the residual kernel of that network, the input-scaled Matérn kernel, the Matérn kernel on the network’s last hidden layer, and a convex stack of the cubic, the corrected network, the input-scaled kernel and the feature kernel. Every hyperparameter was copied from development on seed 101 and none was re-tuned. The partition holds out 3,497 states, 15% of the table, drawn with a seed used in no earlier run; the remaining 19,816 states train the families, 1,982 of them serving only as the network’s early-stopping set. The held-out states belong to the same table as every other split, so this evaluation guards against the choice of split and of hypothesis rather than testing generalization to new states.

The hypothesis was that the family with the lowest radiance error does not have the lowest all-band 95th-percentile reflectance error, and that its all-band 95th percentile is strictly larger than that of the residual-corrected network.

The reversal reproduces (Table 8). The convex stack has the lowest radiance error of the six, 0.0687%, and the fourth-lowest reflectance tail; the kernel on learned features has a larger radiance error, 0.0828%, and an all-band 95th percentile smaller by a factor of 6.60, and by 5.54

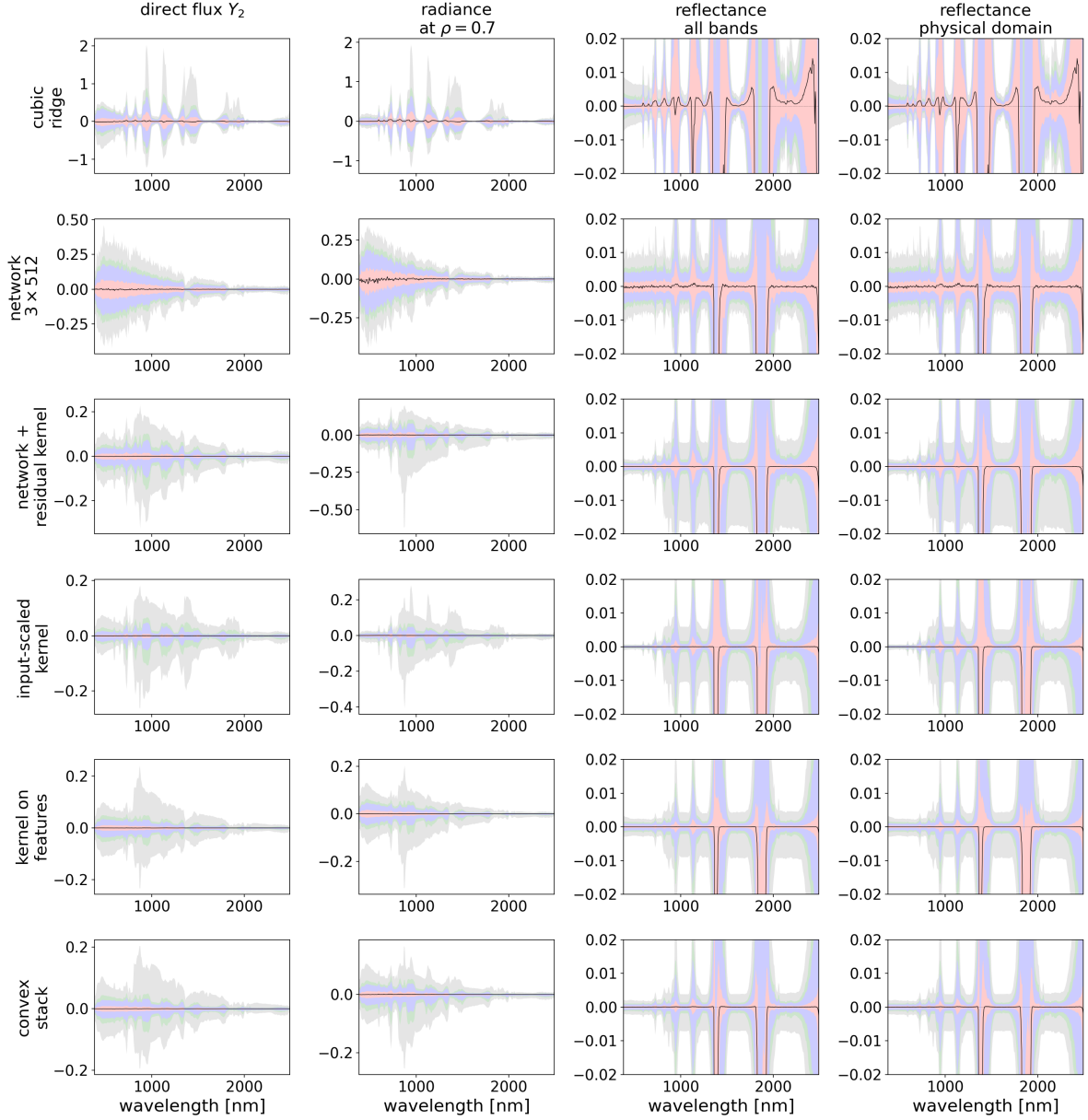


Figure 4: Residual quantile bands across the test set of split 101, band by band, for six families of the main pipeline. The envelopes are nested percentiles of the residual at each wavelength: 25–75 in red, 5–25 and 75–95 in blue, 2.5–5 and 95–97.5 in green, 0.5–2.5 and 97.5–99.5 in grey, with the median as a thin black line. From the left: the direct-flux residual $Y_2 - \widehat{Y}_2$ in physical units; the radiance residual at $\rho = 0.7$; the reflectance residual over all bands; and the reflectance residual on the physical domain with the floor $\tau_0 = 10^{-12}$. The last two columns share a vertical scale of ± 2 points.

family	radiance rel. L_2 [%]	all bands [pp]			screened [pp]	
		median	p_{95}	p_{99}	median	p_{95}
cubic ridge	0.7544	0.475	70.72	129.1	0.468	70.67
network 3×512	0.1719	0.106	41.83	84.1	0.105	28.07
network + residual kernel	0.0978	0.061	26.07	80.5	0.061	16.87
input-scaled Matérn kernel	0.0900	0.028	26.89	189.4	0.027	19.38
kernel on learned features	0.0828	0.034	3.91	129.2	0.034	2.94
convex stack	0.0687	0.044	25.82	89.6	0.044	16.30

Table 8: The fresh partition: 3,497 held-out states, evaluated once. The screened columns use $t \geq 10^{-12}$ and $q \geq 0.3$, which covers 99.5327% of the 996,645 entries; no inversion is non-finite in any family. The network here is the 3×512 GELU network of the frozen pipeline, trained with Adam, not the network of Table 1.

training rows	seeds	Cubic ridge	KRR, isotropic	KRR, per-input	FC-DNN	DNN + residual KRR	kernel on features
500	10	0.846±0.031	0.639±0.022	0.415±0.024	1.961±0.057	0.815±0.035	0.456±0.020
1,000	10	0.802±0.024	0.519±0.034	0.260±0.066	1.862±0.015	0.590±0.027	0.316±0.040
2,000	10	0.780±0.016	0.424±0.051	0.217±0.052	1.232±0.034	0.376±0.041	0.240±0.049
4,000	10	0.769±0.010	0.328±0.028	0.154±0.037	0.830±0.019	0.256±0.021	0.164±0.033
8,000	10	0.763±0.010	0.279±0.020	0.119±0.027	0.584±0.020	0.190±0.014	0.121±0.017
18,884	10	0.760±0.011	0.239±0.016	0.095±0.009	0.387±0.012	0.141±0.006	0.098±0.007

Table 9: Learning curves: radiance relative L^2 error [%] against the number of training rows, mean $\pm$ standard deviation over ten splits. Each rung uses the first n rows of the split’s training block; the validation and test blocks are those of Table 1, so the top rung is that table. The per-input kernel repeats its coordinate search at every rung. Component errors are shown in Figure 5.

on the screened entries. Both factors are close to the 6.13 and 4.99 of the same pipeline on its development split, seed 101. The second clause of the hypothesis fails: the stack’s all-band 95th percentile is 25.82 points and the corrected network’s 26.07, smaller by 0.25 points where the hypothesis required it to be larger. The qualitative statement survives a single read of a fresh partition, while the order among the four intermediate families does not separate at this sample size.

7 Learning curves, output rank and network width

The experiments below vary training size, output rank and network width under the same validation-based selection. Learning curves include cubic ridge, the isotropic and input-scaled kernels, the network, its residual correction and the feature kernel.

Learning curves. Each rung uses the first n rows of the split’s training block, $n = 500, 1,000, 2,000, 4,000, 8,000$ and 18,884, with the validation and test blocks unchanged, so the rungs of one split are nested (Table 9, Figure 5). The cubic error changes little, from 0.85% to 0.76% over a factor of 38 in data, and most of that decrease is reached by a thousand rows. On the first three components every other family decays at close to $n^{-1/2}$: the least-squares slopes of log error against log n over the six rungs lie between -0.38 and -0.61 for the kernel, the network,

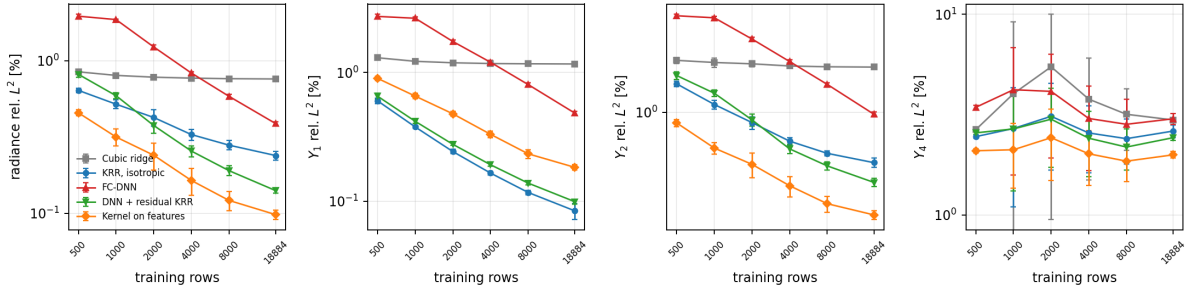


Figure 5: Learning curves on log-log axes for the radiance and for three of the four components, mean and standard deviation over ten splits. The cubic error changes little over the range. The kernel, network, residual-corrected network and feature kernel decrease approximately as $n^{-1/2}$ on the first three components, with fitted slopes in Table 10. The albedo error shows no sustained decrease; unusually large errors on individual splits widen its spread at 1,000 and 2,000 rows.

family	slope over the six rungs					slope of the last step				
	Y_1	Y_2	Y_3	Y_4	radiance	Y_1	Y_2	Y_3	Y_4	radiance
Cubic ridge	-0.03	-0.03	-0.03	-0.02	-0.03	-0.00	-0.01	-0.00	-0.08	-0.01
KRR, isotropic	-0.55	-0.38	-0.40	-0.01	-0.28	-0.38	-0.19	-0.17	0.10	-0.18
KRR, per-input	-0.45	-0.43	-0.43	0.01	-0.40	-0.32	-0.26	-0.29	0.12	-0.26
FC-DNN	-0.50	-0.49	-0.57	-0.09	-0.48	-0.59	-0.60	-0.55	0.07	-0.48
DNN + residual KRR	-0.52	-0.53	-0.61	-0.04	-0.50	-0.39	-0.33	-0.39	0.12	-0.35
Kernel on features	-0.45	-0.44	-0.46	-0.03	-0.43	-0.28	-0.22	-0.31	0.08	-0.25

Table 10: Learning-curve exponents: the least-squares slope of log error against log n over the six rungs of Table 9, and the slope of the last step alone, from 8,000 to 18,884 rows, per family and component.

the corrected network and the feature kernel on Y_1 , Y_2 and Y_3 (Table 10). On the albedo every slope lies between -0.09 and $+0.01$. These are slopes over a finite range of sample sizes; the rates of kernel regression theory require sampling, source and regularization conditions [46, 48] that this experiment does not check.

The isotropic kernel bends at the largest size. Its slope over the first five rungs is -0.59 , -0.44 and -0.47 on the smooth components, and its last step is -0.38 , -0.19 and -0.17 , while the network’s last step is as steep as its average (-0.59 , -0.60 and -0.55). Part of the bend is the tuning protocol: the kernel’s smoothness, scale and nugget are chosen on a 6,000-row subsample at every rung, so above 6,000 rows the hyperparameters belong to a smaller problem, and Section 7.1 shows that tuning with every row recovers much of the lost slope. The ratio of the network’s radiance error to the kernel’s falls from 3.1 at 500 rows to 1.6 at the full block, or to 1.8 with the kernel tuned on every row. The kernel’s margin over the network is a margin at this training size.

The per-input metric changes the slope as well as the level. With its coordinate search repeated at every rung it has slope -0.399 on radiance against the isotropic kernel’s -0.279 , and the ratio of the two errors falls from 0.65 at 500 rows to 0.40 at the full block, with one step against that direction (0.50 at 1,000 rows, 0.51 at 2,000). The selected metric, smoothness and nugget all change with the sample size, so this is not a change of asymptotic dimension. Against the kernel on network features the margin closes as the data grow. The per-input metric has the

PCA rank	variance outside the basis	KRR, isotropic	FC-DNN	DNN + residual KRR	kernel on features
16	7.5×10^{-4}	0.270 $\pm$ 0.015	0.390 $\pm$ 0.015	0.175 $\pm$ 0.007	0.148 $\pm$ 0.009
32	1.5×10^{-5}	0.243 $\pm$ 0.018	0.390 $\pm$ 0.015	0.139 $\pm$ 0.006	0.102 $\pm$ 0.009
64	8.2×10^{-8}	0.242 $\pm$ 0.018	0.390 $\pm$ 0.015	0.143 $\pm$ 0.006	0.100 $\pm$ 0.009
128	5.4×10^{-9}	0.242 $\pm$ 0.018	0.390 $\pm$ 0.015	0.150 $\pm$ 0.005	0.100 $\pm$ 0.009
285	0	0.242 $\pm$ 0.018	0.390 $\pm$ 0.015	0.172 $\pm$ 0.004	0.100 $\pm$ 0.009

Table 11: Output-rank ablation: radiance relative L^2 error [%] when the kernels and corrections regress the leading r principal coefficients, mean $\pm$ standard deviation over five splits. The second column is the fraction of the standardized variance the basis leaves out for the slowest-decaying component, the spherical albedo; at rank 285 it is zero. The network regresses the 285 bands directly and is the same predictor in every row.

lower mean at all six rungs, but it wins at 10, 9, 10, 7, 5 and 6 of ten splits from 500 rows to the full block, and the gap narrows from 0.415 against 0.456 at 500 rows to 0.095 against 0.098 at the top; the feature kernel’s own slope, -0.435 , is the steeper. On the albedo the per-input slope is $+0.01$.

The corrected network is below both of its parents on radiance at every split from 2,000 rows on, and at no split at 500 or 1,000 rows, where its error lies between the network’s and the kernel’s. On Y_2 and Y_3 it is below both at every split at the two largest sizes; on Y_1 it never is, the isotropic kernel alone being the better head at every size (0.084% against 0.099% at the full block). Its exponent is close to the network’s and its level close to the kernel’s. Section 8 gives the exact residual decomposition behind this behavior.

Output rank. The kernels and corrections regress the leading r principal coefficients of the standardized component, while the network regresses the 285 bands and is the same predictor at every rank. Table 11 refits at $r = 16, 32, 128$ and 285 on five splits beside the rank-64 rows of those splits. The kernel changes little above rank 32: its radiance error reads 0.243, 0.242, 0.242 and 0.242 at 32, 64, 128 and 285, and 0.270 at 16, where the truncation leaves 7.5×10^{-4} of the albedo’s variance outside the basis. The kernel on features behaves the same way. The residual correction does not: it is best at 32 (0.139), slightly worse at 64 (0.143), and worse again at 128 and 285 (0.150 and 0.172). The two regressions have different targets. The direct kernel fits the component, whereas the correction fits the projected network residual, and a higher rank exposes the correction to more directions of network error even where the target has little variance. The projection identity of Section 8 separates the two error terms.

Width. A network of three hidden layers of 2,000 units, trained for up to 500 epochs with the optimizer, learning rate and early stopping of the 512-unit network, was run on all ten splits with the rest of the pipeline built on it (Table 12). It is a worse emulator on every split, 0.75% against 0.39% on radiance, and early stopping fired between 42 and 210 of the 500 epochs allowed. Width and training schedule change together here, so this describes the trained networks rather than width as such. The exact kernel on its last-layer features reaches 0.079% on radiance against 0.098% on the 512-unit features; it is below the per-input kernel’s 0.095% on all ten splits and below the stack of Table 1 on nine, and on the two flux components, at 0.13% and 0.08%, it is at or below every row of that table. The convex stack over the wide pipeline’s heads has the lowest mean radiance error of the ten-split comparisons, 0.078%, with a median reflectance error of 0.043 points, below the stack of Table 1 on nine of ten splits; the per-coordinate selection over

network	head	component rel. L^2 [%]				radiance	reflectance
		Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $ \hat{\rho} - \rho $ [pp]
3×512	network	0.48 ± 0.02	0.97 ± 0.04	0.41 ± 0.02	3.02 ± 0.18	0.387 ± 0.012	0.245 ± 0.008
	network + residual KRR	0.10 ± 0.00	0.30 ± 0.02	0.16 ± 0.01	2.43 ± 0.08	0.141 ± 0.006	0.089 ± 0.008
	kernel on its features	0.18 ± 0.01	0.17 ± 0.01	0.10 ± 0.01	1.99 ± 0.08	0.098 ± 0.007	0.054 ± 0.004
	per-coordinate selection	0.19 ± 0.01	0.44 ± 0.10	0.13 ± 0.02	2.25 ± 0.18	0.193 ± 0.022	0.123 ± 0.023
	convex stack	0.07 ± 0.00	0.16 ± 0.01	0.09 ± 0.00	1.94 ± 0.06	0.087 ± 0.006	0.048 ± 0.003
3×2000	network	0.93 ± 0.20	2.11 ± 0.49	0.76 ± 0.19	3.27 ± 0.19	0.748 ± 0.116	0.508 ± 0.110
	network + residual KRR	0.12 ± 0.01	0.35 ± 0.03	0.19 ± 0.03	2.35 ± 0.13	0.160 ± 0.016	0.106 ± 0.010
	kernel on its features	0.13 ± 0.02	0.13 ± 0.02	0.08 ± 0.01	2.10 ± 0.34	0.079 ± 0.009	0.045 ± 0.006
	per-coordinate selection	0.14 ± 0.02	0.30 ± 0.26	0.09 ± 0.01	2.55 ± 0.26	0.187 ± 0.057	0.099 ± 0.033
	convex stack	0.08 ± 0.01	0.13 ± 0.02	0.08 ± 0.00	2.00 ± 0.24	0.078 ± 0.011	0.043 ± 0.005

Table 12: The pipeline of Section 4 with a 3×2000 network in place of the 3×512 one, on the same ten splits: the network (one member, up to 500 epochs under the same optimizer, learning rate and early stopping), its residual correction, the exact kernel on its last-layer features, and the per-coordinate selection and convex stack over the wide pipeline’s four distinct predictors (the isotropic kernel, network, correction and feature kernel; with a single network the ensemble slots repeat the network), beside the corresponding rows of Table 1; mean $\pm$ standard deviation over the ten splits.

the same heads again does not help. The residual correction of the wider network is slightly worse than that of the narrower one. A 3×1000 network on the same ten splits sits between the two: the network at 0.72%, the kernel on its features at 0.088%. From 512 to 2,000 units the network’s own error rises with width while the error of a kernel on its features falls. At 4,000 units, on three splits, the network is worse again at 1.10% and the kernel on its features turns back up to 0.099%, so the gain in the representation stops by 2,000 units.

Concatenated features. If the hidden layer is a good kernel metric, several of them together may be better. On all ten splits we solved one exact kernel on the concatenated last-layer features of the five width-512 networks, 2,560 coordinates in all. It reaches $0.076 \pm 0.007\%$ on radiance against $0.098 \pm 0.007\%$ for the features of one network, lower on every split by 0.022 ± 0.003 points, and 0.039 against 0.054 points of median reflectance error. Its mean within-split 95th-percentile reflectance error is 5.93 points against 7.75. One kernel solve on concatenated features therefore beats the six-head stack of Table 1 on radiance, 0.076% against 0.087%, and has a far smaller reflectance tail (Table 3). On three splits the same construction with three width-2,000 networks, 6,000 coordinates, reaches 0.074% against 0.083% for a single member’s features on those splits, lower on all three, and 0.043 against 0.051 reflectance points; there it is level with the six-head stack fitted on the same splits (0.074%).

7.1 Full-data tuning and gradient-based metrics

We compare hyperparameter selection on a 6,000-row subsample with selection on all training rows, and coordinate search with a metric read off the gradients of a cubic surrogate (Table 13).

Tuning with every row in the solve. The kernel families of Section 4 choose smoothness, scale and nugget by the validation error of a solve on a 6,000-row subsample and refit the winner on all 18,884 rows. Tuning with the full solve instead lowers the isotropic kernel’s radiance error

kernel	component rel. L^2 [%]				radiance	reflectance
	Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $ \hat{\rho} - \rho $ [%]
isotropic, tuned on 6,000 rows	0.08±0.01	0.42±0.03	0.20±0.00	2.62±0.18	0.239±0.016	0.141±0.018
isotropic, tuned on every row	0.08±0.00	0.36±0.02	0.17±0.01	2.58±0.14	0.210±0.008	0.107±0.006
per-input search, tuned on 6,000 rows	0.08±0.01	0.16±0.02	0.08±0.01	2.18±0.16	0.095±0.009	0.036±0.004
per-input search, re-tuned on every row	0.08±0.00	0.16±0.02	0.08±0.01	2.09±0.12	0.096±0.009	0.036±0.004
per-input from the cubic's gradients, $p = 1$	0.34±0.01	0.19±0.02	0.14±0.01	2.17±0.14	0.159±0.007	0.078±0.004
per-input from the cubic's gradients, $p = 1/2$	0.11±0.00	0.19±0.02	0.09±0.01	2.27±0.15	0.125±0.008	0.055±0.004
per-input from the cubic's gradients, $p = 1/4$	0.08±0.00	0.25±0.02	0.14±0.01	2.38±0.20	0.157±0.015	0.077±0.013

Table 13: Where the kernel’s hyperparameters are chosen, and how its metric is obtained, at the full training block and the same ten splits. The first pair tunes smoothness, scale and nugget on a 6,000-row subsample and on every training row; the last three rows replace the coordinate search by a metric read off the cubic surrogate’s gradients, at three exponents. Selecting the exponent on validation, per split, chooses $p = 1/2$ on all ten, so that row would repeat the $p = 1/2$ row and is omitted. Mean $\pm$ standard deviation over the ten splits.

from 0.239% to 0.210% and its median reflectance error from 0.141 to 0.107 points, better on nine of ten splits, with the ratio averaging 0.882 and never above 1.001. The selected scales differ: over the forty component-split cells the full block picks a shorter length scale than the subsample in 26, the same in 12 and a longer one in 2. A smaller tuning fit prefers more smoothing.

Below 6,000 rows the tuning subsample is the whole training block, so those rungs are already fully tuned; at 8,000 rows it is three quarters of the block and only at the top rung is it a third. Applying the measured ratio to the top rung of Table 9 takes the kernel’s last step on radiance from -0.183 to -0.334 , against a slope of -0.305 over the five lower rungs, and on the two flux components from -0.187 and -0.171 to -0.352 and -0.358 , against -0.437 and -0.468 . Most of the bend of Section 7 is the tuning protocol, and with full tuning the kernel’s margin over the network at the full block is 1.8 rather than 1.6.

Full-data tuning barely changes the input-scaled kernel: 0.096% against 0.095%, better on four of ten splits, ratio 1.005. Its coordinate search already arrives at the scale the full block selects.

Gradient-based metric. The coordinate search costs a sweep over the six inputs at five multipliers, twice, per component and split. The alternative fixes the metric before any kernel is fitted: the multiplier of coordinate j is the root-mean-square gradient of the fitted cubic surrogate along j over the training rows, raised to a power p , normalized to geometric mean one and clipped to the search’s range $[1/16, 4]$, so that weakly varying coordinates receive longer length scales. It costs one cubic fit and six finite differences.

At $p = 1$ the radiance error is 0.159%, below the isotropic kernel on every split, by a factor 0.669 on average, and above the searched metric on every split, by a factor 1.680. The dependence on p is not monotone: $p = 1/2$ gives 0.125% and $p = 1/4$ gives 0.157%. Choosing p like every other hyperparameter here, by the lowest mean validation error over the four components, selects $p = 1/2$ on all ten splits. At that exponent the gradient metric closes 79% of the gap between the isotropic and searched kernels on radiance and 82% on the median reflectance error, for one cubic fit and a choice among three numbers.

The components prefer different exponents: Y_2 and Y_3 are best at $1/2$, the albedo at 1 (2.18% against 2.27%), and the path radiance at $1/4$. On path radiance the isotropic kernel already reads 0.084% and the searched metric 0.085%, while the proportional metric reads 0.34%, four

times the isotropic error. The gradient of the surrogate along a coordinate measures how fast the map moves with it, not how much that coordinate helps predict, and on this component the two disagree. Softening the exponent to $1/4$ brings Y_1 back to 0.085%, at the cost of Y_2 and Y_3 . Coordinate search can adjust each component separately, whereas the gradient construction uses one exponent for all four.

8 Residual correction and error alignment

Fix the training design, neural mean and selected kernel hyperparameters. The following identities hold without sampling assumptions. We first work with one output coordinate; the vector-valued statements follow coordinatewise.

8.1 A frozen regression operator

Fix a design $X_n = (x_1, \dots, x_n)$, a positive-semidefinite kernel with Gram matrix K , and $\lambda > 0$. Define

$$(Sg)(x) = k(x, X_n)(K + \lambda I)^{-1}g(X_n), \quad R = I - S. \quad (8)$$

Here λ is the diagonal regularizer in the linear system. The code uses $\lambda = n\gamma$, where γ is the reported nugget; the distinction matters when the training size changes. Although a selected kernel can depend on the targets, S is linear in the response only *after that selection is frozen*.

Proposition 2 (Frozen residual correction). *Let f be the target and h any fitted mean, and set $\hat{f}_c = h + S(f - h)$. Then:*

$$f - \hat{f}_c = R(f - h), \quad (9)$$

$$(Rg)(X_n) = \lambda(K + \lambda I)^{-1}g(X_n). \quad (10)$$

If $Ku_j = \mu_j u_j$, the design residual in direction u_j is multiplied by $\lambda/(\mu_j + \lambda) \in (0, 1]$. The factor is one when $\mu_j = 0$ and strictly below one when $\mu_j > 0$; this is shrinkage, not a hard spectral projection.

Suppose, in addition, that $\mathcal{S}$ is a normed vector space containing f and h and that the following bound holds uniformly on that space for the realized operator:

$$\|Rg\|_{L^2(P)} \leq C_n \|g\|_{\mathcal{S}} \quad (g \in \mathcal{S}).$$

Then

$$\|f - \hat{f}_c\|_{L^2(P)} \leq C_n \|f - h\|_{\mathcal{S}}. \quad (11)$$

Consequently, $\|f - h\|_{\mathcal{S}} < \|f\|_{\mathcal{S}}$ makes this displayed upper bound smaller than $C_n \|f\|_{\mathcal{S}}$ when $C_n > 0$. It is neither a necessary nor a sufficient condition for the corrected predictor's actual error to be smaller than that of Sf .

Proof in Appendix A.

Why the bounds do not order the errors. On a two-point design take $K = \text{diag}(1, 99)$ and $\lambda = 1$, so $R = \text{diag}(1/2, 1/100)$ and $C_n = 1/2$ is its Euclidean operator norm. With $f = (1, 0)$ and $f - h = (0, 2)$, the residual norm is twice the target norm, yet $\|R(f - h)\| = 0.02 < 0.50 = \|Rf\|$. Conversely, with $f = (0, 1)$ and $f - h = (1/2, 0)$, the residual norm is smaller, but $\|R(f - h)\| = 0.25 > 0.01 = \|Rf\|$. These examples also show why source-norm accuracy alone does not order corrected prediction errors.

8.2 An exact criterion for improvement

The relevant comparison can be made exactly in any Hilbert error metric. This includes a finite weighted squared error and $L^2(P)$, but not directly the mean of per-sample relative norms or the nonlinear radiance and reflectance metrics in Table 1.

Proposition 3 (Error alignment). *For the same frozen operator S , put $a = Rf$ and $b = Rh$. Then*

$$\|f - \hat{f}_c\|^2 - \|f - Sf\|^2 = \|b\|^2 - 2\langle a, b \rangle. \quad (12)$$

Thus full correction improves on Sf exactly when $2\langle a, b \rangle > \|b\|^2$. More generally, the interpolation $p_t = Sf + tRh$, $0 \leq t \leq 1$, has minimum squared error at

$$t_* = \min\{1, \max\{0, \langle a, b \rangle / \|b\|^2\}\} \quad \text{if } b \neq 0; \quad (13)$$

if $b = 0$, every t is equivalent. If the direct fit uses S_0 and the correction uses S_1 , (12) instead holds with $a = (I - S_0)f$ and $b = (S_1 - S_0)f + (I - S_1)h$.

Proof in Appendix A. The criterion depends on the unknown target error. Replacing its inner products by validation errors gives a validation-optimal interpolation, not a guarantee on new data. The reported experiments use separately tuned direct and residual kernels, so their comparison must use the final, two-operator form with their actual selected operators. Neither the criterion nor the training-design contraction establishes that correction will beat both parents at every sample size.

8.3 What the principal-component implementation computes

The pipeline trains a network on all standardized bands but projects its predictions into the retained principal-component basis before correction. Thus its corrected predictor is based on the projected network mean. To state the implementation precisely, let f and h now denote the target and network output after the common fitted standardization and PCA centering. Let Π be the orthogonal projection onto the retained spectral coordinates. In these centered coordinates, the implemented prediction is

$$\hat{f}_{c,\Pi} = \Pi h + S\Pi(f - h).$$

The output mean and diagonal band scaling are restored afterwards. Since the same scalar kernel operator acts on each retained output coordinate,

$$f - \hat{f}_{c,\Pi} = (I - \Pi)f + (I - S)\Pi(f - h). \quad (14)$$

The terms on the right are orthogonal in the standardized spectral Euclidean norm at each query, hence their squared norms add there. They need not be orthogonal in physical units after unequal band rescaling. This decomposition separates truncation error from residual regression and explains why output rank is a genuine part of the model specification. It does not by itself prove the observed optimum near rank 32. Comparisons with the unprojected network include the effect of projection as well as residual fitting.

Conditional rate bound. Kernel learning-rate results impose source, spectral, sampling and regularization conditions [46, 47, 48]. If a uniform residual bound holds with $C_n = O(n^{-\alpha})$ and $\|f - h_n\|_{\mathcal{S}} = O(n^{-\beta})$, then (11) gives $\|f - \hat{f}_c\|_{L^2(P)} = O(n^{-(\alpha+\beta)})$. The source-norm bound and the uniformity required for a data-dependent residual are additional hypotheses. They do not follow from empirical network loss. The experiments measure finite-range error slopes, not these source norms or an asymptotic rate.

The code checks the residual identities, regularizer convention, semidefinite boundary, alignment formula, counterexamples and projected implementation on explicit and randomized finite designs.

9 Retrieval-error transfer and the sensitivity of learned corrections

9.1 Sensitivity of the correction

For a common frozen operator $R = I - S_\lambda$ and the norm of Section 8, the exact criterion $2\langle Rf, Rh \rangle > \|Rh\|^2$ is necessary and sufficient for strict improvement of the corrected predictor over the standalone kernel, since $\|Rf - Rh\|^2 < \|Rf\|^2$ is that inequality rearranged; the comparison of smoothness norms is neither necessary nor sufficient. Write the design values of the targets, of the fitted mean and of the corrected predictor as y , h_X and $\hat{y}$, with $\hat{y} = h_X + S_\lambda(y - h_X)$ and $S_\lambda = K(K + \lambda I)^{-1}$; in the nugget convention of the experiments, $\lambda = n\gamma$. The mean is trained on the same labels, so $h_X = h_X(y)$. In the following Jacobian calculation, K and λ are held fixed; label-dependent feature or kernel selection would add terms involving derivatives of S_λ .

Proposition 4. *1. If h_X is differentiable in y , the Jacobian of this fixed-kernel pipeline is $J_{\hat{y}} = S_\lambda + (I - S_\lambda)J_h$. In particular the correction stage alone does not bound the sensitivity of the pipeline: with $K = I$, $\lambda = 1$ and $h_X(y) = 2y$ one has $\hat{y} = \frac{3}{2}y$ although $\|S_\lambda\|_{\text{op}} = \frac{1}{2}$; and if the mean interpolates the training labels, $h_X(y) = y$, then $\hat{y} = y$ and the correction changes nothing on the design.*

2. With the mean and the kernel frozen, $y \mapsto \hat{y}$ is affine with linear part S_λ , its effective degrees of freedom are $\text{tr } S_\lambda = \sum_j \mu_j / (\mu_j + \lambda)$, and $\|S_\lambda\|_{\text{op}} = \max_j \mu_j / (\mu_j + \lambda) < 1$.

3. Suppose the mean, the kernel and the ridge are fixed independently of the correction noise, conditional on the design, and the correction is fitted on an independent sample with $y = f_X + \varepsilon$, $\mathbb{E}\varepsilon = 0$, $\mathbb{E}[\varepsilon\varepsilon^\top] = \sigma^2 I$. With $r = f_X - h_X$ and r_j its coordinates in the eigenbasis of K ,

$$\frac{1}{n} \mathbb{E}_\varepsilon \|\hat{y} - f_X\|^2 = \frac{1}{n} \sum_j \left(\frac{\lambda}{\mu_j + \lambda} \right)^2 r_j^2 + \frac{\sigma^2}{n} \sum_j \left(\frac{\mu_j}{\mu_j + \lambda} \right)^2.$$

4. At a new input x , with $a_\lambda(x) = (K + \lambda I)^{-1}k(X_n, x)$, the noise variance of the correction is $\sigma^2 \|a_\lambda(x)\|_2^2$.

Proof in Appendix A. Item (iii) is what the ridge-path experiments report: the residual's spectral energy $\sum_j r_j^2$ by eigendirection, and the two sums as functions of λ , together with the out-of-sample factor of item (iv). A ridge chosen from fits that depend on the correction noise is outside the hypothesis of (iii).

9.2 A finite-class selection guarantee

A finite comparison on a genuinely independent selection sample has a simple risk bound [82, 83, 84]. The independence assumption is stronger than reusing the same validation rows for training decisions and combination weights.

Proposition 5. *Let $p_1, \dots, p_M$ be predictors frozen before an independent selection sample of m i.i.d. units, let the loss take values in $[0, B]$, and let $\hat{p}$ minimise the empirical risk on that sample. Then with probability at least $1 - \delta$,*

$$R(\hat{p}) \leq \min_{1 \leq j \leq M} R(p_j) + 2B \sqrt{\frac{\log(2M/\delta)}{2m}}.$$

Proof in Appendix A. The squared constrained-retrieval loss is bounded by R^2 . Thus the proposition can compare a finite set of frozen corrections or stacks, with bands and prescribed reflectances averaged within each independent state. For example, a finite grid of damped corrections $h + \alpha k(x, X_n)(K + \lambda I)^{-1}(y - h_X)$ can include $\alpha = 0$; the standalone kernel must be added as a separate candidate when it is not already in the grid. To apply the result to stacks, one first fits their weights on development rows and then compares a frozen finite collection on an independent state-level sample. In the experiments below the same validation rows serve early stopping, kernel tuning and combination, so the proposition describes the protocol under which such a comparison is certified rather than a property of the reported stacks.

9.3 Transfer of component error into retrieval error

Section 3.1 controls the inverse where the denominator stays away from zero. The theorem below describes a whole evaluation distribution, with entries of vanishing transmission included, with the constant one, and shows that the resulting loss of rate is not an artefact of the argument. Let P be the evaluation distribution over entries (a state, a band, and the surface reflectance when it is varied), in fixed physical units; expectations are conditional on the fitted emulator. Assume

$$0 \leq \rho \leq R \leq 1, \quad t > 0, \quad 0 \leq s \leq S, \quad RS < 1. \quad (15)$$

The emulator predicts $(\hat{a}, \hat{t}, \hat{s})$. The *constrained retrieval* replaces $\hat{t}$ by $\max\{\hat{t}, 0\}$ and $\hat{s}$ by its projection onto $[0, S]$, neither of which increases $|e_t|$ or $|e_s|$, and sets

$$\bar{\rho} = \Pi_{[0, R]} \left(\frac{L - \hat{a}}{\hat{t} + \hat{s}(L - \hat{a})} \right) \quad (16)$$

when the denominator is positive, and otherwise returns a prescribed value in $[0, R]$ and records a failure. It is an estimator in its own right and is reported beside the unconstrained inverse and its failure counts, never in place of them. The *retrieval-weighted component error*, computed after the projections, is

$$e_R = |e_a| + R|e_t| + R^2 t |e_s|, \quad \varepsilon_R^2 = \mathbb{E}_P[e_R^2], \quad (17)$$

which is at most $e = |e_a| + |e_t| + t|e_s|$ since $R \leq 1$; ε_R is not the mean relative radiance error of the tables. Put $F_t(\tau) = \mathbb{P}(t \leq \tau)$.

Theorem 6 (error transfer with constant one). *Under (15), for the constrained retrieval (16):*

1. $|\bar{\rho} - \rho| \leq \min\{R, e_R/t\} \leq \min\{R, e/t\}$ at every entry, and no universal constant below one is possible.

2. For every $\tau > 0$,

$$\mathbb{E}_P |\bar{\rho} - \rho|^2 \leq \mathbb{E}_P \min\left\{R^2, \frac{e_R^2}{t^2}\right\} \leq R^2 F_t(\tau) + \mathbb{E}_P \left[\frac{e_R^2}{t^2} \mathbf{1}\{t > \tau\}\right] \leq R^2 F_t(\tau) + \frac{\varepsilon_R^2}{\tau^2}.$$

3. If $F_t(u) \leq Au^\beta$ for $0 < u \leq u_0$ with some $\beta > 0$, then for $\varepsilon_R \leq u_0^{(\beta+2)/2}$, $\mathbb{E}_P |\bar{\rho} - \rho|^2 \leq (R^2 A + 1) \varepsilon_R^{2\beta/(\beta+2)}$; a component-error bound $\varepsilon_{R,n} = O(n^{-r})$ yields $\|\bar{\rho}_n - \rho\|_{L^2(P)} = O(n^{-r\beta/(\beta+2)})$.

4. If $t \geq t_0 > 0$ P -almost surely, then $\|\bar{\rho} - \rho\|_{L^2(P)} \leq \varepsilon_R/t_0$.

Proof in Appendix A. The middle expression in (ii) keeps the association between the errors and the transmission and is the one the tables evaluate; the last needs no power law. Both are computable on a finite test table; (iii) is a population statement whose tail hypothesis a finite table cannot establish down to $t = 0$. The exponent in (iii) cannot be improved from its hypotheses. Item (i) holds entry by entry, so it passes to quantiles: on any set of entries, a single band for instance, every quantile of $|\bar{\rho} - \rho|$ is at most the same quantile of $\min\{R, e_R/t\}$, which depends only on the component errors and the true transmission.

Proposition 7 (the exponent is attained). *For every $\beta > 0$, every $0 < c < r_0 < R$ and every $\tau \in (0, 1)$ there are an evaluation distribution and a predictor satisfying (15) with $S = 0$ and $F_t(u) = u^\beta$ on $(0, 1)$, namely $\rho = r_0$, $a = s = 0$, $\hat{t} = t$, $\hat{s} = 0$ and $\hat{a} = ct \mathbf{1}\{t \leq \tau\}$, for which $\bar{\rho} = r_0 - c \mathbf{1}\{t \leq \tau\}$,*

$$\begin{aligned} \varepsilon_R^2 &= \frac{c^2 \beta}{\beta + 2} \tau^{\beta+2}, & \mathbb{E}_P |\bar{\rho} - \rho|^2 &= c^2 \tau^\beta, \\ \mathbb{E}_P |\bar{\rho} - \rho|^2 &= c^{4/(\beta+2)} \left(\frac{\beta+2}{\beta}\right)^{\beta/(\beta+2)} \varepsilon_R^{2\beta/(\beta+2)}. \end{aligned}$$

Proof in Appendix A. This is sharpness of the transfer inequality for the physical inverse, not a minimax statement over retrieval algorithms. What decides the rate is where the errors sit relative to the transmission.

Theorem 8 (conditional error profile). *Under (15), suppose $\mathbb{E}_P[e_R^2 | t] \leq \sigma^2 t^{2\nu}$ P -almost surely for some $0 \leq \nu \leq 1$. Then*

$$\mathbb{E}_P |\bar{\rho} - \rho|^2 \leq \mathbb{E}_P \min\{R^2, \sigma^2 t^{-2(1-\nu)}\}.$$

Under the tail hypothesis of Theorem 6(iii) and $\nu < 1$, as $\sigma \rightarrow 0$,

$$\mathbb{E}_P |\bar{\rho} - \rho|^2 = \begin{cases} O(\sigma^{\beta/(1-\nu)}), & \beta < 2(1-\nu), \\ O(\sigma^2 \log(1/\sigma)), & \beta = 2(1-\nu), \\ O(\sigma^2), & \beta > 2(1-\nu), \end{cases}$$

and at $\nu = 1$, $\mathbb{E}_P |\bar{\rho} - \rho|^2 \leq \sigma^2$ with no tail hypothesis. In the model of Proposition 7 the predictor $\hat{a} = a + \min\{ct, \sigma t^\nu\}$ has $e_R \leq \sigma t^\nu$ and $|\bar{\rho} - \rho| = \min\{c, \sigma t^{\nu-1}\}$, and attains each order.

Proof in Appendix A. At $\nu = 0$ the theorem recovers the uniform-error regimes under the weaker hypothesis of a bounded conditional second moment; at $\nu = 1$ an emulator whose conditional root-mean-square error decreases proportionally to the transmission loses no rate. The central diagnostic is therefore the profile $t \mapsto \mathbb{E}[e_R^2 \mid t]$ and not only the tail $t \mapsto F_t(t)$: the same overall error concentrated in the opaque entries gives the worst-case loss, and proportional to the transmission gives none.

9.4 Perturbing a learned-mean kernel predictor

Fix the training sample and write a corrected predictor as $p(x) = h(x) + k_x^\top H^{-1}(Y - M)$ with $H = K + \lambda I$ positive definite, $Y \in \mathbb{R}^{n \times q}$ the targets and $M \in \mathbb{R}^{n \times q}$ the training-row means used as correction labels. Let a second fitted system be $\tilde{p}(x) = \tilde{h}(x) + \tilde{k}_x^\top \tilde{H}^{-1}(Y - \tilde{M})$ with $\tilde{H}$ symmetric positive definite; both systems may depend on Y through the mean, the features, the selected ridge and any kernel approximation. Put $B = H^{-1/2}(\tilde{H} - H)H^{-1/2}$, whose least eigenvalue exceeds -1 , $Q = (I + B)^{-1}$, $\gamma_Q = \|Q\|_{\text{op}} = 1/(1 + \lambda_{\min}(B))$, $\eta_Q = \|Q - I\|_{\text{op}} = \max_i |b_i|/(1 + b_i)$ over the eigenvalues b_i of B , and

$$g = H^{-1/2}k_x, \quad \tilde{g} = H^{-1/2}\tilde{k}_x, \quad v = \tilde{g} - g, \quad W = H^{-1/2}(Y - M), \quad U = H^{-1/2}(\tilde{M} - M).$$

A conditioning argument bounds such a difference by the condition number of H times a norm of the change [109, 107, 110]. The statement below instead keeps the residual action W and the label shift U separate, which is what makes it readable on data.

Proposition 9 (same-data perturbation). *1. The exact difference is*

$$\tilde{p}(x) - p(x) = \tilde{h}(x) - h(x) + v^\top W - (Q\tilde{g})^\top (BW + U).$$

2. $\|\tilde{p}(x) - p(x)\|_2 \leq \|\tilde{h}(x) - h(x)\|_2 + \|W^\top v\|_2 + \|Q\tilde{g}\|_2 \|BW + U\|_{\text{op}}.$
3. $\|\tilde{p}(x) - p(x)\|_2 \leq \|\tilde{h}(x) - h(x)\|_2 + \|W\|_F (\gamma_Q \|v\|_2 + \eta_Q \|g\|_2) + \gamma_Q \|U\|_F \|\tilde{g}\|_2.$
4. *If $\tilde{H} = H + sI$ with $s \geq 0$ and the kernel and the means unchanged, then $\|\tilde{p}(x) - p(x)\|_2 \leq \|g\|_2 \|W\|_F s / (\lambda_{\min}(H) + s).$*

Proof in Appendix A. Only positive definiteness of $\tilde{H}$ is needed. A large positive eigenvalue of B , as can arise from increasing the ridge, does not invalidate the bound. The bound can nonetheless be numerically large, especially when the least eigenvalue approaches -1 . Item (ii) keeps the interaction between the perturbation and the residual: with $v = 0$, $U = 0$ and $BW = 0$ the kernel correction does not move, however large $\|B\|$ is; the full predictor is unchanged as well when $\tilde{h}(x) = h(x)$. The bounds are stated in the residual W , not in Y , they use no independence between the representation, the mean, the ridge and Y , and they transfer between fitted predictors without being generalisation statements.

For component j the physical target at band b is $f_{j,b} = \mu_{j,b} + \sum_\ell (A_j)_{b\ell} z_{j,\ell} + \pi_{j,b}$, with A_j the standardiser scale times the retained PCA basis and π_j the truncation residual, and the pipeline predicts the retained coordinates. For a frozen realised system $p_\ell(x) = h_\ell(x) + a_x^\top (y_\ell - h_{\ell,X})$ with $y_\ell = f_{\ell,X} + \varepsilon_\ell$, the conditional RKHS evaluation bound [8, 12] gives

$$|p_\ell(x) - f_\ell(x)| \leq B_\ell \tilde{P}_\lambda(x) + |a_x^\top \varepsilon_\ell|, \tag{18}$$

where, writing $k_x = k(X_n, x)$ and $a_x = (K + \lambda I)^{-1}k_x$,

$$\tilde{P}_\lambda(x)^2 = k(x, x) - 2a_x^\top k_x + a_x^\top K a_x = k(x, x) - k_x^\top (K + \lambda I)^{-1} k_x - \lambda \|a_x\|_2^2.$$

This follows by applying the RKHS Cauchy–Schwarz inequality to the residual and $k(x, \cdot) - \sum_i (a_x)_i k(x_i, \cdot)$. Here B_ℓ is a bound on the residual RKHS norm, a hypothesis and not an observed certificate, and the second term zero for deterministic simulator targets, which is the form used for EMIT. A perturbed system adds the bound $\Delta_\ell(x)$ of Proposition 9. If $|\hat{z}_{j,\ell} - z_{j,\ell}| \leq u_{j,\ell}$, then

$$|\hat{f}_{j,b} - f_{j,b}| \leq |\pi_{j,b}| + \sum_\ell |(A_j)_{b\ell}| u_{j,\ell}, \quad (19)$$

and a change of basis or standardiser between two systems adds its own reconstruction term. Assembling $e_R = |e_a| + R|e_t| + R^2 t|e_s|$ from (19) for the three physical components and applying Theorem 6 closes the chain. If the inverse is evaluated at $L_{\text{obs}} = L + \eta$, the proof of Theorem 6 goes through with e_a replaced by $e_a - \eta$, so $|\bar{\rho} - \rho| \leq \min\{R, (|e_a - \eta| + R|e_t| + R^2 t|e_s|)/t\}$. The theorem is an exact-arithmetic statement. In floating point the added term $\rho t/q$ can be lost when it is small relative to the spacing of representable numbers near a , in a way that depends on ρ , q , the magnitudes and the rounding rather than on a single threshold such as $t < 2^{-52}a$; a rounded radiance may then give zero for $L - a$ even with the tabulated components. A count of numerical violations is therefore a check on the implementation, not a test of the inequality.

9.5 The stacking objective

Affine fitting predictions and projected retrieval. Let w denote simplex weights, with one simplex for each component being combined. Write $\tilde{a}_w$, $\tilde{t}_w$ and $\tilde{s}_w$ for the affine combinations used in fitting, and $\tilde{e}_a = \tilde{a}_w - a$, $\tilde{e}_t = \tilde{t}_w - t$, $\tilde{e}_s = \tilde{s}_w - s$ for their errors. Retrieval uses $\hat{t}_w = \max\{\tilde{t}_w, 0\}$ and $\hat{s}_w = \Pi_{[0,S]}\tilde{s}_w$, followed by (16). Base predictions may instead be projected before combination, provided the values entering the fitting objective remain affine in w . In either case, projection during retrieval does not increase flux or albedo error under (15).

For a fixed finite threshold $\tau > 0$, the total-flux raw-error surrogate is

$$J_\tau(w) = \mathbb{E}_P \left[\frac{\tilde{e}_a^2 + R^2 \tilde{e}_t^2 + R^4 t^2 \tilde{e}_s^2}{(t \vee \tau)^2} \right]. \quad (20)$$

It is a convex quadratic: the errors are affine in the weights and the entry weights are nonnegative and independent of w . Its empirical version replaces the expectation by the validation average. Theorem 6 and Cauchy–Schwarz imply

$$\mathbb{E}_P |\bar{\rho}_w - \rho|^2 \leq R^2 F_t(\tau) + 3J_\tau(w). \quad (21)$$

Indeed, projection gives $e_R \leq |\tilde{e}_a| + R|\tilde{e}_t| + R^2 t|\tilde{e}_s|$; split the expectation at $t = \tau$ and apply the three-term square bound on $\{t > \tau\}$. This inequality holds for every feasible w , not only an optimizer.

The implemented separable surrogate. The diagnostic driver `code/joint_stack.py` combines Y_2 and Y_3 with separate weight vectors. It projects each base albedo onto $[0, S]$ before fitting but leaves the base fluxes unprojected. With $\tilde{e}_j = \tilde{Y}_j - Y_j$ for $j = 2, 3$, its finite-threshold fitting objective is

$$J'_\tau(w) = \mathbb{E}_P \left[\frac{\tilde{e}_a^2 + R^2 \tilde{e}_2^2 + R^2 \tilde{e}_3^2 + R^4 t^2 \tilde{e}_s^2}{(t \vee \tau)^2} \right]. \quad (22)$$

This is also a convex quadratic. It omits the cross term $2R^2\tilde{e}_2\tilde{e}_3$ present in J_τ . The flux floor is applied after fitting, to the combined total flux when the retrieval is scored, and is not part of the fitted loss. Since $t \geq 0$,

$$|\max\{\tilde{Y}_2 + \tilde{Y}_3, 0\} - t| \leq |\tilde{e}_2 + \tilde{e}_3| \leq |\tilde{e}_2| + |\tilde{e}_3|.$$

The same argument now uses four squared terms and gives

$$\mathbb{E}_P|\bar{\rho}_w - \rho|^2 \leq R^2 F_t(\tau) + 4J'_\tau(w). \quad (23)$$

The factor-three statement is therefore not the bound for the reported separable fit. Both statements require the physical hypotheses on the distribution being averaged; full-table statistics that include entries outside those hypotheses are not certificates of either bound.

Putting projection *inside* a squared fitting loss is a different problem and need not preserve convexity. On $0 \leq w \leq 1$, for example, $g(w) = (\max\{2w - 1, 0\} - 1)^2$ satisfies $g(1/2) = 1 > (g(0) + g(1))/2 = 1/2$. This counterexample does not apply to the raw-error quadratic actually fitted by the driver.

Forward-optimal weights. A stack fitted on a forward objective weights every band alike, while the retrieval divides the error at each entry by its transmission. The following example shows how far apart the two optima can be.

Proposition 10 (forward-optimal and retrieval-optimal weights). *Let $0 < \rho < R \leq 1$. For a convex combination of heads with weights w , let $F(w)$ be the mean squared radiance error at ρ and $E(w)$ the mean squared error of the constrained retrieval (16), and let w_f minimize F and w_r minimize E .*

1. *Take two equally weighted entries with $a = s = 0$ and transmissions $t_1 = 1$, $t_2 = \delta \in (0, 1]$, and two heads that predict t and s exactly and the path radiance with errors $e^A = (0, c\delta)$ and $e^B = (b, 0)$, where $0 < b, c \leq \min\{\rho, R - \rho\}$; w is the weight on A . Then*

$$F(w) = \frac{1}{2}[(1-w)^2b^2 + w^2c^2\delta^2], \quad E(w) = \frac{1}{2}[(1-w)^2b^2 + w^2c^2],$$

$$w_f = b^2/(b^2 + c^2\delta^2), \quad w_r = b^2/(b^2 + c^2), \quad F(w_f) \leq F(w_r), \quad \text{and}$$

$$\frac{E(w_f)}{E(w_r)} = \frac{(b^2 + c^2\delta^4)(b^2 + c^2)}{(b^2 + c^2\delta^2)^2},$$

which tends to $1 + c^2/b^2$ as $\delta \rightarrow 0$ and equals $(1 + \delta^2)^2/(4\delta^2)$ at $c = b/\delta$. For every $\tau \leq \delta$ the transmission-weighted objective J_τ of (20) equals E .

2. *Suppose instead that $s = 0$ at every entry, that every head predicts t and s exactly, that $t_{\min} \leq t \leq t_{\max}$ P -almost surely, and that the retrieval of every combination lies in $[0, R]$. Then*

$$E(w_f) \leq \left(\frac{t_{\max}}{t_{\min}}\right)^2 E(w_r).$$

Proof in Appendix A. The weights that minimize the forward error put almost all their mass on the head whose error sits at the opaque entry, because there it costs little radiance; the retrieval divides that error by δ and pays for it in full. With c/b large the forward-optimal stack is better

than either head in radiance and worse than the retrieval-optimal stack by any factor in the inverse, which is the pattern of Table 3 in miniature. Item (ii) bounds the loss by the squared dynamic range of the transmission, and item (i) at $c = b/\delta$ comes within a factor of four of the bound. On the physical domain of the EMIT table the transmission spans about twenty-four decades, and thirteen above the floor $10^{-12}T_{\text{train}}$ of Section 6.2, so the bound gives no control of the retrieval for weights fitted on a forward loss; the bound (21) for J_τ does not involve the range.

Numerical weights and threshold convention. The driver solves a nonnegative least-squares problem with a heavily weighted row penalizing $\sum_h w_h - 1$ and then normalizes the nonnegative weights to sum to one. This gives feasible simplex weights, though not an exact solution of the equality-constrained quadratic program, and the mean-relative-norm arm runs six iteratively reweighted least-squares sweeps from the squared-loss solution. The weights reported below are therefore approximate minimizers.

A finite threshold is written $\tau = uT_{\text{train}}$, where $T_{\text{train}} > 0$ is a flux scale of the training data, and Table 16 reports u . The unweighted fit is a plain squared-error baseline on the physical components, not the limit of (22) as $\tau \rightarrow \infty$. The ladder of thresholds is a sensitivity analysis; a deployed threshold would be fixed in advance by a coverage rule. The stack weights here are fitted on validation rows that also serve the other model decisions, so Proposition 5 does not apply to them.

10 Retrieval error and the transmission profile

The transfer theorem makes the ratio of component error to transmission the natural retrieval diagnostic, and the conditional-profile theorem asks for the profile $t \mapsto \mathbb{E}[e_R^2 \mid t]$. This section measures both on finite test sets; a slope fitted to a finite profile is a description of that profile, not an asymptotic rate.

Table 14 uses the three-member width-2000 pipeline on splits 104–107, with float64 predictions of nine heads on 2,331 test states and 285 bands, or 664,335 state–band entries per split. The retrieval uses $\rho = 0.7$, $R = 0.9$ and an albedo bound S taken from the training data.

The physical domain in these splits. S is the largest finite training albedo not exceeding one, and the training blocks hold 227 or 454 entries above one depending on the split. On split 104, 228 of the 664,335 test entries exceed S , and none do on the other three splits; there are 203, 6, 3 and 3 nonpositive transmissions on splits 104, 105, 106 and 107. The pointwise check covers the entries with $s \leq S$, and the expectations and ratios in the table are averages over all entries. Section 6 scores the main pipeline on the physical domain itself.

Observed error profiles. The feature kernel has $\varepsilon_R = 0.0514$ and the single network 0.1650. The mean fitted profile slopes 2ν range from 0.16 to 0.28. The raw 95th-percentile error is 0.075 for the single-member feature kernel and 0.060 for concatenated features, against 0.143 for the stack, and the constrained 95th percentiles are 0.076, 0.061 and 0.151. The ratio of the middle bound to the mean squared error lies between about 3.4 and 5.3 across heads.

The middle term of Theorem 6(ii), $\mathbb{E} \min\{R^2, e_R^2/t^2\}$, orders the nine heads by their retrieval tails. Taking the four-split means, its rank correlation with the raw 95th percentile is exactly one (Spearman and Kendall), and with the constrained 95th percentile 0.98 (Spearman), while the

A. Errors and finite-range profiles

head	ε_R	2ν (fit)	raw p95	constrained p95
			raw p99	constrained p99
isotropic kernel	0.0886±0.0331	0.16±0.11	0.696±0.000 0.97±0.02 0.670±0.026	0.266±0.018 0.70±0.00 0.268±0.111
network	0.1650±0.0177	0.20±0.16	0.99±0.14 0.625±0.013	0.70±0.00 0.239±0.078
network ensemble	0.1085±0.0254	0.18±0.14	0.86±0.08 0.489±0.054	0.70±0.01 0.200±0.000
network + residual	0.0698±0.0373	0.21±0.14	0.89±0.01 0.464±0.033	0.70±0.00 0.200±0.000
ensemble + residual	0.0690±0.0381	0.22±0.14	0.88±0.01 0.075±0.014	0.70±0.00 0.076±0.015
feature kernel	0.0514±0.0454	0.26±0.16	0.89±0.05 0.060±0.004	0.70±0.00 0.061±0.004
concatenated features	0.0634±0.0680	0.28±0.16	0.92±0.03 0.098±0.031	0.70±0.00 0.099±0.032
coordinate selection	0.0721±0.0648	0.25±0.15	0.85±0.03 0.143±0.015	0.70±0.00 0.151±0.015
convex stack	0.0600±0.0661	0.25±0.14	1.09±0.05	0.70±0.00

B. Failure counts and bound diagnostics

head	failure [%]	$\mathbb{E} \min\{R^2, \frac{e_R^2}{t^2}\}$	violations on $s \leq S$	diagnostic
	upper clip [%]			bound / MSE
isotropic kernel	1.09±0.03	0.064±0.000	0	3.9±0.2
	3.56±0.07			
network	1.31±1.00	0.056±0.003	0	3.4±0.7
	2.93±0.43			
network ensemble	0.61±0.33	0.051±0.002	0	3.7±0.9
	2.56±0.92			
network + residual	0.82±0.02	0.044±0.001	0	4.0±1.2
	2.51±0.12			
ensemble + residual	0.85±0.01	0.043±0.001	0	3.9±1.3
	2.43±0.07			
feature kernel	0.71±0.02	0.031±0.001	0	5.1±2.3
	1.53±0.10			
concatenated features	0.72±0.02	0.030±0.000	0	5.3±2.4
	1.49±0.02			
coordinate selection	0.73±0.02	0.033±0.002	0	4.7±2.0
	1.61±0.13			
convex stack	0.84±0.05	0.036±0.001	0	4.4±2.0
	1.78±0.08			

Table 14: Transmission diagnostics for the three-member width-2000 pipeline, splits 104–107. Each cell with a spread is a mean $\pm$ sample standard deviation over the four splits. All reflectance errors in this table are in reflectance units, not percentage points. Panel A gives the retrieval-weighted error, the fitted profile slope and the raw and constrained percentiles; panel B gives failure and clipping rates and the bound diagnostics. The violation count tests the pointwise bound on the entries with $s \leq S$; on entries with nonpositive transmission or an albedo outside $[0, 1)$ the bound expressions are diagnostics rather than bounds.

unweighted ε_R reaches only 0.78 and 0.81. The stack, for instance, has a smaller ε_R than the kernel on concatenated features (0.060 against 0.063) but a larger middle term (0.036 against 0.030) and a constrained tail more than twice as large. What orders the retrievals is the component error weighted by the transmission, as the theorem says, not the component error itself.

About five percent of the entries have transmission below 3×10^{-4} of the training flux scale, and among positive transmissions the one-percent quantile is of order 10^{-10} of that scale, while the share of squared retrieval-weighted component error on entries below 0.03 of the scale is under a thousandth for every head. Small absolute errors are amplified by weak transmission. The heads still differ in the inverse tail, because the location, sign and dependence of their errors matter, as the exact identities show.

The table’s own transmission distribution is extremely heavy near zero: relative to the median positive flux, its 1%, 3%, 5% and 10% quantiles are 1.1×10^{-10} , 3.6×10^{-6} , 2.2×10^{-4} and 1.7×10^{-2} , a log-log slope of $\beta \approx 0.12$ – 0.13 for F_t over the lowest tenth of the entries. With the measured profile slopes $2\nu \approx 0.16$ – 0.28 , Theorem 8 places the all-band retrieval in its slow regime, $\beta < 2(1 - \nu)$, where the mean squared error scales like $\sigma^{\beta/(1-\nu)}$ with exponent 0.13–0.15: halving the component error would shrink it by less than a tenth. On a passband $t \geq t_0 > 0$ fixed in advance the bound of item (iv) is linear in ε_R instead. A slope fitted over a finite range of transmissions does not establish the tail law down to zero or an envelope uniform over training sizes, so this reading describes the table’s regime rather than proving a rate. Scaling a fixed error profile down buys little over all bands; moving the error away from the opaque entries, as the feature kernels do, buys much more.

11 Representation, correction and combination

Concatenated features. Table 15 repeats the three-member width-2000 experiment of Section 7 on splits 104–107, with the networks trained on one GPU.

The concatenated-feature kernel has mean radiance error 0.0688%, against 0.0764% for the features of one network; the paired gain is 0.0076 ± 0.0053 percentage points, positive on three splits and -0.0002 on the fourth. Its mean within-split 95th-percentile reflectance error is 5.99 points, against 7.51 for single-member features and 14.35 for the stack. With the ten width-512 splits of Section 7 this makes three independent settings in which concatenation lowers both the radiance error and the reflectance tail.

Fitting objective against retrieval metric. Table 16 compares stacks fitted under different objectives on five splits: the wide pipeline at seed 101 and the three-member pipeline on splits 104–107. The objectives and the projection sequence are those of Section 9.5.

The mean-relative-norm stack has radiance error 0.0754% and constrained 95th-percentile error 0.141, and the row-relative squared-error stack 0.1130% and 0.180. The finite-threshold J'_τ fits give radiance errors of about 0.22–0.23% for $u \leq 0.1$, 0.1491% at $u = 0.3$ and 0.1175% at $u = 1$, and the unweighted baseline 0.1262%. Weighting by the transmission moves the stack the way Proposition 10 predicts: at $u = 0.003$ the constrained 95th percentile falls from 0.141 to 0.103 while the radiance error rises from 0.075% to 0.216%. It does not reach the kernel on features alone (0.077) or on concatenated features (0.061). The fitting loss and the reported metric need not favor the same combination, and J'_τ is a mean-square criterion while the table reports a quantile; the factor-four transfer bound holds for every feasible weight vector on a distribution satisfying its hypotheses.

Kernel choice on the state and on learned features. Table 17 covers four components on seeds 101–103. The spectral mixture and the Matérn-5/2 kernel have similar mean errors on Y_1 (0.089%), and the spectral mixture has the lowest mean errors of the raw-state kernels on Y_2 and Y_3 (0.237% and 0.099%). The first-order additive kernel is much less accurate in this specification. Albedo errors for the nonadditive kernels lie between about 2.3% and 3.5%.

The Matérn-5/2 feature kernel gives 0.105%, 0.211%, 0.114% and 2.303% on Y_1 – Y_4 , better than the best raw-state kernel on some components and not on others. The convex stack has the lowest component means in this comparison (0.066%, 0.156%, 0.077%, 1.956%), which is consistent with its large reflectance tails elsewhere: the component metric does not weight errors the way the inverse does.

Network regularization. Table 18 compares a control, a weight-decay grid and an early-stopping grid on three seeds. The validation-selected weight decay matches the control’s component mean (2.27%), and the selected early stopping is less accurate and more variable ($2.72 \pm 0.45\%$). Residual correction brings the component means to 1.88–1.95% and feature kernels to 0.95–0.99%, with radiance errors near 0.15%, whichever regularization the network used. The reference row is the isotropic kernel fitted in the same runs. Under this budget, regularizing the network changes little; the kernel stage does the work.

Perturbation and ridge-path diagnostics. The tables in Appendix B give the perturbation diagnostics for the state inputs and the learned features. The residual-action bound of Proposition 9 is often much smaller than the spectrum-only bound, yet it can still exceed the actual movement by a factor of tens, thousands or more: positive definiteness makes the bound applicable without making it a useful tolerance.

Table 21 reports a 4,000-row fit of four components at seed 101. At the smallest ridge on input-space Y_1 the effective degrees of freedom are 3,841 and the noise factor 0.924. The same-half and split-half principal-component errors are 0.079% and 2.465%, and the physical-unit test errors of the corrected and standalone kernels 0.2168% and 0.1754%; these columns use different metrics and samples and are not to be subtracted. At $\gamma = 10^{-2}$ the corresponding test errors are 0.5095% and 5.7578%: the fitted mean keeps its predictive information when the kernel correction is strongly regularized.

12 A synthetic check of the transfer theorem

Table 22 checks the construction of Proposition 7 on manufactured data, two million samples for each $\beta \in \{0.5, 1, 2, 3, 4\}$. The average-error slopes are 0.200, 0.333, 0.498, 0.597 and 0.664, against the limits 0.200, 0.333, 0.500, 0.600 and 0.667. For the uniform family the finite-ladder slope is compared with the analytic curve on the same ladder: at $\beta = 2$ the measured slope is 1.721 and the analytic one 1.724, while the limit is two. The largest relative Monte Carlo discrepancy is about 1.01% for the uniform family at $\beta = 2$ and below 0.053% for the profile family against its quadrature reference, and no pointwise violation occurs among the ten million sampled transmissions.

13 The matched benchmark and the transfer contrast

Correction-coefficient benchmark. Table 23 scores our families under the standard and out-of-distribution protocols of [4], with and without the low-fidelity 6S coefficients as inputs, against the published values of the reference model, which we did not retrain.

The state-only residual-corrected network reaches standard-split RMSE 0.00581 ± 0.00004 , MAE 0.00151, uniform-average $R^2 = 0.9952$ and SMAPE 3.54%, against published values 0.00619, 0.00186, 0.9947 and 3.91%; with 6S inputs its means are 0.00578, 0.00148, 0.9954 and 3.19%. On the out-of-distribution split the state-only means are 0.00620, 0.00252, 0.9967 and 3.37%, against published 0.01022, 0.00544, 0.9942 and 5.40%, and with 6S inputs 0.00596, 0.00232, 0.9969 and 2.92%. The spreads describe three seeds on the standard protocol and three initializations on the out-of-distribution protocol.

The R^2 convention matters here. The reference weights the three coefficients equally, whereas a variance-weighted R^2 is the default of many libraries. We report the uniform average, computed from per-coefficient target scales that are recovered from the per-coefficient RMSE of the heads and checked against the variance-weighted value. The ensemble without correction is close to the corrected network, while the raw-state kernel is much less accurate in these configurations.

OCO-2 coefficient and radiance objectives. Table 24 covers three bands and seeds 0–2. The feature kernel trained under the flat coefficient loss has coefficient errors of about 4.1%, 16.3% and 8.1% on O_2 , weak CO_2 and strong CO_2 , and radiance errors of 0.076%, 0.114% and 0.108%. A radiance-weighted feature construction gives about 0.031%, 0.048% and 0.064% in radiance, at the price of larger coefficient errors. The reconstruction map and the evaluation metric decide which fitted representation is preferred.

The radiance stack minimizes a row-relative squared radiance loss, a convex quadratic because the reconstruction is affine, and its solver reaches KKT gaps between 4.17×10^{-6} and 1.52×10^{-5} . Its mean radiance errors are 0.0264%, 0.0391% and 0.0511%, against 0.0291%, 0.0404% and 0.0536% for the squared-loss stack on reduced coordinates, while its coefficient errors rise to 18.59% and 37.43% on O_2 and weak CO_2 and to 8.87% on strong CO_2 . All rules use 2,000 validation and 2,000 test rows.

14 Discussion

On this table, input scaling and learned features both pay. The input-scaled Matérn kernel reduces radiance error well below the isotropic kernel, the cubic and the width-512 network; residual fitting cuts the network’s error by almost a factor of three; and a kernel on learned features matches the input-scaled kernel with one network and beats it with several. The width experiment separates the two roles of a network: the width-2,000 network is the worse emulator and the better feature map. Concatenating the features of an ensemble and solving one kernel on them gives the lowest radiance error and the smallest reflectance tail of the ten-partition comparisons, with one solve in place of a stack.

Forward and inverse rankings still differ. The input-scaled kernel has the smallest median reflectance error of the main comparison, while the feature kernels have the smallest 95th-percentile errors, and both the narrow and the wide stack increase that tail at every split relative to their feature kernels. For the wide model a mean radiance gain of 0.0006 points accompanies an increase from 7.57 to 22.71 points in the mean within-split 95th-percentile reflectance error.

Neither a single forward score nor a reflectance median describes this. Section 6 shows that the effect is a property of the emulators: it holds on the physical domain of the retrieval, with coverage and failures reported, [[TQ-DISCUSSION]] and on a fresh partition read once under a hypothesis fixed beforehand, where the stack-to-feature-kernel ratios of the tail are 6.60 over all bands and 5.54 on the admissible domain.

Theorem 6 explains why such a reversal is possible. On the physical domain the constrained retrieval error is at most e_R/t , capped at R , with constant one, and a distributional form separates the mass of small transmissions from the error relative to transmission on the rest. Proposition 7 shows that the resulting exponent is attained, and Theorem 8 that a conditional root-mean-square component error proportional to the transmission loses no rate. What decides the retrieval tail is where the component errors sit relative to the transmission, and a stack that lowers the average forward error can move error into the entries with small transmission. The data agree: the transmission-weighted middle term of the bound ranks nine heads exactly as their raw retrieval tails (Section 10), while the unweighted component error does not.

Proposition 10 isolates the mechanism for stacks. Weights that minimize a forward error move mass onto the head whose error sits where the transmission is small, since there it costs little radiance, and the retrieval divides that error by the transmission; the forward-optimal combination can then be worse than the retrieval-optimal one by any factor while winning on radiance. The transmission-weighted quadratic J_τ of Section 9.5 targets the retrieval directly: it bounds the mean squared constrained retrieval error for every feasible weight vector, with factor three (four in its separable form), and in the example of the proposition it is the retrieval error itself. Its weights still come from validation rows, and the bound becomes a guarantee only on an independent selection sample (Proposition 5).

For residual correction, the alignment identity and the principal-component decomposition show which parts of the error the correction can remove and which remain, and they explain why the optimal output rank for the correction is lower than for the direct kernel. A learning rate for the corrected predictor would need a uniform regression bound and control of the residual in a source norm; the learning curves measure slopes over a finite range. Proposition 9 bounds the movement between two fitted systems; its residual-action form is often much smaller than the spectral bound, though still large relative to the observed movement.

Across the further problems in the appendices the comparison stays metric- and task-dependent. On OCO-2 the coefficient and radiance losses select different representations, on the correction-coefficient benchmark the residual-corrected network improves on the published reference values, and on the ten further configurations the convex stack is at or near the best member of its pool.

Accuracy has a cost. A dense kernel matrix needs order n^2 storage and its factorization order n^3 operations; one $18,884 \times 18,884$ double-precision matrix occupies 2.85 GB before workspaces. A feature kernel needs a network evaluation and a kernel prediction against the training design, and a stack evaluates every head, so a matched throughput comparison with operational emulators such as sRTMnet [1] remains to be made.

The evidence has limits that bear on its reading. The families were developed on this table and the ten splits overlap, so the split-to-split spread describes one table rather than independent data; the fresh partition guards against the choice of split and of hypothesis, not against the table itself. The table is an adaptively explored lookup table, and an operational evaluation would fix its passband and coverage in advance, separate physical from numerical failures, and report both raw and constrained retrievals, as Section 6 does for the present emulators. Within

that scope, neural features and exact kernels complement one another, and a small forward gain can come with a much larger inverse-error tail.

seed	head	radiance [%]	median [pp]	p95 [pp]	gain [pp]
104	feature kernel	0.0924	0.044	9.25	–
104	concatenated features	0.0810	0.025	6.25	+0.0114
104	convex stack	0.0781	0.024	12.77	–
104	network + residual	0.1843	0.115	54.71	–
105	feature kernel	0.0611	0.031	7.31	–
105	concatenated features	0.0508	0.028	6.02	+0.0103
105	convex stack	0.0475	0.024	14.50	–
105	network + residual	0.1349	0.087	42.61	–
106	feature kernel	0.0789	0.043	7.67	–
106	concatenated features	0.0702	0.035	6.28	+0.0087
106	convex stack	0.0687	0.034	16.29	–
106	network + residual	0.1608	0.108	51.84	–
107	feature kernel	0.0731	0.042	5.82	–
107	concatenated features	0.0732	0.044	5.41	-0.0002
107	convex stack	0.0833	0.043	13.84	–
107	network + residual	0.1476	0.103	46.58	–
mean	feature kernel	0.0764±0.0130	0.040±0.006	7.51±1.41	–
mean	concatenated features	0.0688±0.0128	0.033±0.009	5.99±0.40	0.0076±0.0053
mean	convex stack	0.0694±0.0158	0.031±0.009	14.35±1.48	–
mean	network + residual	0.1569±0.0211	0.103±0.012	48.93±5.40	–

Table 15: Kernel on concatenated features, three width-2000 networks, splits 104–107. Radiance is a mean relative spectral error in percent; reflectance median and 95th-percentile errors are in percentage points. The last column is the paired radiance gain of concatenated features over the features of one network. The mean row averages the four within-split quantities.

head / fit	radiance [%]		constrained p95	
			ε_R constrained p99	raw p95
network	0.8609±0.1051	0.1664±0.0156	0.254±0.101 0.70±0.00 0.200±0.000	0.677±0.027
network + residual	0.1560±0.0184	0.0654±0.0338	0.70±0.00 0.077±0.013	0.486±0.047
feature kernel	0.0755±0.0114	0.0472±0.0404	0.70±0.00 0.266±0.018	0.076±0.012
isotropic kernel	0.2454±0.0106	0.0886±0.0331	0.70±0.00 0.239±0.078	0.696±0.000
network ensemble	0.5050±0.0592	0.1085±0.0254	0.70±0.01 0.200±0.000	0.625±0.013
ensemble + residual	0.1379±0.0157	0.0690±0.0381	0.70±0.00 0.061±0.004	0.464±0.033
concatenated features	0.0688±0.0128	0.0634±0.0680	0.70±0.00 0.141±0.033	0.060±0.004
stack, mean relative norm	0.0754±0.0116	0.0573±0.0575	0.70±0.00 0.180±0.044	0.133±0.029
stack, row-relative square	0.1130±0.0454	0.0518±0.0414	0.70±0.00 0.135±0.060	0.281±0.122
stack, J'_τ , $u = 0.001$	0.2203±0.1042	0.0633±0.0298	0.69±0.01 0.103±0.055	0.213±0.227
stack, J'_τ , $u = 0.003$	0.2159±0.1100	0.0736±0.0522	0.69±0.01 0.107±0.053	0.182±0.238
stack, J'_τ , $u = 0.01$	0.2303±0.1023	0.0761±0.0508	0.59±0.22 0.107±0.053	0.185±0.236
stack, J'_τ , $u = 0.03$	0.2304±0.1024	0.0761±0.0508	0.59±0.22 0.110±0.052	0.185±0.235
stack, J'_τ , $u = 0.1$	0.2298±0.1013	0.0739±0.0463	0.59±0.22 0.117±0.048	0.188±0.235
stack, J'_τ , $u = 0.3$	0.1491±0.0508	0.0615±0.0468	0.69±0.03 0.156±0.032	0.183±0.196
stack, J'_τ , $u = 1.0$	0.1175±0.0422	0.0553±0.0469	0.70±0.00 0.168±0.039	0.152±0.032
stack, unweighted square	0.1262±0.0406	0.0559±0.0420	0.70±0.00	0.194±0.095

Table 16: Stacks under different fitting objectives at $\rho = 0.7$, five splits. The finite-threshold rows use the separable quadratic J'_τ of (22), with $u = \tau/T_{\text{train}}$; the unweighted row is a component squared-error baseline. Radiance is in percent and the reflectance percentiles in reflectance units. Weights are fitted approximately, as described in Section 9.5.

head	Y_1 [%]	Y_2 [%]	Y_3 [%]	Y_4 [%]
Matérn-1/2	0.179±0.008	0.672±0.003	0.418±0.012	2.326±0.103
Matérn-3/2	0.092±0.002	0.391±0.008	0.207±0.016	2.481±0.043
Matérn-5/2	0.089±0.017	0.434±0.007	0.203±0.003	2.746±0.082
Gaussian	0.092±0.002	0.525±0.113	0.228±0.051	3.502±0.501
spectral mixture (4)	0.089±0.010	0.237±0.014	0.099±0.013	3.468±0.263
NNGP, ReLU	0.175±0.001	0.674±0.015	0.350±0.009	2.548±0.057
NTK, ReLU	0.263±0.002	1.055±0.015	0.621±0.014	2.321±0.048
additive Matérn-5/2	4.091±0.093	20.266±1.035	11.430±0.237	4.413±0.077
ARD by kernel flow	0.322±0.015	0.257±0.019	0.199±0.005	2.418±0.114
ARD by marginal likelihood	0.281±0.039	0.341±0.042	0.183±0.011	2.702±0.027
convex kernel sum	0.087±0.007	0.253±0.020	0.099±0.013	2.557±0.099
network	0.819±0.027	2.289±0.019	0.910±0.057	3.482±0.085
network + residual kernel	0.523±0.007	1.337±0.026	0.588±0.035	3.055±0.121
features, Matérn-5/2	0.105±0.006	0.211±0.016	0.114±0.008	2.303±0.216
features, Gaussian	0.126±0.007	0.262±0.018	0.138±0.014	2.426±0.214
features, NTK	0.152±0.005	0.307±0.015	0.156±0.007	2.150±0.059
features, ARD	0.116±0.007	0.214±0.014	0.111±0.004	2.217±0.207
coordinate selection	0.090±0.010	0.360±0.008	0.105±0.010	2.797±0.447
convex stack of the heads	0.066±0.003	0.156±0.010	0.077±0.004	1.956±0.037
ridge stack of the heads	0.087±0.005	0.288±0.031	0.087±0.005	2.310±0.071

Table 17: Ten input-space kernels, feature kernels and combination rules on the EMIT state inputs, with rank-64 outputs. Entries are component mean relative spectral errors in percent, averaged over seeds 101–103 with sample standard deviations. The network has width 512 and depth 3 with a 150-epoch budget. Kernels include the Matérn family, Gaussian, spectral mixture [72], NNGP [93, 91], NTK [94], additive kernel [73], and learned input scales [58, 8].

head	alone component / radiance	residual correction component / radiance	feature kernel component / radiance
isotropic kernel	0.941±0.109		
	0.284±0.078		
network, control	2.266±0.031	1.903±0.055	0.955±0.038
	0.809±0.029	0.617±0.033	0.153±0.052
network, weight decay	2.269±0.027	1.880±0.072	0.948±0.048
	0.820±0.014	0.618±0.040	0.154±0.048
network, early stopping	2.716±0.445	1.949±0.073	0.986±0.032
	1.293±0.373	0.789±0.127	0.147±0.054

Table 18: Regularization comparison, seeds 101–103. Each cell gives the mean component relative error [percent] / radiance relative error [percent], with sample standard deviations. The width-512, depth-3 network has a 150-epoch cap, and validation selects among three runs per arm. The weight-decay grid is 10^{-5} , 10^{-4} , 10^{-3} and the early-stopping patience grid 3, 5 and 10 epochs [87, 88]. The reference row is the isotropic kernel fitted in the same runs.

A Proofs

The statements are as numbered in the text.

Proof of Proposition 1. Subtract the two forward maps and use $1/\hat{q} - 1/q = \rho e_s/(q\hat{q})$ to obtain (4). Since $u = \rho D$, $\hat{D} = D + e_t + u e_s - \hat{s} e_a$ and $\hat{\rho} - \rho = (u - e_a - \rho \hat{D})/\hat{D}$, which gives (5). The triangle inequality yields $|\hat{D} - D| \leq B$ and then (6). $\square$

Proof of Proposition 2. Subtract $h + S(f - h)$ from f to obtain (9). On the design, $I - K(K + \lambda I)^{-1} = \lambda(K + \lambda I)^{-1}$, which proves (10) and its spectral form. Apply the assumed uniform inequality to $g = f - h$ for (11). The examples below distinguish this bound from actual error comparisons. $\square$

Proof of Proposition 3. For a common operator the corrected error is $a - b$, so expansion of its squared norm gives (12). The error of p_t is $a - tb$; minimize the resulting quadratic on $[0, 1]$ to obtain (13). For distinct operators, $f - h - S_1(f - h) = (I - S_0)f - [(S_1 - S_0)f + (I - S_1)h]$, after which the same expansion applies. $\square$

Proof of Proposition 4. (i) is the chain rule applied to $\hat{y} = S_\lambda y + (I - S_\lambda)h_X(y)$, and the two examples are direct substitutions. (ii) With the mean and kernel frozen the map is affine with linear part S_λ ; the trace is the sum of the eigenvalues $\mu_j/(\mu_j + \lambda)$, and the operator norm of a symmetric positive semidefinite matrix is its largest eigenvalue. (iii) $\hat{y} - f_X = -(I - S_\lambda)r + S_\lambda \varepsilon$; the cross term has zero mean, and both matrices are diagonal in the eigenbasis of K . (iv) The correction at x is $a_\lambda(x)^\top (y - h_X)$, linear in ε with coefficient vector $a_\lambda(x)$. $\square$

Proof of Proposition 5. Let $\hat{\mathcal{R}}_j$ and $\mathcal{R}_j$ be the empirical and population risks. Hoeffding's inequality [108] and a union bound give $|\hat{\mathcal{R}}_j - \mathcal{R}_j| \leq B\sqrt{\log(2M/\delta)/(2m)}$ simultaneously for all candidates with probability at least $1 - \delta$. Apply this once to the empirical minimizer and once to a population minimizer, and use empirical optimality between the two inequalities. $\square$

Proof of Theorem 6. Write $q = 1 - \rho s$, $u = L - a = \rho t/q$, $D = t/q$, $\hat{D} = \hat{t} + \hat{s}(L - \hat{a})$ and $z = \bar{\rho}$. Suppose first $\hat{D} > 0$. The map $F(v) = \hat{D}v - (L - \hat{a})$ is increasing and affine, z is the projection of its root onto $[0, R]$ and $\rho \in [0, R]$, so $(z - \rho)F(z) \leq 0$: at an interior root $F(z) = 0$; at $z = R$ the root exceeds R , so $F(R) < 0$ and $z - \rho \geq 0$; at $z = 0$ the root is negative, so $F(0) > 0$ and $z - \rho \leq 0$. Since $L - \hat{a} = u - e_a$ and, by (5), $\hat{D} = D + e_t + u e_s - \hat{s} e_a$ with $u = \rho D$,

$$F(z) = D(z - \rho) + (1 - z\hat{s})e_a + z e_t + z u e_s,$$

hence $D(z - \rho)^2 \leq |z - \rho|((1 - z\hat{s})|e_a| + z|e_t| + zu|e_s|)$ and, with $D = t/q$,

$$|z - \rho| \leq q(1 - z\hat{s})\frac{|e_a|}{t} + qz\frac{|e_t|}{t} + z\rho|e_s|.$$

Now $q(1 - z\hat{s})$ is positive and at most 1 under the hypotheses, which is all that is used here, $qz \leq R$ and $z\rho \leq R^2$, so $|z - \rho| \leq e_R/t$; and $|z - \rho| \leq R$ because both lie in $[0, R]$. If $\hat{D} \leq 0$, then $\hat{t} \leq \hat{s}(e_a - u) \leq \hat{s}|e_a|$ and $t = \hat{t} - e_t \leq |e_t| + \hat{s}|e_a|$, so $Rt \leq R|e_t| + R\hat{s}|e_a| \leq R|e_t| + |e_a| \leq e_R$; the fallback value and ρ lie in $[0, R]$, so $|z - \rho| \leq R = \min\{R, e_R/t\}$. For optimality take $s = \hat{s} = 0$, $\hat{t} = t$, $\rho = r_0$ and $\hat{a} = a + ct$ with $0 < c < r_0 < R$: then $|\bar{\rho} - \rho| = c = e_R/t$ exactly, with a positive denominator and no projection active. This proves (i). For (ii) integrate (i), split at $t = \tau$ and bound e_R^2/t^2 by e_R^2/τ^2 on $\{t > \tau\}$. For (iii), $\varepsilon_R = 0$ gives zero error by (i). Otherwise take $\tau = \varepsilon_R^{2/(\beta+2)} \leq u_0$; then $R^2 A \tau^\beta + \varepsilon_R^2/\tau^2 = (R^2 A + 1)\varepsilon_R^{2\beta/(\beta+2)}$. (iv) is (i) with $t \geq t_0$. $\square$

Proof of Proposition 7. $L = r_0 t$ and $\hat{\rho} = (r_0 t - \hat{a})/t$; $e_R = |\hat{a}| = ct \mathbf{1}\{t \leq \tau\}$; with density $\beta u^{\beta-1}$ on $(0, 1)$, $\mathbb{E}_P[t^2 \mathbf{1}\{t \leq \tau\}] = \beta \tau^{\beta+2}/(\beta + 2)$; the retrieval error equals c exactly on $\{t \leq \tau\}$, an event of probability τ^β . Eliminate τ . $\square$

Proof of Theorem 8. If $R = 0$ the statement is immediate; assume $R > 0$. Conditional on t the squared error is at most both R^2 and e_R^2/t^2 by Theorem 6(i), so $\mathbb{E}[|\bar{\rho} - \rho|^2 | t] \leq \min\{R^2, \mathbb{E}[e_R^2 | t]/t^2\} \leq \min\{R^2, \sigma^2 t^{2\nu-2}\}$; integrate. With $t_* = (\sigma/R)^{1/(1-\nu)}$, $\mathbb{E}_P \min\{R^2, \sigma^2 t^{2\nu-2}\} = R^2 F_t(t_*) + \sigma^2 \mathbb{E}_P[t^{2\nu-2} \mathbf{1}\{t > t_*\}]$; the first term is at most $R^2 A t_*^\beta = O(\sigma^{\beta/(1-\nu)})$ once $t_* \leq u_0$, and $\mathbb{E}_P[t^{2\nu-2} \mathbf{1}\{t > t_*\}] = \int_0^{t_*^{2\nu-2}} \mathbb{P}(t_* < t < v^{-1/(2-2\nu)}) dv \leq u_0^{2\nu-2} + A \int_{u_0^{2\nu-2}}^{t_*^{2\nu-2}} v^{-\beta/(2-2\nu)} dv$, which is $O(1)$ for $\beta > 2(1-\nu)$, $O(\log(1/\sigma))$ for $\beta = 2(1-\nu)$ and $O(t_*^{\beta-2(1-\nu)})$ for $\beta < 2(1-\nu)$. At $\nu = 1$ the bound is σ^2 before any tail enters. In the model, $c^{-2} \mathbb{E}_P |\bar{\rho} - \rho|^2 = \mathbb{E}_P \min\{1, (\sigma/c)^2 t^{2\nu-2}\}$, the same integral with equality. $\square$

Proof of Proposition 10. (i) With $a = s = 0$ and exact t and s , the stack's path-radiance error is $e(w) = ((1-w)b, wc\delta)$, and the radiance at ρ is predicted with the same error, since $q = \hat{q} = 1$; this gives F . The constrained retrieval evaluates $(L - \hat{a})/t = \rho - e_i/t_i$ at entry i , with denominator $t_i > 0$, and $|e_i|/t_i \leq \max\{b, c\} \leq \min\{\rho, R - \rho\}$ keeps it in $[0, R]$, so the projection is inactive and $\bar{\rho} - \rho = -((1-w)b, wc)$; this gives E . Both are convex quadratics in w , and setting their derivatives to zero gives w_f and w_r . Substituting,

$$E(w_f) = \frac{b^2 c^2 (c^2 \delta^4 + b^2)}{2(b^2 + c^2 \delta^2)^2}, \quad E(w_r) = \frac{b^2 c^2}{2(b^2 + c^2)},$$

whose ratio is the one stated. It tends to $(b^2 + c^2)/b^2$ as $\delta \rightarrow 0$; at $c = b/\delta$ the two factors of the numerator are $b^2(1 + \delta^2)$ and $b^2(1 + \delta^2)/\delta^2$ and the denominator is $4b^4$. Since w_f minimizes F , $F(w_f) \leq F(w_r)$. Finally, with $\tilde{e}_t = \tilde{e}_s = 0$ and $\tau \leq \delta \leq 1$, the weights $(t_i \vee \tau)^{-2}$ in (20) are 1 and δ^{-2} , so $J_\tau(w) = \frac{1}{2}[(1-w)^2 b^2 + w^2 c^2 \delta^2 / \delta^2] = E(w)$.

(ii) With $s = \hat{s} = 0$ and $\hat{t} = t$, the predicted radiance at ρ is $\hat{a} + \rho t$, so its error is the path-radiance error r_w of the combination, and the retrieval evaluates $(L - \hat{a})/t = \rho - r_w/t$, which lies in $[0, R]$ by hypothesis. Hence $F(w) = \mathbb{E}_P r_w^2$ and $E(w) = \mathbb{E}_P [r_w^2/t^2]$. Pointwise $r_w^2/t_{\max}^2 \leq r_w^2/t^2 \leq r_w^2/t_{\min}^2$, so $F \leq t_{\max}^2 E$ and $E \leq t_{\min}^{-2} F$ for every w , and

$$E(w_f) \leq t_{\min}^{-2} F(w_f) \leq t_{\min}^{-2} F(w_r) \leq \left(\frac{t_{\max}}{t_{\min}}\right)^2 E(w_r).$$

In the example of (i), $t_{\max}/t_{\min} = 1/\delta$, and the ratio at $c = b/\delta$ is at least $\delta^{-2}/4$. $\square$

Proof of Proposition 9. $\tilde{H} = H^{1/2}(I + B)H^{1/2}$ gives $\tilde{H}^{-1} = H^{-1/2}QH^{-1/2}$ and $Q = I - QB$. Hence $\tilde{k}_x^\top \tilde{H}^{-1}(Y - \tilde{M}) - k_x^\top H^{-1}(Y - M) = \tilde{g}^\top Q(W - U) - g^\top W = \tilde{g}^\top W - \tilde{g}^\top QBW - \tilde{g}^\top QU - g^\top W = v^\top W - (Q\tilde{g})^\top (BW + U)$, which is (i) since Q is symmetric; (ii) bounds its two terms. For (iii) write the same difference as $v^\top QW + g^\top (Q - I)W - \tilde{g}^\top QU$ and use $\|Q\|_{\text{op}} = \gamma_Q$ and $\|Q - I\|_{\text{op}} = \|QB\|_{\text{op}} = \eta_Q$. For (iv), $v = 0$, $U = 0$ and $B = sH^{-1}$ has eigenvalues $s/\lambda_i(H) \geq 0$, so $\eta_Q = \max_i (s/\lambda_i)/(1 + s/\lambda_i) = s/(\lambda_{\min}(H) + s)$. $\square$

B Extended diagnostic tables

This appendix collects the perturbation, ridge-path, synthetic and external-benchmark tables referred to in Sections 11–13, and the record of the runs behind Section 6.3.

width	seed	machine	raw	admissible	matched	validation	test
3×512	101	DGX	18,884	15,287	15,287	2,098	2,331

Table 19: The runs of the training-target experiment: for each split and width, the machine it ran on and the number of training states in each arm. Validation and test blocks are the same in the three arms. The admissible and matched arms have the same size by construction.

Table 20: Same-data perturbation diagnostics for Proposition 9. Movement is a Euclidean prediction change in retained coordinates. The four ratio columns are *medians of entrywise bound/movement ratios*, not ratios of medians. The fourteen state-input rows use the direct flux Y_2 at seed 101 with 2,000 fitting rows; the feature rows use four component records at seed 101 and four at seed 104, each with 4,000 fitting and 1,000 query rows. The signed minimum eigenvalue is rounded: a displayed -1.000 does not assert exact equality to -1 . A dash means the corresponding bound is not applicable.

source	perturbation	$\lambda_{\min}(B)$	$\ B\ $	median movement	residual-action ratio	signed-spectrum ratio	$\ B\ < 1$ ratio	ridge ratio
state	Nyström 50	-0.941	0.941	2.01	57.3	4.05e+03	4.05e+03	–
state	Nyström 100	-0.872	0.872	1.05	82.1	3.05e+03	3.05e+03	–
state	Nyström 200	-0.642	0.642	0.498	136	1.65e+03	1.65e+03	–
state	Nyström 400	-0.411	0.411	0.2	178	1.54e+03	1.54e+03	–
state	Nyström 800	-0.193	0.193	0.0702	264	1.47e+03	1.47e+03	–
state	Nyström 1600	-0.078	0.078	0.0103	408	3.51e+03	3.51e+03	–
state	ridge $\times 0.5$	-0.500	0.500	0.304	517	1.34e+03	1.34e+03	–
state	ridge $\times 2.0$	+0.000	0.999	0.372	835	550	1.32e+06	550
state	ridge $\times 10.0$	+0.002	8.993	1.36	1.98e+03	271	–	271
state	ridge $\times 100.0$	+0.018	98.918	3.67	7.38e+03	111	–	111
state	PCA inputs 2	-0.998	3.600	10.5	14.3	4.81e+04	–	–
state	PCA inputs 3	-0.998	2.523	9.8	14.2	4.49e+04	–	–
state	PCA inputs 4	-0.998	1.581	5.24	25.3	6.52e+04	–	–
state	PCA inputs 5	-0.997	0.997	1.9	64.3	1.39e+05	1.39e+05	–
s101, Y_1	PCA features 1000	-0.192	0.192	0.00383	4.01e+03	3.8e+04	3.8e+04	–
s101, Y_1	PCA features 500	-0.503	0.503	0.0107	4.41e+03	5.82e+04	5.82e+04	–
s101, Y_1	PCA features 250	-0.764	0.764	0.0185	4.3e+03	1.1e+05	1.1e+05	–
s101, Y_1	PCA features 125	-0.917	0.917	0.0279	3.88e+03	2.6e+05	2.6e+05	–
s101, Y_1	PCA features 64	-0.965	0.965	0.0407	3.55e+03	4.78e+05	4.78e+05	–
s101, Y_1	PCA features 32	-0.991	0.991	0.0595	3.13e+03	1.59e+06	1.59e+06	–
s101, Y_1	PCA features 16	-0.999	0.999	0.0923	2.48e+03	1.3e+07	1.3e+07	–
s101, Y_1	Nyström 100	-1.000	1.000	0.465	637	8.85e+07	8.85e+07	–
s101, Y_1	Nyström 400	-1.000	1.000	0.223	1.27e+03	7.11e+07	7.11e+07	–
s101, Y_1	Nyström 1600	-1.000	1.000	0.0746	2.58e+03	2.74e+07	2.74e+07	–
s101, Y_1	Nyström 3200	-0.997	0.997	0.0196	5.22e+03	9.75e+06	9.75e+06	–
s101, Y_1	ridge $\times 0.5$	-0.490	0.490	0.0102	9.61e+03	5.74e+04	5.74e+04	–
s101, Y_1	ridge $\times 2.0$	+0.000	0.979	0.0115	1.7e+04	2.62e+04	2.5e+06	2.62e+04
s101, Y_1	ridge $\times 10.0$	+0.000	8.813	0.0407	4.32e+04	1.34e+04	–	1.34e+04
s101, Y_1	ridge $\times 100.0$	+0.000	96.942	0.0889	2.17e+05	6.78e+03	–	6.78e+03
s101, Y_2	PCA features 1000	-0.122	0.122	0.00143	6.41e+03	4.95e+04	4.95e+04	–
s101, Y_2	PCA features 500	-0.418	0.418	0.00425	6.8e+03	8.59e+04	8.59e+04	–

Continued on next page

Table 20 (continued)

source	perturbation	$\lambda_{\min}(B)$	$\ B\ $	median movement	residual-action ratio	signed-spectrum ratio	$\ B\ < 1$ ratio	ridge ratio
s101, Y_2	PCA features 250	-0.726	0.726	0.00802	6.45e+03	1.7e+05	1.7e+05	–
s101, Y_2	PCA features 125	-0.868	0.868	0.0117	6.33e+03	2.99e+05	2.99e+05	–
s101, Y_2	PCA features 64	-0.977	0.977	0.0184	5.76e+03	1.3e+06	1.3e+06	–
s101, Y_2	PCA features 32	-0.989	0.989	0.0267	5.78e+03	2.3e+06	2.3e+06	–
s101, Y_2	PCA features 16	-0.999	0.999	0.0419	4.83e+03	1.49e+07	1.49e+07	–
s101, Y_2	Nyström 100	-1.000	1.000	0.345	731	1.01e+08	1.01e+08	–
s101, Y_2	Nyström 400	-1.000	1.000	0.122	1.97e+03	8.48e+07	8.48e+07	–
s101, Y_2	Nyström 1600	-1.000	1.000	0.0823	2.38e+03	5.34e+07	5.34e+07	–
s101, Y_2	Nyström 3200	-0.998	0.998	0.0161	7.05e+03	1.86e+07	1.86e+07	–
s101, Y_2	ridge $\times 0.5$	-0.491	0.491	0.00632	1.71e+04	7.71e+04	7.71e+04	–
s101, Y_2	ridge $\times 2.0$	+0.000	0.982	0.00687	3.14e+04	3.64e+04	3.96e+06	3.64e+04
s101, Y_2	ridge $\times 10.0$	+0.000	8.836	0.0249	7.8e+04	1.82e+04	–	1.82e+04
s101, Y_2	ridge $\times 100.0$	+0.000	97.199	0.0545	3.92e+05	9.17e+03	–	9.17e+03
s101, Y_3	PCA features 1000	-0.170	0.170	0.00292	6.17e+03	4.56e+04	4.56e+04	–
s101, Y_3	PCA features 500	-0.475	0.475	0.00766	7.02e+03	7.73e+04	7.73e+04	–
s101, Y_3	PCA features 250	-0.738	0.738	0.0136	6.44e+03	1.38e+05	1.38e+05	–
s101, Y_3	PCA features 125	-0.898	0.898	0.0211	5.61e+03	2.87e+05	2.87e+05	–
s101, Y_3	PCA features 64	-0.964	0.964	0.028	5.45e+03	7.06e+05	7.06e+05	–
s101, Y_3	PCA features 32	-0.990	0.990	0.0406	4.9e+03	2.09e+06	2.09e+06	–
s101, Y_3	PCA features 16	-0.999	0.999	0.0725	3.51e+03	1.63e+07	1.63e+07	–
s101, Y_3	Nyström 100	-1.000	1.000	0.35	876	4.97e+07	4.97e+07	–
s101, Y_3	Nyström 400	-1.000	1.000	0.191	1.55e+03	4.53e+07	4.53e+07	–
s101, Y_3	Nyström 1600	-0.999	0.999	0.0709	3.33e+03	1.45e+07	1.45e+07	–
s101, Y_3	Nyström 3200	-0.997	0.997	0.0331	4.54e+03	6.39e+06	6.39e+06	–
s101, Y_3	ridge $\times 0.5$	-0.492	0.492	0.00865	1.65e+04	7.26e+04	7.26e+04	–
s101, Y_3	ridge $\times 2.0$	+0.000	0.985	0.00949	3e+04	3.38e+04	4.43e+06	3.38e+04
s101, Y_3	ridge $\times 10.0$	+0.000	8.864	0.0357	7.18e+04	1.63e+04	–	1.63e+04
s101, Y_3	ridge $\times 100.0$	+0.000	97.500	0.0821	3.44e+05	7.81e+03	–	7.81e+03
s101, Y_4	PCA features 1000	-0.006	0.006	0.00138	538	4.64e+03	4.64e+03	–
s101, Y_4	PCA features 500	-0.032	0.032	0.00463	712	7.83e+03	7.83e+03	–
s101, Y_4	PCA features 250	-0.114	0.114	0.0104	826	1.35e+04	1.35e+04	–
s101, Y_4	PCA features 125	-0.269	0.269	0.0191	897	2.12e+04	2.12e+04	–
s101, Y_4	PCA features 64	-0.594	0.594	0.0286	963	5.8e+04	5.8e+04	–
s101, Y_4	PCA features 32	-0.741	0.741	0.0436	983	8.06e+04	8.06e+04	–
s101, Y_4	PCA features 16	-0.902	0.902	0.0769	987	1.65e+05	1.65e+05	–
s101, Y_4	Nyström 100	-0.998	0.998	0.603	304	1.2e+06	1.2e+06	–

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Table 20 (continued)

source	perturbation	$\lambda_{\min}(B)$	$\ B\ $	median movement	residual-action ratio	signed-spectrum ratio	$\ B\ < 1$ ratio	ridge ratio
s101, Y_4	Nyström 400	-0.985	0.985	0.35	500	2.11e+05	2.11e+05	–
s101, Y_4	Nyström 1600	-0.860	0.860	0.135	916	4.86e+04	4.86e+04	–
s101, Y_4	Nyström 3200	-0.762	0.762	0.0397	1.71e+03	8.46e+04	8.46e+04	–
s101, Y_4	ridge $\times 0.5$	-0.494	0.494	0.063	3.15e+03	1.61e+04	1.61e+04	–
s101, Y_4	ridge $\times 2.0$	+0.000	0.989	0.0599	6.6e+03	8.63e+03	1.56e+06	8.63e+03
s101, Y_4	ridge $\times 10.0$	+0.000	8.901	0.191	1.86e+04	4.89e+03	–	4.89e+03
s101, Y_4	ridge $\times 100.0$	+0.000	97.908	0.36	1.07e+05	2.86e+03	–	2.86e+03
s104, Y_1	PCA features 1000	-0.184	0.184	0.00397	4.68e+03	3.85e+04	3.85e+04	–
s104, Y_1	PCA features 500	-0.471	0.471	0.0104	5.01e+03	5.82e+04	5.82e+04	–
s104, Y_1	PCA features 250	-0.724	0.724	0.0175	4.8e+03	1.04e+05	1.04e+05	–
s104, Y_1	PCA features 125	-0.901	0.901	0.0267	4.25e+03	2.45e+05	2.45e+05	–
s104, Y_1	PCA features 64	-0.959	0.959	0.0373	3.87e+03	4.76e+05	4.76e+05	–
s104, Y_1	PCA features 32	-0.985	0.985	0.055	3.43e+03	1.07e+06	1.07e+06	–
s104, Y_1	PCA features 16	-0.997	0.997	0.0893	2.61e+03	3.51e+06	3.51e+06	–
s104, Y_1	concatenation	-0.749	76.009	0.0427	2.3e+05	1.99e+05	–	–
s104, Y_1	member 1	-0.973	175.948	0.049	2.34e+05	2.24e+06	–	–
s104, Y_1	member 2	-0.967	126.717	0.0541	3.81e+05	2.02e+06	–	–
s104, Y_1	Nyström 100	-1.000	1.000	0.861	379	9.61e+07	9.61e+07	–
s104, Y_1	Nyström 400	-1.000	1.000	0.247	1.25e+03	6.69e+07	6.69e+07	–
s104, Y_1	Nyström 1600	-1.000	1.000	0.112	2.01e+03	4.7e+07	4.7e+07	–
s104, Y_1	Nyström 3200	-1.000	1.000	0.0249	4.43e+03	5.86e+07	5.86e+07	–
s104, Y_1	ridge $\times 0.5$	-0.493	0.493	0.011	1.06e+04	5.96e+04	5.96e+04	–
s104, Y_1	ridge $\times 2.0$	+0.000	0.986	0.0124	1.89e+04	2.7e+04	3.93e+06	2.7e+04
s104, Y_1	ridge $\times 10.0$	+0.000	8.877	0.0445	4.73e+04	1.36e+04	–	1.36e+04
s104, Y_1	ridge $\times 100.0$	+0.000	97.646	0.102	2.27e+05	6.54e+03	–	6.54e+03
s104, Y_2	PCA features 1000	-0.108	0.108	0.00163	7.72e+03	5.02e+04	5.02e+04	–
s104, Y_2	PCA features 500	-0.370	0.370	0.00553	8.84e+03	7.23e+04	7.23e+04	–
s104, Y_2	PCA features 250	-0.668	0.668	0.0107	8.77e+03	1.29e+05	1.29e+05	–
s104, Y_2	PCA features 125	-0.834	0.834	0.0166	7.71e+03	2.12e+05	2.12e+05	–
s104, Y_2	PCA features 64	-0.955	0.955	0.0234	6.59e+03	6.62e+05	6.62e+05	–
s104, Y_2	PCA features 32	-0.983	0.983	0.0326	6.12e+03	1.46e+06	1.46e+06	–
s104, Y_2	PCA features 16	-0.998	0.998	0.0503	5.04e+03	1.05e+07	1.05e+07	–
s104, Y_2	concatenation	-0.734	96.255	0.0279	6.7e+05	3.68e+05	–	–
s104, Y_2	member 1	-0.974	200.331	0.0328	8.4e+05	5.59e+06	–	–
s104, Y_2	member 2	-0.968	266.674	0.035	9.16e+05	3.4e+06	–	–
s104, Y_2	Nyström 100	-1.000	1.000	0.713	451	1.85e+08	1.85e+08	–

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Table 20 (continued)

source	perturbation	$\lambda_{\min}(B)$	$\ B\ $	median movement	residual-action ratio	signed-spectrum ratio	$\ B\ < 1$ ratio	ridge ratio
s104, Y_2	Nyström 400	-1.000	1.000	0.119	2.5e+03	4.07e+07	4.07e+07	–
s104, Y_2	Nyström 1600	-0.999	0.999	0.0322	6.59e+03	1.8e+07	1.8e+07	–
s104, Y_2	Nyström 3200	-0.992	0.992	0.0104	1.25e+04	8.1e+06	8.1e+06	–
s104, Y_2	ridge $\times 0.5$	-0.498	0.498	0.00788	2.13e+04	8.49e+04	8.49e+04	–
s104, Y_2	ridge $\times 2.0$	+0.000	0.995	0.00862	3.89e+04	3.91e+04	1.66e+07	3.91e+04
s104, Y_2	ridge $\times 10.0$	+0.000	8.958	0.0323	9.36e+04	1.88e+04	–	1.88e+04
s104, Y_2	ridge $\times 100.0$	+0.000	98.534	0.0733	4.53e+05	9.13e+03	–	9.13e+03
s104, Y_3	PCA features 1000	-0.176	0.176	0.00305	6.26e+03	4.48e+04	4.48e+04	–
s104, Y_3	PCA features 500	-0.474	0.474	0.00801	7.06e+03	7.26e+04	7.26e+04	–
s104, Y_3	PCA features 250	-0.738	0.738	0.0145	6.18e+03	1.26e+05	1.26e+05	–
s104, Y_3	PCA features 125	-0.882	0.882	0.0189	6.1e+03	2.67e+05	2.67e+05	–
s104, Y_3	PCA features 64	-0.957	0.957	0.0259	5.75e+03	6.3e+05	6.3e+05	–
s104, Y_3	PCA features 32	-0.990	0.990	0.0371	5.12e+03	2.16e+06	2.16e+06	–
s104, Y_3	PCA features 16	-0.999	0.999	0.0648	3.65e+03	2.51e+07	2.51e+07	–
s104, Y_3	concatenation	-0.749	12.229	0.0228	6.41e+04	2.52e+05	–	–
s104, Y_3	member 1	-0.987	28.650	0.0374	1.01e+05	5.18e+06	–	–
s104, Y_3	member 2	-0.978	32.067	0.0391	3.02e+04	2.59e+06	–	–
s104, Y_3	Nyström 100	-1.000	1.000	0.777	389	1.27e+08	1.27e+08	–
s104, Y_3	Nyström 400	-1.000	1.000	0.203	1.42e+03	5.4e+07	5.4e+07	–
s104, Y_3	Nyström 1600	-1.000	1.000	0.134	1.72e+03	4.38e+07	4.38e+07	–
s104, Y_3	Nyström 3200	-0.999	0.999	0.029	4.4e+03	1.58e+07	1.58e+07	–
s104, Y_3	ridge $\times 0.5$	-0.487	0.487	0.00833	1.62e+04	7.25e+04	7.25e+04	–
s104, Y_3	ridge $\times 2.0$	+0.000	0.975	0.00917	2.93e+04	3.42e+04	2.65e+06	3.42e+04
s104, Y_3	ridge $\times 10.0$	+0.000	8.771	0.0329	7.36e+04	1.73e+04	–	1.73e+04
s104, Y_3	ridge $\times 100.0$	+0.000	96.479	0.0765	3.48e+05	8.23e+03	–	8.23e+03
s104, Y_4	PCA features 1000	-0.003	0.003	0.000765	496	4.96e+03	4.96e+03	–
s104, Y_4	PCA features 500	-0.019	0.019	0.00303	682	8.81e+03	8.81e+03	–
s104, Y_4	PCA features 250	-0.088	0.088	0.00823	778	1.58e+04	1.58e+04	–
s104, Y_4	PCA features 125	-0.259	0.259	0.0153	851	3.11e+04	3.11e+04	–
s104, Y_4	PCA features 64	-0.406	0.406	0.0238	906	4e+04	4e+04	–
s104, Y_4	PCA features 32	-0.790	0.790	0.0427	859	1.27e+05	1.27e+05	–
s104, Y_4	PCA features 16	-0.883	0.883	0.0733	851	1.62e+05	1.62e+05	–
s104, Y_4	concatenation	-0.658	58.594	0.0605	1.49e+04	1.97e+05	–	–
s104, Y_4	member 1	-0.897	23.688	0.103	5.19e+03	3.18e+05	–	–
s104, Y_4	member 2	-0.919	155.529	0.108	1.74e+04	8.07e+05	–	–
s104, Y_4	Nyström 100	-0.993	0.993	0.517	378	3.93e+05	3.93e+05	–

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Table 20 (continued)

source	perturbation	$\lambda_{\min}(B)$	$\ B\ $	median movement	residual-action ratio	signed-spectrum ratio	$\ B\ < 1$ ratio	ridge ratio
s104, Y_4	Nyström 400	-0.961	0.961	0.241	772	1.39e+05	1.39e+05	–
s104, Y_4	Nyström 1600	-0.850	0.850	0.0775	1.66e+03	9.85e+04	9.85e+04	–
s104, Y_4	Nyström 3200	-0.707	0.707	0.0213	2.57e+03	1.5e+05	1.5e+05	–
s104, Y_4	ridge $\times 0.5$	-0.491	0.491	0.0482	4.89e+03	2.63e+04	2.63e+04	–
s104, Y_4	ridge $\times 2.0$	+0.000	0.983	0.0525	8.97e+03	1.24e+04	1.43e+06	1.24e+04
s104, Y_4	ridge $\times 10.0$	+0.000	8.846	0.179	2.36e+04	6.6e+03	–	6.6e+03
s104, Y_4	ridge $\times 100.0$	+0.000	97.301	0.393	1.18e+05	3.32e+03	–	3.32e+03

Table 21: Ridge-path diagnostics, seed 101, 4,000 fitting and 1,000 query rows. The bias and noise factor refer to Proposition 4(iii); the factor multiplies a hypothetical noise variance and is not an estimate of simulator noise. Corrected and kernel-only test columns are mean relative errors of a physical spectral component, in percent. The final column uses retained PCA coordinates on the held-out half and compares same-half fitting with split-half fitting; it is a different metric and sample. No ridge is selected on these test values.

component	kernel	γ	$\text{tr}S_\lambda$	bias	noise factor	median $\ a_\lambda\ ^2$	median $\tilde{P}_\lambda$	corrected [%]	kernel only [%]	same / split half PCA error [%]
Y1	input	1e-08	3841	0.000226	0.924	0.87	0.0324	0.2168	0.1754	0.079 / 2.465
Y1	input	1e-06	1624	0.0712	0.26	0.226	0.0395	0.2376	0.2420	1.145 / 2.214
Y1	input	0.0001	212	0.282	0.0269	0.0237	0.0997	0.4132	0.9768	2.221 / 2.395
Y1	input	0.01	19	0.413	0.00194	0.00197	0.315	0.5095	5.7578	2.716 / 2.728
Y1	feature	1e-08	2541	0.0012	0.51	0.481	0.00637	0.2027	0.2111	0.280 / 1.171
Y1	feature	1e-06	689	0.0225	0.111	0.0668	0.0151	0.3130	0.3275	1.085 / 1.464
Y1	feature	0.0001	103	0.156	0.0137	0.00825	0.0529	0.4546	0.6396	2.105 / 2.187
Y1	feature	0.01	9	0.418	0.001	0.000845	0.185	0.5326	5.9929	2.783 / 2.786
Y2	input	1e-08	3884	0.000107	0.943	0.523	0.0369	0.4559	0.6885	0.043 / 1.784
Y2	input	1e-06	1554	0.061	0.219	0.158	0.0416	0.4849	0.9183	0.886 / 1.632
Y2	input	0.0001	132	0.211	0.0143	0.0127	0.0845	0.6325	4.4719	1.516 / 1.580
Y2	input	0.01	10	0.273	0.00115	0.00117	0.227	0.8581	26.9965	1.773 / 1.777
Y2	feature	1e-08	2339	0.000906	0.451	0.44	0.00562	0.1431	0.1439	0.184 / 0.738
Y2	feature	1e-06	564	0.0159	0.0893	0.0618	0.0143	0.2768	0.2696	0.662 / 0.899
Y2	feature	0.0001	86	0.134	0.0118	0.00829	0.0512	0.5764	1.0187	1.275 / 1.326
Y2	feature	0.01	9	0.27	0.000998	0.000909	0.188	0.8905	15.5377	1.738 / 1.741
Y3	input	1e-08	3308	0.00797	0.7	0.412	0.0132	0.2382	0.3731	0.324 / 2.147
Y3	input	1e-06	558	0.283	0.0644	0.0523	0.0192	0.2627	0.9135	1.656 / 1.951
Y3	input	0.0001	45	0.455	0.00546	0.00501	0.0457	0.3432	5.4172	2.173 / 2.155
Y3	input	0.01	5	0.554	0.000642	0.000643	0.152	0.4914	19.3748	2.715 / 2.702
Y3	feature	1e-08	2371	0.00158	0.461	0.458	0.00569	0.0999	0.1197	0.221 / 0.903
Y3	feature	1e-06	590	0.0255	0.094	0.063	0.0145	0.1650	0.1866	0.829 / 1.136
Y3	feature	0.0001	90	0.267	0.0123	0.00842	0.0513	0.3145	0.6620	1.790 / 1.882
Y3	feature	0.01	9	0.525	0.001	0.000885	0.189	0.4913	7.4936	2.612 / 2.624
Y4	input	1e-08	3998	9.17e-07	0.999	0.473	0.269	2.1826	2.3706	0.007 / 12.844
Y4	input	1e-06	3792	0.00736	0.9	0.437	0.27	2.1557	2.3495	0.629 / 12.779
Y4	input	0.0001	1074	1.16	0.125	0.0912	0.319	1.9633	2.4907	7.986 / 11.573
Y4	input	0.01	51	2.3	0.00312	0.00306	0.619	2.3401	8.9051	12.340 / 12.554
Y4	feature	1e-08	3823	0.00807	0.917	0.975	0.031	2.4285	2.4310	0.626 / 14.459
Y4	feature	1e-06	1652	0.794	0.272	0.24	0.0392	1.8004	1.8078	6.037 / 11.377
Y4	feature	0.0001	229	1.83	0.0293	0.0257	0.103	1.8454	1.9424	9.636 / 10.497
Y4	feature	0.01	19	2.39	0.0019	0.00181	0.323	2.4706	4.7750	12.907 / 12.992

β	average-family measured slope	$\beta/(\beta + 2)$	uniform-family measured slope	analytic slope same ladder
0.5	0.200	0.200	0.494	0.494
1	0.333	0.333	0.973	0.973
2	0.498	0.500	1.721	1.724
3	0.597	0.600	1.963	1.963
4	0.664	0.667	1.995	1.995

β	uniform-family maximum relative gap	profile-family maximum relative gap	violations
0.5	5.7e-04	4.4e-04	0
1	1.7e-03	3.9e-04	0
2	1.0e-02	3.1e-04	0
3	1.4e-03	5.2e-04	0
4	2.8e-04	2.8e-04	0

Table 22: Manufactured transfer families, seed zero, two million samples for each β . The first panel compares finite-ladder slopes with the analytic curve on the same ladder. The second reports the largest relative discrepancies between the Monte Carlo mean squared error and the analytic value (uniform family) or the quadrature value (profile family), and the total number of pointwise violations.

Table 23: Matched correction-coefficient benchmark [4]. The standard split uses 35,000 / 7,500 / 7,500 states. OOD test states exceed the 0.85 aerosol or water-vapor quantile; remaining states are split 85/15. Cells average three seeds or initializations; the reference rows are the published values. RMSE, MAE and SMAPE are pooled; R^2 is the uniform average over the three coefficients. Model keys: dkr, feature kernel; krr, input kernel; mlp, network; ens, ensemble; resid, residual correction; lowfi, the low-fidelity mean. The categorical inputs follow the benchmark protocol.

split	inputs	head	RMSE	MAE	R^2	SMAPE [%]
OOD	state only	dkr	0.01408±0.00177	0.00709±0.00083	0.99226±0.00106	5.714±0.441
OOD	state only	krr	0.11283±0.00069	0.05430±0.00197	0.66968±0.01490	30.762±1.160
OOD	state only	mlp	0.00750±0.00044	0.00355±0.00036	0.99586±0.00012	4.500±0.249
OOD	state only	mlp_ens	0.00624±0.00015	0.00258±0.00017	0.99667±0.00004	3.470±0.370
OOD	state only	mlp_resid	0.00620±0.00015	0.00252±0.00015	0.99670±0.00005	3.372±0.308
OOD	state only	ridge	0.11944±0.00000	0.06195±0.00000	0.49009±0.00000	37.228±0.000
OOD	with 6S inputs	dkr	0.00735±0.00036	0.00336±0.00033	0.99597±0.00022	4.101±0.474
OOD	with 6S inputs	krr	0.02904±0.00015	0.01432±0.00005	0.97160±0.00020	11.238±0.033
OOD	with 6S inputs	lowfi_resid	0.02901±0.00016	0.01430±0.00006	0.97172±0.00021	11.416±0.047
OOD	with 6S inputs	mlp	0.00741±0.00080	0.00347±0.00063	0.99602±0.00048	4.406±0.366
OOD	with 6S inputs	mlp_ens	0.00611±0.00030	0.00246±0.00023	0.99682±0.00013	3.009±0.093
OOD	with 6S inputs	mlp_resid	0.00596±0.00031	0.00232±0.00022	0.99687±0.00012	2.921±0.067
OOD	with 6S inputs	ridge	0.08048±0.00000	0.04396±0.00000	0.79274±0.00000	29.064±0.000
standard	state only	dkr	0.00874±0.00121	0.00387±0.00071	0.99379±0.00053	5.342±0.323
standard	state only	krr	0.11288±0.00001	0.05157±0.00002	0.68464±0.00028	31.178±0.032
standard	state only	mlp	0.00657±0.00026	0.00242±0.00030	0.99465±0.00007	5.136±0.550
standard	state only	mlp_ens	0.00591±0.00007	0.00170±0.00005	0.99513±0.00010	3.873±0.132
standard	state only	mlp_resid	0.00581±0.00004	0.00151±0.00004	0.99520±0.00010	3.543±0.066
standard	state only	ridge	0.11959±0.00000	0.06067±0.00000	0.48572±0.00000	41.647±0.000
standard	with 6S inputs	dkr	0.00627±0.00025	0.00218±0.00021	0.99502±0.00028	4.429±0.169
standard	with 6S inputs	krr	0.02514±0.00010	0.01155±0.00003	0.97447±0.00010	10.184±0.037
standard	with 6S inputs	lowfi_resid	0.02513±0.00010	0.01155±0.00003	0.97452±0.00010	10.178±0.041
standard	with 6S inputs	mlp	0.00632±0.00018	0.00224±0.00017	0.99504±0.00014	4.496±0.160
standard	with 6S inputs	mlp_ens	0.00587±0.00014	0.00163±0.00012	0.99531±0.00011	3.407±0.130
standard	with 6S inputs	mlp_resid	0.00578±0.00012	0.00148±0.00008	0.99535±0.00010	3.185±0.097
standard	with 6S inputs	ridge	0.06981±0.00000	0.03631±0.00000	0.83813±0.00000	28.300±0.000
standard	published	pKANrtm	0.00619	0.00186	0.99472	3.912
OOD	published	pKANrtm	0.01022	0.00544	0.99418	5.399

Table 24: OCO-2 selection comparison, seeds 0–2 on each of three bands. Each record uses 2,000 validation and 2,000 test rows. Both reported columns are *mean relative norms* in percent; the stack fitting objectives are row-relative *squared* errors. The radiance quadratic uses the exact affine reconstruction Gram matrix. Model keys: dkr, feature kernel; mean, trained network; ridge, linear readout; raw/ard, input kernels. Flat, radx and wnum are the three training losses. Selection is per coordinate, with the radiance variant using greedy coordinate updates, not exhaustive global search.

band	head / rule	reduced error [%]	radiance error [%]
o2	select_radiance	4.08±0.05	0.0283±0.0017
o2	select_reduced	4.09±0.06	0.0283±0.0017
o2	single dkr_flat	4.07±0.06	0.0757±0.0012
o2	single dkr_radx	20.14±0.07	0.0351±0.0006
o2	single dkr_wnum	21.70±0.60	0.0309±0.0017
o2	single kernel_ard	30.02±0.03	0.2706±0.1030
o2	single kernel_raw	40.53±0.09	0.2353±0.0047
o2	single mean_flat	4.40±0.08	0.1756±0.0111
o2	single mean_radx	58.72±0.60	0.0510±0.0049
o2	single mean_wnum	49.22±0.34	0.0437±0.0002
o2	single ridge_flat	4.67±0.09	0.2205±0.0124
o2	single ridge_radx	44.83±0.43	0.0538±0.0044
o2	single ridge_wnum	45.55±0.73	0.0527±0.0016
o2	stack_radiance	18.59±0.25	0.0264±0.0007
o2	stack_reduced	4.10±0.07	0.0291±0.0029
sco2	select_radiance	8.09±0.02	0.0598±0.0013
sco2	select_reduced	8.09±0.02	0.0598±0.0013
sco2	single dkr_flat	8.09±0.02	0.1083±0.0114
sco2	single dkr_radx	15.27±0.25	0.0780±0.0094
sco2	single dkr_wnum	15.48±0.31	0.0638±0.0022
sco2	single kernel_ard	28.72±0.02	0.6023±0.0097
sco2	single kernel_raw	41.01±0.09	0.5229±0.0081
sco2	single mean_flat	8.23±0.01	0.2107±0.0194
sco2	single mean_radx	46.08±0.91	0.0936±0.0082
sco2	single mean_wnum	36.64±0.38	0.0750±0.0020
sco2	single ridge_flat	8.34±0.02	0.2568±0.0244
sco2	single ridge_radx	41.14±1.78	0.0967±0.0081
sco2	single ridge_wnum	40.25±0.24	0.0887±0.0034
sco2	stack_radiance	8.87±0.15	0.0511±0.0016
sco2	stack_reduced	8.09±0.01	0.0536±0.0020
wco2	select_radiance	16.26±0.07	0.0510±0.0086
wco2	select_reduced	16.26±0.07	0.0510±0.0086
wco2	single dkr_flat	16.26±0.07	0.1135±0.0069
wco2	single dkr_radx	48.28±0.28	0.0475±0.0055
wco2	single dkr_wnum	48.60±0.38	0.0516±0.0086
wco2	single kernel_ard	52.45±0.07	0.2231±0.0014
wco2	single kernel_raw	58.63±0.20	0.4431±0.0063
wco2	single mean_flat	16.42±0.06	0.2254±0.0161
wco2	single mean_radx	67.92±0.46	0.0605±0.0075
wco2	single mean_wnum	62.39±0.42	0.0700±0.0087
wco2	single ridge_flat	16.49±0.07	0.2571±0.0234
wco2	single ridge_radx	61.03±0.31	0.0663±0.0080
wco2	single ridge_wnum	61.33±0.05	0.0778±0.0095
wco2	stack_radiance	37.43±0.20	0.0391±0.0026
wco2	stack_reduced	16.28±0.07	0.0404±0.0034

C Comparison with OCO-2

The companion paper [12] applies the same construction to the OCO-2 radiative-transfer emulation problem of Lamminpää et al. [3]: per spectral band (O_2 , weak CO_2 , strong CO_2), a reduced atmospheric state of twenty or twenty-four coordinates maps to forty reduced radiance coefficients, with errors on the standardized coefficients and on the reconstructed radiance, under the ten-split protocol of Section 3. It gives a comparison at a higher input dimension.

There a kernel on the state trails the network by a factor of nine, four and five on the reduced coefficients: the isotropic Matérn reaches 40.6%, 58.6% and 43.4% on the three bands against 4.4%, 16.4% and 8.2% for the network, and residual correction adds little. The coordinate search that gives a kernel on the EMIT state its own length scales is not the strongest construction there either. Four learned metrics were run on the same ten splits on all three bands: a kernel-flow metric [58], a Mahalanobis metric under the same flow, the gradient outer product of the recursive feature machine [97], and per-input scales fitted by empirical Bayes. The last is the best of the four on every band, taking the isotropic kernel’s 40.6%, 58.6% and 44.3% on the coefficients to 18.7%, 25.3% and 16.9%, and on O_2 its radiance from 0.233% to 0.150%; the two flow metrics agree within half a point of each other at 26.9%, 29.8% and 23.9%, the recursive feature machine reads 28.2%, 47.9% and 31.3%, and an additive kernel 38.2%, 46.9% and 47.7%. (The isotropic baseline on strong CO_2 reads 44.3% in the metric experiment and 43.4% in the baseline experiment; this band has a spread of ± 3.4 because two of its ten splits land three points below the other eight.) The feature kernel remains more accurate, by a margin that depends on the band: the exact Matérn regression on the network’s frozen features reads 4.1%, 16.3% and 8.1% on the same splits, so the best state metric is 4.6 times that on O_2 but only 1.6 and 2.1 times it on the two CO_2 bands. The empirical-Bayes metric is bimodal across splits on O_2 , eight of ten landing near 17.2% and two at 24.7%, so its spread of ± 3.2 is thirty times the isotropic kernel’s ± 0.1 . The exact Matérn regression on the network’s frozen features beats a ridge readout refitted on the same features on every split, every band and every training loss, and a per-coefficient choice among heads trained under different losses has lower mean error than the release’s own emulator under both reported metrics: it wins the reduced-coefficient comparison on all ten splits on each band, and the radiance comparison on ten, nine and ten splits on O_2 , weak CO_2 and strong CO_2 .

The release accompanying [3] also contains 976 retrieval records, each with an atmospheric state, the full-physics monochromatic radiance at that state, its rank-40 reconstruction and the prediction of the release’s Gaussian-process emulator. Every one of the 976 reduced states matches a test input of the release within 6.67×10^{-16} , mapping to 974 distinct test rows with the same nearest index in all three bands, so these records belong to the test set; they supply a different target, the full-physics radiance, at those states rather than an independent evaluation pool.

D Ten further emulation configurations

Table 25 extends the comparison to ten further configurations, each with validation-based selection; the comparison concerns the families under a common protocol, not the best method available for each problem.

The corpora are the advection equation with discontinuous initial data from the operator suite of [103], which is also the source of the structural-mechanics problem of [12]; Burgers’

equation at three viscosities from PDEBench [104]; three one-step maps from The Well [105]: a two-dimensional turbulent radiative layer, an active-matter system and a Helmholtz staircase; the ClimSim atmospheric physics emulation task [106]; and the paired correction-coefficient corpus of the physics-guided radiative-transfer work of [4], a third radiative-transfer emulation problem and the largest here at 487,790 rows. That corpus is run in both of its configurations: from the atmospheric state alone, and with the low-fidelity 6S coefficients as additional inputs, which is a different and easier problem.

Advection is the 200-point input/output pair of the operator suite at full resolution. Burgers is PDEBench’s `1D_Burgers_Sols_Nu` release, the map from the initial condition to the solution at $t = 1$ on a grid subsampled by four to 256 cells, with the input used raw and the output reduced to 128 principal coefficients. The three Well datasets are one-step maps of the full multi-channel state with 256 principal coefficients in and out: the turbulent radiative layer on its 128×384 grid (density, pressure, velocity), active matter on 256×256 (concentration, velocity and the two tensor fields), and the Helmholtz staircase on its two pressure channels. ClimSim is the low-resolution release, 124 inputs to 128 tendencies, with 100,000 training rows drawn from the 10.1 million available and validation and test taken from the release’s own validation and scoring files. The correction-coefficient corpus is the paired 6S/libRadtran Sentinel-2 set [14, 15]: wavelength, solar zenith, view zenith, relative azimuth, aerosol optical depth, water vapor, ozone and elevation map to three libRadtran coefficients, split by atmospheric state so that all bands of a state stay together; the multi-fidelity configuration adds the three 6S coefficients, for input dimensions eight and eleven. For these two configurations the error is

$$E_{0.05} = \frac{1}{N} \sum_{i=1}^N \frac{\|Y_i - \hat{Y}_i\|_2}{\max\{\|Y_i\|_2, 0.05\}}, \quad (24)$$

and the same floor enters validation selection and the kernel-flow objective, which limits the weight of coefficient vectors of small norm.

The families are those of Section 4 with two changes forced by scale. The reference is a ridge regression linear in the inputs, since a cubic expansion of a 256-dimensional input is not tractable. Kernel solves are exact when the training block has at most 20,000 rows and use a Nyström approximation [60] on 6,000 uniformly drawn landmarks above that, which applies to ClimSim, the Helmholtz staircase at 20,384 rows and the two correction-coefficient configurations. The convex stack and the per-coordinate selection are formed from the kernels, the networks, the residual correction and the feature kernel; the ridge is a reference row, not one of their members. Advection, Burgers, ClimSim and the two correction-coefficient configurations have three to five random splits. The Well’s releases carry their own training, validation and test files, which are used unchanged, so the three turbulent-radiative-layer runs share one split and differ only in network initialization (their spread, of order 0.01 points, measures initialization), and the active-matter and Helmholtz-staircase rows are single runs.

The relative errors vary widely with the problem. Where the map is smooth and the sample small, Burgers at $\nu = 0.1$, the exact kernel is at 2.81% against the network’s 20.34%, a factor of seven, of the same kind as the five- to six-fold gap between the per-input kernel and the network on the EMIT table’s smooth components; as the viscosity falls and the solutions roughen the gap closes to 23.40% against 36.65% and then to 31.66% against 39.98%. Where the sample is large and the map rough, on ClimSim and on the correction-coefficient corpus at nearly half a million rows, the network leads instead, 25.10% against 27.20% and 3.75% against 62.93%, which is the OCO-2 regime. On advection, where the operator is linear, the linear ridge is the lowest row and

every other family lands within 0.4 points of it.

Learned metrics help most on the correction-coefficient corpus. From the state alone the isotropic kernel is at 62.93 ± 0.51 and a kernel-flow metric [58] takes it to 3.97 ± 0.42 , a factor of sixteen, which brings a kernel on the state within six percent of the network on a corpus where the plain kernel was seventeen times worse. With the low-fidelity coefficients as inputs the same pair reads 15.76 and 4.60, a factor of three: the easier the problem is made for the plain kernel, the less the learned metric adds.

The convex stack is the lowest row on eight of the ten configurations and the second lowest on the other two; the residual-corrected network is among the lowest two on five. Both exceptions are the linear problems, and on both the row that beats the stack is the ridge, which the stack does not contain. On advection the ridge is lower by 0.07 points while the stack, at 11.36%, matches its best member, the isotropic kernel, on which it puts weight 0.87, to two decimals. On the Helmholtz staircase the ridge reaches 0.004%, sixteen times below the next row and about a hundred times below the weakest member of the pool, and the stack reaches 0.066%, slightly below its best member, the learned-metric kernel at 0.067%, on which it puts weight 0.94. In both cases the stack does what it is asked within its pool and cannot recover a family the pool excludes. A combination is a good default when no member of its pool dominates it, and it can be read against the whole comparison only when the pool contains the families being compared.

These rows use one training budget per corpus and one architecture family, and several of the corpora have published results from methods tuned for them. With three to five runs, and one on two rows, the spreads support the large orderings above and not fine distinctions between adjacent families. Every kernel column also inherits the tuning protocol of Section 7.1, with the length scale chosen on a 6,000-row subsample and the winner refit on every row. On the EMIT table that costs the isotropic kernel a factor 0.882; on ClimSim, where the subsample is a sixteenth of the solve, re-choosing the scale against the full solve halves it and lowers the kernel’s error by 3.6%; on a 20,000-row corpus it changes nothing. Retuning can move a comparison in either direction, and the columns of Table 25 describe the stated training and selection procedures.

A. Configurations

corpus	n	d	q	runs
Advection	19,000	200	200	5
Burgers $\nu = 0.1$	7,200	256	128	3
Burgers $\nu = 0.01$	7,200	256	128	3
Burgers $\nu = 0.001$	7,200	256	128	3
Radiative layer	7,200	256	256	3
Active matter	14,000	256	256	1
Helmholtz staircase	20,384	256	256	1
ClimSim	100,000	124	128	5
RT coefficients, state	487,773–487,823	8	3	5
RT coefficients, +6S	487,773–487,823	11	3	5

B. Reference, input kernels and network

corpus	linear ridge	KRR	learned-metric	network
			KRR	
Advection	11.29±0.04	11.36±0.04	11.36±0.04	11.69±0.07
Burgers $\nu = 0.1$	13.89±0.53	2.81±0.28	2.90±0.33	20.34±0.61
Burgers $\nu = 0.01$	33.68±0.28	23.40±0.54	23.58±0.50	36.65±1.95
Burgers $\nu = 0.001$	38.38±0.12	31.66±0.16	33.23±1.73	39.98±0.65
Radiative layer	31.73	30.31	30.26±0.01	32.17±0.02
Active matter	37.24	35.81	35.18	37.31
Helmholtz staircase	0.00	0.26	0.07	0.40
ClimSim	32.24±0.06	27.20±0.05	26.31±0.04	25.10±0.18
RT coefficients, state	71.54±0.59	62.93±0.51	3.97±0.42	3.75±0.25
RT coefficients, +6S	33.83±0.26	15.76±1.98	4.60±0.31	2.79±0.14

C. Ensembles, corrections and combinations

corpus	network +			
	ensemble	residual KRR	feature kernel	stack
Advection	11.50±0.05	11.50±0.05	11.64±0.07	11.36±0.04
Burgers $\nu = 0.1$	19.28±0.09	2.78±0.29	4.92±0.30	2.78±0.29
Burgers $\nu = 0.01$	35.31±0.70	23.12±0.56	29.20±0.59	23.01±0.54
Burgers $\nu = 0.001$	38.89±0.11	31.82±0.11	35.59±0.48	31.34±0.14
Radiative layer	31.58±0.02	30.40±0.01	31.08±0.01	30.12
Active matter	36.93	35.77	36.22	35.05
Helmholtz staircase	0.24	0.23	0.45	0.07
ClimSim	24.43±0.17	24.33±0.13	24.71±0.25	24.25±0.17
RT coefficients, state	3.24±0.11	3.13±0.05	3.52±0.07	3.09±0.05
RT coefficients, +6S	2.40±0.11	2.34±0.09	2.80±0.09	2.32±0.08

Table 25: Ten further configurations: dimensions and run counts (A), reference models (B), and combinations (C). Errors are mean relative spectral norms in percent, with the 0.05 denominator floor of (24) for the two RT-coefficient rows. Spreads are sample standard deviations; the radiative-layer repetitions share one split and vary initialization, and two rows are single runs. The network budget is 200 epochs at width 512, except ClimSim (60 at 512) and RT coefficients (120 at 384). The linear ridge is excluded from the stack; its Helmholtz error is 0.004%, rounded in the table. Kernel solves above 20,000 training rows use 6,000 Nyström landmarks.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital; both entities supported this work and are named above. These entities and the public funders had no role in the study design, analysis, manuscript preparation, or decision to publish.

AI-assisted contributions

Claude Code (Anthropic) and Codex (OpenAI) implemented the data pipeline and the model comparisons, ran the experiments, produced the figures and tables from the run records, and drafted the manuscript. ChatGPT (OpenAI) developed the residual-error and inversion-stability arguments, checked the identities and wrote the conditioning diagnostics and their tests. The author supplied the data, set the questions, and reviewed the results, the proofs and the text.

Data and code availability

The code, the per-split run records, the scripts that generate every table and figure from those records, and this manuscript are at <https://github.com/yspennstate/emit-oco2-kernel-emulators>. Each record carries the SHA-256 digests of the input arrays, the split sizes, the selected hyper-parameters and the test metrics; the records of the training-target experiment also carry the digests of their split indices, and the float64 test predictions of every family in those runs are available on request. The EMIT table is the file `data_EMIT_24k.jld2` (213,761,068 bytes, SHA-256 344a6d0e6d77b9c152de18505befb10c6e7566bcb5ff9bc00e10d260a048c6de), supplied by the Jet Propulsion Laboratory. Its root holds the state matrix $\mathbf{X}$ ($6 \times 23,313$), the four components $\mathbf{Y1}$ – $\mathbf{Y4}$ ($285 \times 23,313$ each), the wavelength grid, the index vectors identifying the lookup table’s input and component columns, and per-component affine vectors. The drivers read NumPy exports of $\mathbf{X}$ and $\mathbf{Y1}$ – $\mathbf{Y4}$, whose digests are recorded in every run. The table is available from the author on request, subject to the permission of the Jet Propulsion Laboratory. The OCO-2 arrays and reference predictions are from the release accompanying [3], and the other configurations use the cited public releases.

References

- [1] P. G. Brodrick, D. R. Thompson, J. E. Fahlen, M. L. Eastwood, C. M. Sarture, S. R. Lundeen, W. Olson-Duvall, N. Carmon, and R. O. Green. Generalized radiative transfer

- emulation for imaging spectroscopy reflectance retrievals. *Remote Sensing of Environment*, 261:112476, 2021.
- [2] D. R. Thompson, V. Natraj, R. O. Green, M. C. Helmlinger, B.-C. Gao, and M. L. Eastwood. Optimal estimation for imaging spectrometer atmospheric correction. *Remote Sensing of Environment*, 216:355–373, 2018.
 - [3] O. Lamminpää, J. Susiluoto, J. Hobbs, J. McDuffie, A. Braverman, and H. Owhadi. Forward model emulator for atmospheric radiative transfer using Gaussian processes and cross validation. *Atmospheric Measurement Techniques*, 18:673–694, 2025.
 - [4] M. A. A. Mazid and N. Rishe. Multi-fidelity emulation of atmospheric correction coefficients with physics-guided Kolmogorov–Arnold networks. *Remote Sensing*, 18(11):1826, 2026.
 - [5] P. Batlle, M. Darcy, B. Hosseini, and H. Owhadi. Kernel methods are competitive for operator learning. *Journal of Computational Physics*, 496:112549, 2024.
 - [6] L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3:218–229, 2021.
 - [7] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.
 - [8] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.
 - [9] C. Mora, A. Yousefpour, S. Hosseinmardi, H. Owhadi, and R. Bostanabad. Operator learning with Gaussian processes. *Computer Methods in Applied Mechanics and Engineering*, 434:117581, 2025.
 - [10] A. G. Wilson, Z. Hu, R. Salakhutdinov, and E. P. Xing. Deep kernel learning. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics*, pages 370–378, 2016.
 - [11] G. R. Yoo and H. Owhadi. Deep regularization and direct training of the inner layers of neural networks with kernel flows. *Physica D: Nonlinear Phenomena*, 426:132952, 2021.
 - [12] Y. Shmalo. Neural means and kernel corrections for operator learning. Preprint, ResearchGate, 2026. Code and per-run summaries at <https://github.com/yspennstate/neural-means-kernel-corrections>.
 - [13] R. O. Green et al. The Earth surface mineral dust source investigation: an Earth science imaging spectroscopy mission. In *IEEE Aerospace Conference*, 2020.
 - [14] E. F. Vermote, D. Tanré, J. L. Deuzé, M. Herman, and J.-J. Morcrette. Second simulation of the satellite signal in the solar spectrum, 6S: an overview. *IEEE Transactions on Geoscience and Remote Sensing*, 35(3):675–686, 1997.

- [15] B. Mayer and A. Kylling. Technical note: the libRadtran software package for radiative transfer calculations — description and examples of use. *Atmospheric Chemistry and Physics*, 5:1855–1877, 2005.
- [16] C. Emde, R. Buras-Schnell, A. Kylling, B. Mayer, J. Gasteiger, U. Hamann, J. Kylling, B. Richter, C. Pause, T. Dowling, and L. Bugliaro. The libRadtran software package for radiative transfer calculations (version 2.0.1). *Geoscientific Model Development*, 9:1647–1672, 2016.
- [17] A. Berk, P. Conforti, R. Kennett, T. Perkins, F. Hawes, and J. van den Bosch. MODTRAN6: a major upgrade of the MODTRAN radiative transfer code. In *Proceedings of SPIE 9088, Algorithms and Technologies for Multispectral, Hyperspectral, and Ultraspectral Imagery XX*, 2014.
- [18] C. D. Rodgers. *Inverse Methods for Atmospheric Sounding: Theory and Practice*. World Scientific, 2000.
- [19] A. Eldering, C. W. O’Dell, P. O. Wennberg, D. Crisp, M. R. Gunson, et al. The Orbiting Carbon Observatory-2: first 18 months of science data products. *Atmospheric Measurement Techniques*, 10:549–563, 2017.
- [20] D. Crisp, H. R. Pollock, R. Rosenberg, L. Chapsky, R. A. M. Lee, et al. The on-orbit performance of the Orbiting Carbon Observatory-2 (OCO-2) instrument and its radiometrically calibrated products. *Atmospheric Measurement Techniques*, 10:59–81, 2017.
- [21] B. J. Connor, H. Boesch, G. Toon, B. Sen, C. Miller, and D. Crisp. Orbiting Carbon Observatory: inverse method and prospective error analysis. *Journal of Geophysical Research: Atmospheres*, 113:D05305, 2008.
- [22] C. W. O’Dell, B. Connor, H. Bosch, D. O’Brien, C. Frankenberg, et al. The ACOS CO₂ retrieval algorithm — part 1: description and validation against synthetic observations. *Atmospheric Measurement Techniques*, 5:99–121, 2012.
- [23] F. Chevallier, F. Chérut, N. A. Scott, and A. Chédin. A neural network approach for a fast and accurate computation of a longwave radiative budget. *Journal of Applied Meteorology*, 37:1385–1397, 1998.
- [24] V. M. Krasnopolsky, M. S. Fox-Rabinovitz, and D. V. Chalikov. New approach to calculation of atmospheric model physics: accurate and fast neural network emulation of longwave radiation in a climate model. *Monthly Weather Review*, 133:1370–1383, 2005.
- [25] M. A. Veerman, R. Pincus, R. Stoffer, C. M. van Leeuwen, D. Podareanu, and C. C. van Heerwaarden. Predicting atmospheric optical properties for radiative transfer computations using neural networks. *Philosophical Transactions of the Royal Society A*, 379:20200095, 2021.
- [26] P. Ukkonen. Exploring pathways to more accurate machine learning emulation of atmospheric radiative transfer. *Journal of Advances in Modeling Earth Systems*, 14:e2021MS002875, 2022.
- [27] S. Rasp, M. S. Pritchard, and P. Gentile. Deep learning to represent subgrid processes in climate models. *Proceedings of the National Academy of Sciences*, 115(39):9684–9689, 2018.

- [28] N. D. Brenowitz and C. S. Bretherton. Prognostic validation of a neural network unified physics parameterization. *Geophysical Research Letters*, 45:6289–6298, 2018.
- [29] J. P. Rivera, J. Verrelst, J. Gómez-Dans, J. Muñoz-Marí, J. Moreno, and G. Camps-Valls. An emulator toolbox to approximate radiative transfer models with statistical learning. *Remote Sensing*, 7(7):9347–9370, 2015.
- [30] J. Verrelst, N. Sabater, J. P. Rivera, J. Muñoz-Marí, J. Vicent, G. Camps-Valls, and J. Moreno. Emulation of leaf, canopy and atmosphere radiative transfer models for fast global sensitivity analysis. *Remote Sensing*, 8(8):673, 2016.
- [31] J. L. Gómez-Dans, P. E. Lewis, and M. Disney. Efficient emulation of radiative transfer codes using Gaussian processes and application to land surface parameter inferences. *Remote Sensing*, 8(2):119, 2016.
- [32] G. Camps-Valls, J. Verrelst, J. Muñoz-Marí, V. Laparra, F. Mateo-Jiménez, and J. Gómez-Dans. A survey on Gaussian processes for earth-observation data analysis. *IEEE Geoscience and Remote Sensing Magazine*, 4(2):58–78, 2016.
- [33] J. Sacks, W. J. Welch, T. J. Mitchell, and H. P. Wynn. Design and analysis of computer experiments. *Statistical Science*, 4(4):409–423, 1989.
- [34] M. C. Kennedy and A. O’Hagan. Predicting the output from a complex computer code when fast approximations are available. *Biometrika*, 87(1):1–13, 2000.
- [35] M. C. Kennedy and A. O’Hagan. Bayesian calibration of computer models. *Journal of the Royal Statistical Society: Series B*, 63(3):425–464, 2001.
- [36] P. Perdikaris, M. Raissi, A. Damianou, N. D. Lawrence, and G. E. Karniadakis. Nonlinear information fusion algorithms for data-efficient multi-fidelity modelling. *Proceedings of the Royal Society A*, 473:20160751, 2017.
- [37] B. Peherstorfer, K. Willcox, and M. Gunzburger. Survey of multifidelity methods in uncertainty propagation, inference, and optimization. *SIAM Review*, 60(3):550–591, 2018.
- [38] K. Cawse-Nicholson, P. A. Townsend, D. Schimel, A. M. Assiri, P. L. Blake, et al. NASA surface biology and geology designated observable: a perspective on surface imaging algorithms. *Remote Sensing of Environment*, 257:112349, 2021.
- [39] N. Aronszajn. Theory of reproducing kernels. *Transactions of the American Mathematical Society*, 68:337–404, 1950.
- [40] G. Kimeldorf and G. Wahba. Some results on Tchebycheffian spline functions. *Journal of Mathematical Analysis and Applications*, 33(1):82–95, 1971.
- [41] G. Wahba. *Spline Models for Observational Data*. SIAM, 1990.
- [42] B. Schölkopf, R. Herbrich, and A. J. Smola. A generalized representer theorem. In *Proceedings of the 14th Annual Conference on Computational Learning Theory*, pages 416–426, 2001.
- [43] A. E. Hoerl and R. W. Kennard. Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970.

- [44] H. W. Engl, M. Hanke, and A. Neubauer. *Regularization of Inverse Problems*. Kluwer, 1996.
- [45] M. L. Stein. *Interpolation of Spatial Data: Some Theory for Kriging*. Springer, 1999.
- [46] A. Caponnetto and E. De Vito. Optimal rates for the regularized least-squares algorithm. *Foundations of Computational Mathematics*, 7(3):331–368, 2007.
- [47] D. J. Sutherland. Fixing an error in Caponnetto and de Vito (2007). arXiv:1702.02982v2, 2021.
- [48] S. Fischer and I. Steinwart. Sobolev norm learning rates for regularized least-squares algorithms. *Journal of Machine Learning Research*, 21(205):1–38, 2020.
- [49] S. Smale and D.-X. Zhou. Learning theory estimates via integral operators and their approximations. *Constructive Approximation*, 26:153–172, 2007.
- [50] B. Schölkopf and A. J. Smola. *Learning with Kernels*. MIT Press, 2002.
- [51] I. Steinwart and A. Christmann. *Support Vector Machines*. Springer, 2008.
- [52] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning*. Springer, second edition, 2009.
- [53] C. A. Micchelli and M. Pontil. On learning vector-valued functions. *Neural Computation*, 17(1):177–204, 2005.
- [54] M. A. Álvarez, L. Rosasco, and N. D. Lawrence. Kernels for vector-valued functions: a review. *Foundations and Trends in Machine Learning*, 4(3):195–266, 2012.
- [55] H. Owhadi. Bayesian numerical homogenization. *Multiscale Modeling & Simulation*, 13(3):812–828, 2015.
- [56] H. Owhadi and C. Scovel. *Operator-Adapted Wavelets, Fast Solvers, and Numerical Homogenization*. Cambridge University Press, 2019.
- [57] Y. Chen, B. Hosseini, H. Owhadi, and A. M. Stuart. Solving and learning nonlinear PDEs with Gaussian processes. *Journal of Computational Physics*, 447:110668, 2021.
- [58] H. Owhadi and G. R. Yoo. Kernel flows: from learning kernels from data into the abyss. *Journal of Computational Physics*, 389:22–47, 2019.
- [59] B. Hamzi and H. Owhadi. Learning dynamical systems from data: a simple cross-validation perspective, part I: parametric kernel flows. *Physica D: Nonlinear Phenomena*, 421:132817, 2021.
- [60] C. K. I. Williams and M. Seeger. Using the Nyström method to speed up kernel machines. In *Advances in Neural Information Processing Systems 13*, pages 682–688, 2001.
- [61] P. Drineas and M. W. Mahoney. On the Nyström method for approximating a Gram matrix for improved kernel-based learning. *Journal of Machine Learning Research*, 6:2153–2175, 2005.

- [62] F. Bach. Sharp analysis of low-rank kernel matrix approximations. In *Proceedings of the 26th Annual Conference on Learning Theory*, pages 185–209, 2013.
- [63] A. Rahimi and B. Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems 20*, pages 1177–1184, 2008.
- [64] N. H. Nelsen and A. M. Stuart. The random feature model for input-output maps between Banach spaces. *SIAM Journal on Scientific Computing*, 43(5):A3212–A3243, 2021.
- [65] A. Rudi, L. Carratino, and L. Rosasco. FALKON: an optimal large scale kernel method. In *Advances in Neural Information Processing Systems 30*, 2017.
- [66] G. Meanti, L. Carratino, L. Rosasco, and A. Rudi. Kernel methods through the roof: handling billions of points efficiently. In *Advances in Neural Information Processing Systems 33*, 2020.
- [67] J. Quiñonero-Candela and C. E. Rasmussen. A unifying view of sparse approximate Gaussian process regression. *Journal of Machine Learning Research*, 6:1939–1959, 2005.
- [68] M. K. Titsias. Variational learning of inducing variables in sparse Gaussian processes. In *Proceedings of the 12th International Conference on Artificial Intelligence and Statistics*, pages 567–574, 2009.
- [69] J. Hensman, N. Fusi, and N. D. Lawrence. Gaussian processes for big data. In *Proceedings of the 29th Conference on Uncertainty in Artificial Intelligence*, pages 282–290, 2013.
- [70] A. G. Wilson and H. Nickisch. Kernel interpolation for scalable structured Gaussian processes. In *Proceedings of the 32nd International Conference on Machine Learning*, pages 1775–1784, 2015.
- [71] J. R. Gardner, G. Pleiss, D. Bindel, K. Q. Weinberger, and A. G. Wilson. GPyTorch: blackbox matrix-matrix Gaussian process inference with GPU acceleration. In *Advances in Neural Information Processing Systems 31*, 2018.
- [72] A. G. Wilson and R. P. Adams. Gaussian process kernels for pattern discovery and extrapolation. In *Proceedings of the 30th International Conference on Machine Learning*, pages 1067–1075, 2013.
- [73] D. Duvenaud, H. Nickisch, and C. E. Rasmussen. Additive Gaussian processes. In *Advances in Neural Information Processing Systems 24*, 2011.
- [74] D. Duvenaud, J. R. Lloyd, R. Grosse, J. B. Tenenbaum, and Z. Ghahramani. Structure discovery in nonparametric regression through compositional kernel search. In *Proceedings of the 30th International Conference on Machine Learning*, pages 1166–1174, 2013.
- [75] G. R. G. Lanckriet, N. Cristianini, P. Bartlett, L. El Ghaoui, and M. I. Jordan. Learning the kernel matrix with semidefinite programming. *Journal of Machine Learning Research*, 5:27–72, 2004.
- [76] M. Gönen and E. Alpaydm. Multiple kernel learning algorithms. *Journal of Machine Learning Research*, 12:2211–2268, 2011.

- [77] N. Cristianini, J. Shawe-Taylor, A. Elisseeff, and J. Kandola. On kernel-target alignment. In *Advances in Neural Information Processing Systems 14*, pages 367–373, 2002.
- [78] C. Cortes, M. Mohri, and A. Rostamizadeh. Algorithms for learning kernels based on centered alignment. *Journal of Machine Learning Research*, 13:795–828, 2012.
- [79] J. M. Bates and C. W. J. Granger. The combination of forecasts. *Operational Research Quarterly*, 20(4):451–468, 1969.
- [80] D. H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992.
- [81] L. Breiman. Stacked regressions. *Machine Learning*, 24:49–64, 1996.
- [82] M. Stone. Cross-validated choice and assessment of statistical predictions. *Journal of the Royal Statistical Society: Series B*, 36(2):111–147, 1974.
- [83] G. H. Golub, M. Heath, and G. Wahba. Generalized cross-validation as a method for choosing a good ridge parameter. *Technometrics*, 21(2):215–223, 1979.
- [84] S. Arlot and A. Celisse. A survey of cross-validation procedures for model selection. *Statistics Surveys*, 4:40–79, 2010.
- [85] K. Hornik, M. Stinchcombe, and H. White. Multilayer feedforward networks are universal approximators. *Neural Networks*, 2(5):359–366, 1989.
- [86] D. P. Kingma and J. Ba. Adam: a method for stochastic optimization. In *International Conference on Learning Representations*, 2015.
- [87] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.
- [88] L. Prechelt. Early stopping — but when? In *Neural Networks: Tricks of the Trade*, volume 1524 of *Lecture Notes in Computer Science*, pages 55–69. Springer, 1998.
- [89] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15:1929–1958, 2014.
- [90] R. M. Neal. *Bayesian Learning for Neural Networks*. Springer, 1996.
- [91] J. Lee, Y. Bahri, R. Novak, S. S. Schoenholz, J. Pennington, and J. Sohl-Dickstein. Deep neural networks as Gaussian processes. In *International Conference on Learning Representations*, 2018.
- [92] A. G. de G. Matthews, M. Rowland, J. Hron, R. E. Turner, and Z. Ghahramani. Gaussian process behaviour in wide deep neural networks. In *International Conference on Learning Representations*, 2018.
- [93] Y. Cho and L. K. Saul. Kernel methods for deep learning. In *Advances in Neural Information Processing Systems 22*, pages 342–350, 2009.
- [94] A. Jacot, F. Gabriel, and C. Hongler. Neural tangent kernel: convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems 31*, 2018.

- [95] S. Arora, S. S. Du, W. Hu, Z. Li, R. Salakhutdinov, and R. Wang. On exact computation with an infinitely wide neural net. In *Advances in Neural Information Processing Systems 32*, 2019.
- [96] M. Belkin, S. Ma, and S. Mandal. To understand deep learning we need to understand kernel learning. In *Proceedings of the 35th International Conference on Machine Learning*, pages 541–549, 2018.
- [97] A. Radhakrishnan, D. Beaglehole, P. Pandit, and M. Belkin. Mechanism for feature learning in neural networks and backpropagation-free machine learning models. *Science*, 383(6690):1461–1467, 2024.
- [98] A. N. Kolmogorov. On the representation of continuous functions of several variables by superpositions of continuous functions of one variable and addition. *Doklady Akademii Nauk SSSR*, 114:953–956, 1957.
- [99] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, and M. Tegmark. KAN: Kolmogorov–Arnold networks. arXiv:2404.19756, 2024.
- [100] N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural operator: learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.
- [101] L. Lu, X. Meng, S. Cai, Z. Mao, S. Goswami, Z. Zhang, and G. E. Karniadakis. A comprehensive and fair comparison of two neural operators (with practical extensions) based on FAIR data. *Computer Methods in Applied Mechanics and Engineering*, 393:114778, 2022.
- [102] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [103] M. V. de Hoop, D. Z. Huang, E. Qian, and A. M. Stuart. The cost-accuracy trade-off in operator learning with neural networks. *Journal of Machine Learning*, 1(3):299–341, 2022.
- [104] M. Takamoto, T. Praditia, R. Leiteritz, D. MacKinlay, F. Alesiani, D. Pflüger, and M. Niepert. PDEBench: an extensive benchmark for scientific machine learning. In *Advances in Neural Information Processing Systems 35*, 2022.
- [105] R. Ohana, M. McCabe, L. Meyer, R. Morel, F. Agocs, M. Beneitez, M. Berger, B. Burkhart, S. Dalziel, D. Fielding, et al. The Well: a large-scale collection of diverse physics simulations for machine learning. In *Advances in Neural Information Processing Systems 37*, pages 44989–45037, 2024.
- [106] S. Yu, W. M. Hannah, L. Peng, J. Lin, M. A. Bhourri, R. Gupta, B. Lütjens, J. C. Will, G. Behrens, et al. ClimSim: a large multi-scale dataset for hybrid physics-ML climate emulation. In *Advances in Neural Information Processing Systems 36*, 2023.
- [107] G. H. Golub and C. F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, fourth edition, 2013.
- [108] W. Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

- [109] G. W. Stewart and J.-G. Sun. *Matrix Perturbation Theory*. Academic Press, 1990.
- [110] N. J. Higham. *Accuracy and Stability of Numerical Algorithms*. SIAM, second edition, 2002.